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Investigating sequence-dependent DNA mechanics and nucleosome stability through coarse-grained simulations

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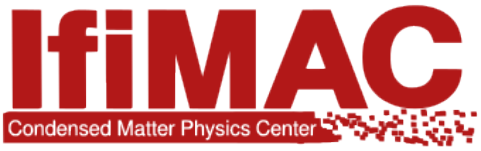
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Abstract

DNA sequence encodes genetic information, but also influences DNA shape, deformability and interactions with proteins. In nucleosomes, these properties affect how readily DNA accommodates the curved path imposed by the histone octamer and therefore contribute to sequence-dependent nucleosome stability.

In this thesis, sequence-dependent nucleosome behaviour was investigated using coarse-grained molecular dynamics simulations combining MADna for DNA with CALVADOS 3 for the histone octamer. Thermodynamic Integration calculations in DNA-only and histone-associated systems were used to determine relative nucleosome-binding free energies for controlled A-rich variants, randomized controls, Widom 601 and two chromosome XIII sequences. The thermodynamic results were then interpreted using analyses of DNA geometry, nucleosomal wrapping, DNA–histone proximity and sequence-dependent crookedness.

The clearest effect was observed for the A-rich series: progressively extending the central adenine tract increased the relative nucleosome-binding preference for 1KX5 over the A-rich variants. This trend correlated strongly with the change in crookedness and was accompanied by greater DNA opening and departures from the compact superhelical geometry. The randomized sequences and Widom 601 produced smaller differences that did not support a robust thermodynamic ordering given the numerical and sampling uncertainty indicated by the consistency test. For the chromosome XIII pair, the preference for $+920$ over -397 agreed qualitatively with the experimental occupancy ordering and was only weakly affected by linker DNA.

Overall, the results show that the coarse-grained framework captures the response to strong, controlled sequence perturbations and connects the calculated relative nucleosome-binding free energies with changes in DNA geometry and nucleosomal wrapping. Intrinsic limitations of the coarse-grained models must also be considered when interpreting the molecular origin of subtler sequence-dependent effects. The approach therefore provides a computationally efficient framework for investigating how DNA sequence and mechanics contribute to nucleosome stability, while identifying the sampling and numerical limitations that constrain subtler sequence-dependent effects.

List of Acronyms

Acronym	Meaning
AFM	atomic force microscopy
ARG	arginine
BBK	Brünger–Brooks–Karplus algorithm
bp	base pair(s)
CALVADOS	coarse-grained protein model
CG	coarse-grained
DNA	deoxyribonucleic acid
dsDNA	double-stranded DNA
ENM	elastic network model
GPU	graphics processing unit
IBWLC	intrinsically bent worm-like chain
LAMMPS	Large-scale Atomic/Molecular Massively Parallel Simulator
LYS	lysine
MADna	Mechanically Accurate DNA
MD	molecular dynamics
NUC	histone-associated nucleosome system
NVT	constant number of particles, volume and temperature
PDB	Protein Data Bank
RMSD	root-mean-square deviation
RMSE	root-mean-square error
SD	standard deviation
SHL	superhelical location
SLURM	Simple Linux Utility for Resource Management
TI	thermodynamic integration
TSS	transcription start site
UAMMD	Universally Adaptable Multiscale Molecular Dynamics
VLMP	Virtual Lab Modeling Platform
VMD	Visual Molecular Dynamics
WCA	Weeks–Chandler–Andersen
WLC	worm-like chain

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Chapter 1

Introduction

1.1 Context and motivation

DNA is commonly introduced as the molecule in which genetic information is stored. This information is encoded in the order of its nucleotide bases and transmitted whenever cells divide or organisms reproduce. However, the information encoded in its sequence represents only one aspect of DNA. DNA is also a physical object. Its bases and sugar–phosphate backbone form a three-dimensional structure that interacts with the surrounding environment and responds to mechanical forces. DNA can bend, twist, stretch and undergo local deformations, and these properties depend on the underlying sequence. Different nucleotide arrangements can therefore give rise to different preferred shapes and elastic responses, influencing how DNA interacts with proteins.

This mechanical dimension is particularly important in eukaryotic genomes, where DNA is not freely exposed in solution but is packaged into chromatin. Its fundamental repeating unit is the nucleosome, in which approximately 147 base pairs of DNA wrap around a histone octamer. Nucleosome formation requires the double helix to adopt a strongly curved conformation while interacting with the histone surface. Its stability depends not only on the electrostatic attraction between the negatively charged DNA backbone and the positively charged histones, but also on how readily the preferred shape and mechanical response of a given sequence can accommodate the nucleosomal geometry. Several studies have shown that DNA sequence can influence nucleosome positioning, stability and accessibility. Sequence-dependent effects can arise from local structural features, such as minor groove width, electrostatic potential and base-pair step geometry, as well as from more global mechanical properties, such as intrinsic curvature and bendability.

An important example of this connection is protein–DNA recognition. Although recognition is often associated with direct contacts between amino acids and specific bases, proteins can also recognize structural features produced by the sequence. Rohs et al. showed that A/T-rich regions are often associated with narrow minor grooves, where the negative electrostatic potential is enhanced, favouring interactions with positively charged amino acids, particularly arginine residues [1]. The sequence therefore defines not only the chemical identity of the bases, but also the three-dimensional and electrostatic landscape encountered by proteins.

In the nucleosomal context, sequence-dependent mechanics affects how easily DNA can adapt to the histone surface. Since histones interact mainly with the sugar–phosphate backbone, the effect of sequence is often indirect: different base orders change the preferred geometry and deformability of the double helix, and therefore the energetic cost required to place the DNA backbone along the nucleosomal path. Chua et al. showed that nucleosome stability depends on the interplay between sequence-dependent DNA conformation and flexibility and the geometrical constraints imposed by the histone core [2]. Along similar lines, Freeman et al. showed that the intrinsic shape of DNA can

strongly influence sequence-dependent nucleosome affinity [3]. In this view, nucleosome formation is not favoured simply by generic flexibility, but by the compatibility between the preferred geometry of a given DNA sequence and the curved conformation imposed by the histone core.

More recent high-throughput experiments have extended this picture to genome-scale measurements of DNA mechanics. Loop-seq measures the cyclizability of DNA fragments, providing an experimental measure of their propensity to form loops. Using this approach, Basu et al. showed that DNA bendability varies systematically with sequence and identified characteristic mechanical patterns around nucleosomes and transcription start sites [4]. Subsequent measurements showed how the occurrence and spatial distribution of short sequence motifs influence DNA bendability, supporting the idea of a mechanical code of the genome: a set of sequence-dependent physical properties with potential consequences for chromatin organization and DNA accessibility [5].

Among sequence motifs with distinctive mechanical properties, A-rich tracts are particularly relevant. A-tracts have long been associated with intrinsic curvature and unusual elastic behaviour, but their effect cannot be described as a simple increase in flexibility. Marin-Gonzalez et al. showed that phased A-tracts can produce long-range bending and different mechanical responses depending on the force regime: intrinsic curvature softens their response at low forces, whereas at higher forces they display considerable stretching rigidity [6]. Their behaviour therefore depends on how the tracts are arranged and on the deformation imposed on the DNA. This makes A-rich sequences useful controlled systems for investigating how sequence-dependent mechanics affects relative nucleosome-binding free energies.

The sequence dependence of DNA mechanics may also influence DNA accessibility within chromatin. The presence of a transcription factor recognition motif does not by itself determine whether binding will occur, because its accessibility depends on its position and structural context within the nucleosome. Mariani et al. showed that transcription factor binding is primarily mediated by the recognition motif but is also regulated by the broader nucleosomal sequence context. They proposed that local DNA bendability contributes to determining where transcription factors can bind within nucleosomes [7]. DNA mechanics may therefore provide an additional regulatory layer connecting the surrounding sequence with the accessibility of protein-binding sites.

These questions require a description that balances molecular detail and computational efficiency. Fully atomistic simulations can describe local chemical interactions and base-pair-step geometries with high resolution, but they are computationally expensive when many sequences or systematic sequence modifications have to be compared. Many simple continuum descriptions, such as elastic polymer models, are efficient and useful at larger length scales, but do not explicitly retain the local sequence dependence that is important for A-tracts, groove geometry and nucleosome positioning. Nucleosomal DNA lies between these two limits: it is long enough for global bending, curvature and wrapping to matter, but short enough that local sequence-dependent features can still influence the behaviour of the whole molecule. For this reason, coarse-grained sequence-dependent models provide a useful compromise. In this thesis, DNA is described using the MADna model, a coarse-grained model in which each nucleotide is represented by three interaction sites corresponding to the sugar, base and phosphate groups. Sequence dependence is introduced through a step-dependent parametrization of the bonded interactions, developed to reproduce the conformational and elastic properties of double-stranded DNA [8]. This makes it possible to study how different sequences affect DNA conformations and relative nucleosome-binding free energies, while keeping the simulations computationally feasible.

Understanding these effects is important because many biological processes depend on the physical response of DNA to deformation and protein binding. During transcription, replication, DNA repair and chromatin remodelling, the double helix is not only read as a sequence, but is also bent, wrapped, locally exposed or mechanically constrained. A better description of sequence-dependent DNA mechanics can therefore help investigate why different DNA regions may differ in their energetic compatibility with nucleosome formation and in their accessibility to regulatory proteins. In this way, DNA mechanics can provide a physical link between local sequence properties and larger-scale features of chromatin organization and gene regulation.

1.2 Aim of the thesis

The aim of this thesis is to investigate how DNA sequence affects relative nucleosome-binding free energies and to identify the molecular structural changes associated with these thermodynamic differences. Two sets of sequences are considered. The first uses the DNA sequence from the nucleosome core particle structure 1KX5 as a structurally characterized reference and compares it with adenine-rich variants, random controls and the Widom 601 nucleosome-positioning sequence. The second consists of genomic sequences from chromosome XIII of *Saccharomyces cerevisiae*, selected from experimental nucleosome-occupancy data to compare the computational predictions with an experimentally observed difference in nucleosome preference [9].

The focus on A-rich variants is motivated by the peculiar mechanical properties of A-tracts discussed above. These motifs can affect DNA shape, intrinsic curvature, bendability and local deformability, but their behaviour cannot be summarized by a simple increase or decrease in flexibility. In this thesis, progressively extending the central adenine tract provides a controlled way of introducing an increasingly long and comparatively rigid segment at the centre of the sequence, while leaving the surrounding sequence unchanged. This makes it possible to investigate how a localized modification of DNA mechanics affects the relative preference for nucleosome formation.

To address these questions, molecular dynamics simulations are performed using the sequence-dependent coarse-grained MADna model. The relative nucleosome-binding free energies calculated with this approach are relevant to nucleosome positioning. Because an intact nucleosome can spontaneously reposition along DNA through thermal fluctuations, differences in relative binding free energy contribute to determining which DNA regions are preferentially occupied at equilibrium.

Finally, the free-energy results are complemented by structural analyses of DNA–histone proximity and of the geometry adopted by DNA within the nucleosome. A central advantage of the simulation framework is that the same sequence can be examined under controlled conditions in both the DNA-only and histone-associated systems. The resulting trajectories show where structural changes occur along the DNA and which conformations are explored over time. This makes it possible to combine relative nucleosome-binding free energies with structural information that would be difficult to obtain simultaneously from experiments alone. The objective is therefore not only to calculate relative nucleosome-binding free energies, but also to determine how sequence-dependent differences are expressed through local geometrical adjustments, changes in DNA–histone proximity and alterations of the overall wrapping of DNA around the histone core.

1.3 Thesis outline

Chapter 2 introduces the theoretical and computational background required for the thesis. It first describes double-stranded DNA as a polymer with sequence-dependent mechanical properties, focusing on its local structure, elastic behaviour, intrinsic curvature and the distinctive properties of A-rich sequences. It then discusses the nucleosome as a mechanically constrained DNA–protein complex and reviews how DNA sequence can influence nucleosome positioning and stability. The final part introduces coarse-grained modelling, the MADna and CALVADOS 3 models, their coupling within the nucleosomal system, molecular dynamics simulations, free-energy calculations and Thermodynamic Integration.

Chapter 3 describes the computational methods used in this work. It presents the DNA sequences considered, including the controlled sequence variants defined relative to 1KX5 and the genomic sequences from chromosome XIII of *S. cerevisiae*. It then describes the construction of the systems used in the simulations, the calculation of relative nucleosome-binding free energies and the structural and sequence-based analyses applied to the resulting trajectories.

Chapter 4 presents and discusses the results. It first assesses the reliability and numerical resolution of the free-energy calculations. The relative nucleosome-binding free energies are then examined for the A-rich variants, the random controls, Widom 601 and the chromosome XIII sequences, including the

effect of linker DNA in the genomic systems. Finally, the thermodynamic trends are compared with sequence-based crookedness and with trajectory-based analyses of local DNA geometry, nucleosomal wrapping and DNA–histone proximity.

Chapter 5 summarizes the main findings and discusses their implications for the relationship between DNA sequence, mechanics and nucleosome stability. It also outlines the limitations of the present computational approach and possible directions for future work.

The Appendix provides supplementary figures and tables, the complete DNA sequences used in the simulations, the Thermodynamic Integration derivative profiles and additional validation and convergence analyses.

Chapter 2

Theoretical and Computational Background

This chapter introduces the theoretical and computational background required for the work presented in the following chapters. The first part reviews the structure and sequence-dependent mechanics of DNA, from its local geometry and elastic response to the distinctive behaviour of A-rich sequences. It then considers how these properties influence the deformation, positioning and relative stability of DNA within the nucleosome. The final part introduces coarse-grained modelling, the MADna model for DNA, the CALVADOS 3 model for the histone proteins and the effective interactions used to couple the two molecular components. It then presents the statistical-mechanical foundations of molecular dynamics, free-energy calculations and Thermodynamic Integration.

2.1 DNA structure and mechanical response

2.1.1 Structure and local geometry of double-stranded DNA

Double-stranded DNA consists of two chains of nucleotides wound around a common axis to form a double helix. Each nucleotide contains a deoxyribose sugar, a phosphate group and one of four nitrogenous bases: adenine, thymine, guanine or cytosine. The nucleotides within each strand are connected through the sugar and phosphate groups, forming the sugar–phosphate backbone.

The two ends of a strand are conventionally labelled $5'$ and $3'$, according to the numbered carbon atoms of the deoxyribose sugar involved in the backbone. This gives each strand a defined chemical orientation. In double-stranded DNA, the two backbones run in opposite directions—one from $5'$ to $3'$ and the other from $3'$ to $5'$—and are therefore described as antiparallel. The bases point towards the interior of the double helix, where they form specific pairs through hydrogen bonds: adenine pairs with thymine, while guanine pairs with cytosine. These canonical pairings are known as Watson–Crick–Franklin base pairs, and their specificity makes the two DNA strands complementary. These structural features are summarized schematically in Figure 2.1.

Under conditions close to those found in cells, double-stranded DNA most commonly adopts the B form. B-DNA is a right-handed helix with an approximate diameter of 2 nm. The helical axis is the reference line passing through the centre of the double helix, and the base-pair planes are oriented approximately perpendicular to it. Consecutive base pairs are separated by about 0.34 nm along the helical axis, and approximately 10–10.5 base pairs complete one turn of the helix, corresponding to a helical pitch of about 3.4 nm. As shown in Figure 2.1, these values describe an average geometry: the actual structure is not perfectly uniform and can vary with sequence and environmental conditions.

The arrangement of the two sugar–phosphate backbones also creates two unequal channels along the helix, known as the major and minor grooves. The major groove is wider, whereas the minor groove is narrower. Both grooves expose portions of the base pairs to the surrounding solvent and provide

surfaces through which proteins can interact with DNA. The width and shape of these grooves can vary locally with nucleotide sequence. This sequence dependence will be discussed in Section 2.2, together with the role of DNA shape in protein recognition.

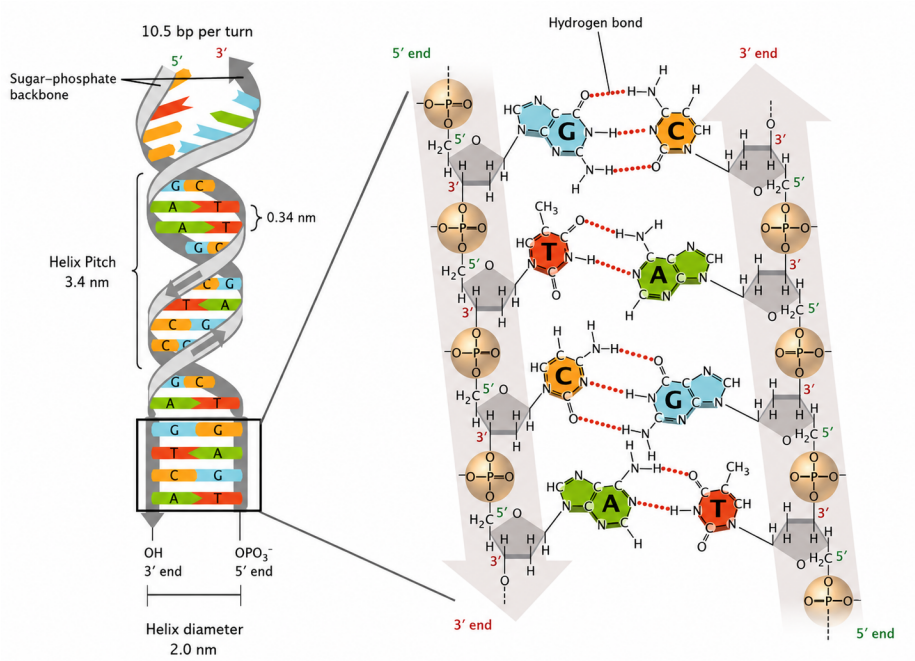


Figure 2.1: Schematic representation of B-form double-stranded DNA. The left-hand view shows the approximate helical dimensions and the sugar–phosphate backbone. The enlarged view illustrates the antiparallel orientation of the two strands, their 5′ and 3′ ends, canonical Watson–Crick base pairing, and the corresponding hydrogen bonds. Source: L. A. Pray, "Discovery of DNA Structure and Function: Watson and Crick," *Nature Education*, 2008 [10].

While these average quantities describe the overall organization of the double helix, its local geometry requires a more detailed description. The geometry of an individual base pair concerns the relative position and orientation of the two paired bases, whereas a base-pair step describes the arrangement of two consecutive base pairs. A base-pair step is characterized by three translational parameters—shift, slide and rise—and three rotational parameters—tilt, roll and twist. Shift and slide describe lateral displacements between consecutive base pairs, while rise measures their separation along the local direction approximately normal to the base-pair planes. Tilt and roll describe bending towards the backbone and the grooves, respectively, whereas twist measures the relative rotation between neighbouring base pairs. Although these parameters describe distinct geometrical components, they should not be regarded as completely independent deformation modes. A change in the larger-scale path of DNA can involve coupled translational and rotational adjustments distributed across several consecutive base-pair steps. Different combinations of the six parameters may therefore contribute to a similar overall deformation. Moreover, because the parameters have different units and characteristic ranges, their numerical amplitudes cannot be compared directly to establish which deformation is mechanically or energetically the most important. Their preferred values are not identical for every base-pair step, providing a first connection between local geometry and nucleotide sequence. The six base-pair-step parameters are illustrated schematically in Figure 2.2.

2.1.2 Elastic response of DNA

The local parameters introduced above describe DNA at the scale of individual base-pair steps. When many consecutive steps are considered together, their translations and rotations combine into larger-scale deformations of the double helix. This section considers how DNA responds when such deformations are imposed over longer portions of the molecule. On length scales of approximately 10–100 nm, double-stranded DNA can be described as a semiflexible polymer: it is neither a rigid rod nor

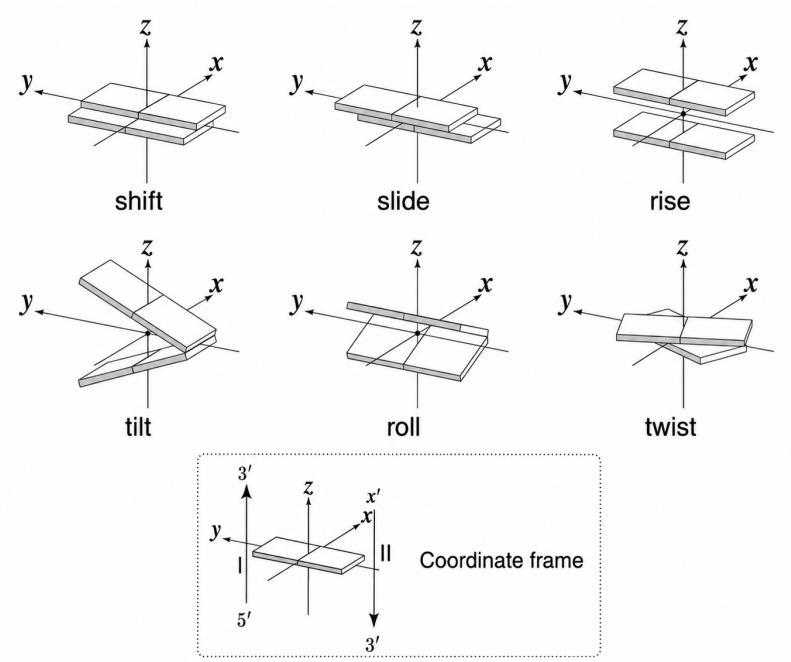


Figure 2.2: The six base-pair-step parameters describing the relative position and orientation of consecutive base pairs. Shift, slide and rise are translations along the local coordinate axes, whereas tilt, roll and twist are the corresponding rotations. Source: A. Capobianco, A. Landi and A. Peluso, "Duplex DNA Retains the Conformational Features of Single Strands: Perspectives from MD Simulations and Quantum Chemical Computations," *International Journal of Molecular Sciences*, 2022 [11].

a freely flexible chain. Thermal fluctuations continuously change its conformation, while its elastic stiffness opposes deformation away from the preferred state.

This mechanical response can be organized into three main deformation modes. Bending changes the direction of the helical axis, twisting changes the accumulated rotation around it, and stretching changes the contour length of the molecule. Because bending is directly involved in the wrapping of DNA around the histone octamer, it provides the natural starting point for describing DNA elasticity in the context of this thesis.

The simplest continuum description of DNA bending is the worm-like chain (WLC) model. The molecule is represented by a continuous curve parameterized by the contour coordinate s, with $\mathbf{t}(s)$ denoting the unit tangent to the helical axis. In its simplest form, the model assigns an energetic cost to changes in the direction of this tangent:

$$E_{\text{bend}} = \frac{\kappa}{2} \int_0^L \left| \frac{d\mathbf{t}(s)}{ds} \right|^2 ds, \tag{2.1}$$

where L is the contour length and κ is the bending rigidity. Strongly curved conformations therefore have a higher bending energy than conformations in which the direction of the helical axis changes more gradually.

The bending rigidity is commonly expressed through the persistence length,

$$L_p = \frac{\kappa}{k_{\text{B}}T}, \tag{2.2}$$

which represents the characteristic distance over which the direction of the polymer remains correlated. In the worm-like chain model, this loss of orientational correlation is described by

$$\langle \mathbf{t}(s) \cdot \mathbf{t}(0) \rangle = e^{-s/L_p}. \tag{2.3}$$

Under typical solution conditions, double-stranded DNA has a persistence length of approximately 50 nm, corresponding to roughly 150 base pairs [12, 13]. The approximately 147 base pairs associated with the histone octamer have a similar contour length, but are wrapped into a nucleosome with a diameter of only about 11 nm. Nucleosome formation therefore requires DNA to undergo pronounced bending on a spatial scale much smaller than its persistence length.

Bending alone, however, does not provide a complete description of the mechanical response of DNA. The other two deformation modes introduced above concern changes in twist and contour length. A torsional deformation occurs when the double helix is overwound or underwound relative to its preferred state. For sufficiently small deformations, the associated energy increases approximately quadratically with the change in twist, and the resistance to this deformation is quantified by the torsional stiffness. This property is relevant whenever DNA is rotationally constrained or forced to adopt a geometry with a twist different from its preferred value.

The stretching response of DNA depends on the applied force. At low forces, extension mainly results from the progressive straightening of thermally fluctuating conformations. This regime is therefore dominated by entropy. Once the molecule is nearly aligned, further extension requires deformation of the double helix itself and is governed by its stretch modulus. Bending, twisting and stretching are useful distinct descriptions, but they are not completely independent: because DNA is a chiral helix, twist can be coupled both to extension [14] and to bending [15].

Together, these elastic properties determine how DNA responds to mechanical constraints. They describe the resistance of the molecule to deformation, but they do not yet specify the conformation that a particular DNA sequence prefers in the absence of external constraints. This complementary aspect—the intrinsic shape of DNA—is introduced in the following section.

2.1.3 Intrinsic shape, curvature and crookedness

DNA shape and mechanical response can be described through related but distinct concepts. Flexibility describes how readily DNA fluctuates away from its preferred conformation and can be characterized at different length scales, from local base-pair-step fluctuations to the effective bending response of the entire molecule. Intrinsic curvature instead refers to the shape adopted by DNA in its minimum-energy state, even in the absence of external forces or protein binding. This preferred geometry can be described locally through curvature and summarized at a more global level through the crookedness parameter.

In the standard WLC model introduced above, the bending energy is lowest when the direction of the DNA axis does not change along the contour, corresponding to a straight conformation. For a planar DNA conformation, the amount by which the axis bends at a given position can be described by its local curvature $c(s)$, which is zero for a straight segment. If the molecule has a preferred curved shape, a second quantity, $c_0(s)$, can be introduced to describe its preferred local curvature. In this case, the bending energy depends on the difference between the curvature actually adopted by the molecule and the curvature that it would prefer to have:

$$E_{\text{bend}} = \frac{\kappa}{2} \int_0^L [c(s) - c_0(s)]^2 \, ds. \tag{2.4}$$

Here, κ is the bending rigidity introduced in the previous section. If $c_0(s) = 0$, the minimum-energy conformation is straight and the standard WLC description is recovered. If $c_0(s) \neq 0$, the energy is instead minimized when the molecule follows its intrinsically preferred curved shape. This extension is commonly referred to as the intrinsically bent worm-like chain model (IBWLC). The scalar formulation above is appropriate for a planar description, such as that used to analyse DNA contours deposited on a surface. In three dimensions, however, the same curvature magnitude can correspond to bending in different directions. The preferred deformation must therefore be specified through directional curvature components defined relative to a local frame. Equation 2.4 should consequently be regarded

as a simplified planar representation of intrinsic curvature rather than as a complete three-dimensional elastic model.

The distinction between flexibility and intrinsic curvature is important when interpreting an observed DNA conformation. A flexible molecule with a straight preferred shape can frequently adopt curved conformations because of thermal fluctuations. Conversely, a molecule can have a strongly curved preferred shape while remaining relatively resistant to fluctuations around that shape. A curved configuration alone therefore does not establish that the DNA is particularly flexible.

This distinction has been demonstrated experimentally for phased A-tract sequences. By analysing DNA conformations obtained through atomic force microscopy (AFM), Marin-Gonzalez et al. found that a standard WLC description could not fully reproduce their conformational statistics. Including an intrinsically curved reference shape allowed them to separate thermal bending from static curvature and showed that the pronounced bending of these sequences originated primarily from their intrinsic shape rather than from increased bending flexibility [6]. The properties of A-tracts will be discussed more specifically in Section 2.2.3.

The preferred geometry of a finite DNA fragment can also be summarized through the crookedness parameter, β, originally introduced by Marin-Gonzalez et al. [16]. Crookedness describes how much the path followed by the base-pair centres departs, as a whole, from a straight configuration. In the convention used in this thesis, $\beta = 0$ represents the limiting case of a straight DNA axis, while larger values indicate a more crooked conformation [17].

Crookedness is a structural descriptor and should not be interpreted as a direct measure of bending flexibility. Nevertheless, previous atomistic and coarse-grained analyses found systematic relationships between equilibrium crookedness and global mechanical properties, including crookedness stiffness and persistence length, connecting the preferred geometry of a DNA fragment with its mechanical response [17, 18]. In this thesis, crookedness is calculated directly from the nucleotide sequence and used as a compact descriptor of the preferred global DNA geometry. The procedure used to estimate β is described in Chapter 3.

These considerations are also relevant to nucleosomal DNA. The cost of wrapping a sequence around the histone octamer depends not only on its resistance to bending, but also on the difference between its preferred shape and the path imposed by the histone core. Intrinsic curvature may reduce the deformation cost when it is compatible with the nucleosomal geometry, whereas an incompatible preferred curvature may oppose wrapping. Crookedness can therefore be investigated as a compact sequence-based proxy for DNA shape and mechanical response. However, because it does not retain information about the position and direction of individual local deformations, it cannot by itself determine whether a sequence is geometrically compatible with the nucleosomal path. Its relation to relative nucleosome preference is therefore tested against the free-energy results in Chapter 4.

2.2 Sequence-dependent DNA shape and mechanics

2.2.1 Sequence dependence at the base-pair-step level

The previous section described the structure and mechanical response of DNA in general terms, without yet considering in detail how these properties vary with nucleotide sequence. To understand the origin of this dependence, we now return to the local scale, where an important part of the sequence-dependent behaviour arises from the interactions between neighbouring base pairs.

As introduced in Section 2.1.1, the geometry of a base-pair step can be described by shift, slide, rise, tilt, roll and twist. In a thermally fluctuating DNA molecule, these parameters do not remain fixed but vary around characteristic equilibrium values. These equilibrium values define the preferred local geometry of the step, while the range of the fluctuations provides information about how easily it can be deformed.

Both aspects depend on sequence. Base stacking and backbone constraints change with the identities

of neighbouring base pairs, so different steps can have different preferred geometries and different local stiffnesses. A step may, for example, favour a particular value of roll while strongly resisting deviations from it. Another step may have a similar preferred roll but explore a broader range of values. Sequence dependence can therefore modify the local shape of DNA, its deformability, or both.

The base-pair step provides a useful local unit for describing these effects, although it should not be interpreted as a completely isolated element. The conformation of a step can also be influenced by its surrounding sequence, and deformations of neighbouring steps may be correlated. Nevertheless, the step-level description provides a natural local connection between the order of the bases and the mechanical properties of the double helix.

The effects associated with an individual step may be small, but they can accumulate along the DNA. Depending on the order and spacing of the steps, local variations in roll, twist and the other geometrical parameters may reinforce or compensate one another. Their combination can therefore produce larger-scale features such as preferred bending directions, changes in helical organization and sequence-dependent groove profiles.

This local picture also provides the physical basis for sequence-dependent models such as MADna. In such models, changing the nucleotide sequence changes the preferred geometry and elastic response encoded by the local interactions. Two DNA fragments of the same length but with different sequences therefore represent different mechanical systems, rather than identical polymers carrying different base labels. The implementation of this sequence dependence in MADna will be described later in this chapter [8].

2.2.2 Sequence-dependent groove geometry and shape readout

Changes in the preferred geometry of individual base-pair steps can combine along the DNA and produce structural effects at different length scales. As discussed in Section 2.1.3, one possible consequence is intrinsic curvature. Another is a local change in the width and shape of the major and minor grooves. Since these grooves provide the main regions through which proteins access the chemical groups of DNA, their sequence-dependent geometry can influence protein recognition. As shown schematically in Figure 2.3, the two grooves expose different portions of the base pairs and provide distinct surfaces for interactions with proteins.

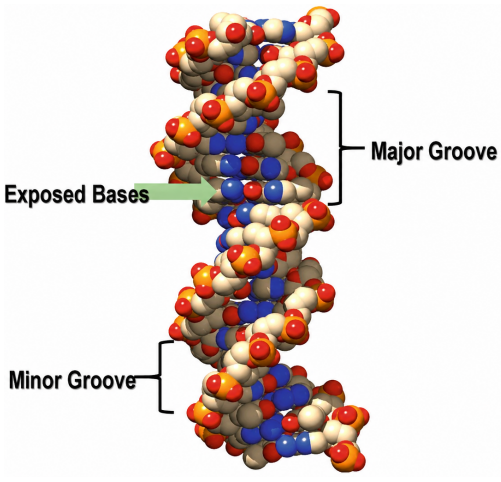


Figure 2.3: Space-filling representation of double-stranded DNA showing the major and minor grooves. The wider major groove exposes the edges of the base pairs more prominently, whereas the minor groove forms a narrower channel along the double helix. Source: S. Roy, "Genes, DNA, and Protein–DNA Interactions," Springer Nature, 2025 [19].

In the major groove, the exposed edge of each base pair presents a characteristic pattern of hydrogen-bond donors and acceptors. Proteins can use these chemical patterns to distinguish between base pairs through direct contacts, a mechanism commonly referred to as direct readout. By contrast, the minor

groove does not present a unique hydrogen-bonding pattern for every base pair, making recognition through base-specific contacts more limited [1].

The minor groove nevertheless contains sequence information through its shape. Its width varies along the double helix, reflecting the combined effects of local base-pair and base-pair-step geometries. A/T-rich regions, particularly A-tracts, are frequently associated with a narrow minor groove, although its precise geometry also depends on the arrangement of the bases within the region and on the surrounding sequence.

Changes in groove width also affect the electrostatic environment encountered by a protein. The phosphate groups give the DNA backbone its negative charge. When the minor groove narrows, the two sides of the backbone are brought closer together in space, enhancing the negative electrostatic potential within the groove without changing the total charge of the molecule. Sequence-dependent shape can therefore create local electrostatic features that are not evident from the linear sequence alone.

Rohs et al. showed that arginine residues are particularly prone to form contacts with narrow minor grooves. The positive charge and geometry of the arginine side chain allow it to interact favourably with the enhanced negative potential found in these regions. These observations show that proteins can recognize the shape and electrostatic environment created by a sequence, rather than relying exclusively on base-specific contacts, a mechanism generally described as shape readout [1].

The same principle is relevant to nucleosomal DNA, where the minor groove periodically faces the histone surface and many contacts involve the sugar–phosphate backbone. Groove width and phosphate positioning can influence how easily favourable contacts are formed. Their specific contribution to nucleosome stability, however, cannot be determined from groove geometry alone and will be considered together with DNA deformation and histone–DNA interactions in Section 2.3.

A-tracts are particularly relevant in this context because they bring together several of the effects described above, including minor-groove narrowing, changes in the local electrostatic environment and intrinsic curvature over longer length scales. They therefore provide a useful model system for examining how a specific sequence motif can influence both local DNA structure and larger-scale mechanical behaviour. Their properties are examined in the following subsection.

2.2.3 Mechanical properties and intrinsic curvature of A-tracts

A-tracts are commonly defined as stretches of at least four consecutive A:T base pairs that do not contain an internal TpA step, meaning a thymine followed by an adenine along the same strand. This definition distinguishes A-tracts from generic A/T-rich regions, which may contain the same overall proportion of adenine and thymine but arrange them differently. A continuous run of adenines on one strand, paired with thymines on the complementary strand, represents the homopolymeric case usually denoted as poly(dA:dT) in the literature. These distinctions are important because DNA mechanics depends not only on base composition, but also on the order in which the bases occur along the sequence [6].

At the local scale, A-tracts are associated with a characteristic geometry that often includes minor-groove narrowing. As discussed in the previous subsection, this narrowing modifies the electrostatic environment within the groove. A-tracts can also introduce a preferred direction of bending in the DNA helical axis. Their structural effect, however, cannot be described by considering only the local geometry within each tract, because the overall curvature also depends on how multiple tracts are positioned relative to one another.

This dependence is a general consequence of the helical organization of DNA rather than a property unique to A-tracts. Since B-DNA completes approximately one turn every 10 to 10.5 base pairs, motifs separated by about one helical turn tend to lie on the same face of the double helix. If A-tracts are repeated with this spacing, their preferred bending directions can become aligned and reinforce one another, producing a pronounced curvature over longer distances. These sequences are generally

described as phased A-tracts. If the same motifs are separated by approximately half a helical turn, they are in opposite rotational phases, meaning that their preferred bending directions point towards opposite sides of the helix and can partially compensate for one another. A-tracts nevertheless provide a useful example because their characteristic local geometry can be arranged at different rotational phases to enhance or reduce the resulting global curvature.

The curvature produced by phased A-tracts should not be interpreted as evidence that these sequences are simply more flexible. As discussed in Section 2.1.3, flexibility determines the amplitude of fluctuations around a preferred conformation, whereas intrinsic curvature describes the conformation itself. Marin-Gonzalez et al. used AFM measurements together with an intrinsically bent worm-like chain model to separate these contributions. Their results showed that the pronounced long-range bending of the examined phased A-tract sequences originated primarily from an intrinsically curved preferred shape, rather than from an increased bending flexibility [6].

Their mechanical response also depends on the deformation regime. At low forces, straightening an intrinsically curved molecule can make it appear more compliant because part of its extension is obtained by removing the pre-existing curvature. This does not imply that the double helix is locally soft. At higher forces, when the molecule is already nearly straight, further extension probes the stretching stiffness of the helix itself, and A-tract-containing DNA can display a comparatively high stretch modulus. Describing A-tracts simply as either flexible or rigid would therefore conceal the distinction between their preferred shape and their resistance to different types of deformation.

Their relevance is not limited to model sequences, since A-tracts are widespread in genomic DNA and frequently occur in regulatory regions, where they can contribute to nucleosome organization [9]. They consequently provide a clear example of how local sequence features can influence DNA structure across several length scales. Minor-groove narrowing is a local property, while the coherent alignment of preferred bending directions can generate curvature over a much longer region. At the nucleosomal scale, the effect of an A-rich sequence will therefore depend not only on its adenine content, but also on tract length, sequence arrangement and compatibility with the curvature and rotational orientation required by the histone-bound DNA path. This point will be developed in Section 2.3.

2.2.4 Cyclizability as a probe of sequence-dependent DNA mechanics

After considering A-tracts as a specific example, a broader question is how sequence-dependent DNA mechanics can be compared experimentally across many different sequences. Cyclizability provides one such experimental proxy. It describes the propensity of a DNA fragment to form a loop by bringing its two ends close together with a compatible orientation. Because loop formation requires both bending and rotational alignment, cyclizability is sensitive to several properties of the molecule, including its bending response, intrinsic shape and twist.

Basu et al. developed loop-seq, a high-throughput assay that makes it possible to compare the cyclizability of many DNA sequences in parallel. In this experiment, a library of DNA fragments containing different central sequences is prepared with complementary single-stranded overhangs at the two ends. If a fragment bends sufficiently, the overhangs can hybridize and close the molecule into a loop. Molecules that remain open retain accessible ends and can be selectively degraded, enriching the sequences that loop more efficiently. Comparing their abundance with that in an untreated control provides a measure of the relative looping propensity of each sequence. By accounting for the effect of the experimental geometry on end orientation, the authors defined a quantity called intrinsic cyclizability [4].

Despite its name, intrinsic cyclizability should not be interpreted as a direct measurement of intrinsic curvature alone, nor as a pure measurement of bending flexibility. A sequence may loop efficiently because it fluctuates easily away from its preferred conformation, because its preferred shape is already curved towards a looped state, or because its twist favours the required alignment of the two ends. Different combinations of these properties can therefore produce similar cyclizability values. Cyclizability is best regarded as an effective observable of the ability of a sequence to adopt a particular

strongly bent conformation.

High-throughput measurements have shown that this property cannot be predicted from nucleotide composition alone. Both the occurrence of particular sequence motifs and their relative positions contribute to cyclizability. For example, A/T-containing motifs can promote or oppose global bending depending on whether their preferred bending directions are aligned or are in opposite rotational phases. This observation extends the principle introduced for phased A-tracts: the mechanical effect of a motif depends not only on its presence, but also on where it occurs along the helical repeat [5].

These results support the idea that genomic sequences contain a mechanical code. Here, the term does not refer to a code analogous to the genetic code. It indicates that the order of nucleotides encodes physical properties, including preferred shape, bendability and rotational organization. These properties can influence biological processes that require DNA deformation, even when the underlying nucleotide sequence is not recognized through direct base-specific contacts.

This mechanical information is particularly relevant to nucleosome organization. Measurements in *Saccharomyces cerevisiae*, a species of budding yeast, identified regions of unusually low intrinsic cyclizability within nucleosome-depleted regions upstream of transcription start sites. Across nucleosome arrays, DNA near nucleosome dyads showed, on average, higher cyclizability than DNA in linker regions. These observations are consistent with the different mechanical requirements of the two environments, since nucleosomal DNA must adopt a strongly bent path around the histone core [4].

Loop formation and nucleosomal wrapping impose different geometrical constraints. Cyclizability therefore supports the broader idea that DNA sequence encodes mechanical information, but it should not be interpreted as a direct measurement of nucleosome stability. The ability to bend is only one contribution: stability also depends on the compatibility between the preferred DNA shape and the nucleosomal path, on groove orientation and on direct histone–DNA interactions. The relative nucleosome-binding free energies calculated in this thesis combine the alchemical transformations performed in the DNA-only and histone-associated systems, allowing these mechanical and interaction contributions to be examined in relation to nucleosome association. To understand the physical basis of this comparison, the following section introduces the structure and sequence-dependent mechanics of nucleosomal DNA.

2.3 Nucleosomal DNA and sequence-dependent stability

2.3.1 Structure of the nucleosome core particle

The previous section showed that DNA sequence can influence preferred shape, deformability and cyclizability. To understand how these properties become relevant to nucleosome stability, it is first necessary to consider the specific geometry that DNA adopts when associated with histones.

In eukaryotic cells, DNA is packaged into chromatin. Its fundamental repeating unit is the nucleosome core particle, which consists of a segment of double-stranded DNA wrapped around a histone octamer. The core particle contains approximately 147 base pairs of DNA. In chromatin, neighbouring core particles are connected by stretches of linker DNA, whose length can vary. The term nucleosome is therefore sometimes used more broadly to include the core particle together with its adjoining linker regions.

Wrapping DNA around histones contributes to genome compaction, but it also limits the direct accessibility of the wrapped sequence to transcription factors and other DNA-binding proteins. Nucleosomal DNA is not permanently inaccessible: DNA can transiently unwrap, while nucleosomes can reposition either spontaneously through thermal fluctuations or through the action of chromatin remodellers. In the spontaneous process, the relative probabilities of alternative positions are governed by their free-energy differences. Nevertheless, the nucleosome acts as a regulated physical barrier between the DNA and its surrounding molecular environment.

Histones are positively charged proteins that associate to form the protein core of the nucleosome.

The histone octamer contains eight histone molecules, corresponding to two copies of each of the four core histones, H2A, H2B, H3 and H4. Two H3–H4 dimers associate to form a central tetramer, while two H2A–H2B dimers bind on either side. Together, these proteins form the surface around which the DNA is wrapped.

In addition to their structured globular regions, the core histones contain flexible terminal regions, particularly N-terminal tails, that extend from the nucleosome core. These tails are rich in positively charged residues and can interact with nucleosomal DNA, linker DNA and, within chromatin, neighbouring nucleosomes. They therefore contribute to electrostatic stabilization and to the regulation of chromatin organization.

The organization of the nucleosome core particle is shown in Figure 2.4, using the crystallographic structure with PDB identifier 1KX5 determined at a resolution of approximately 1.9 Å [20].

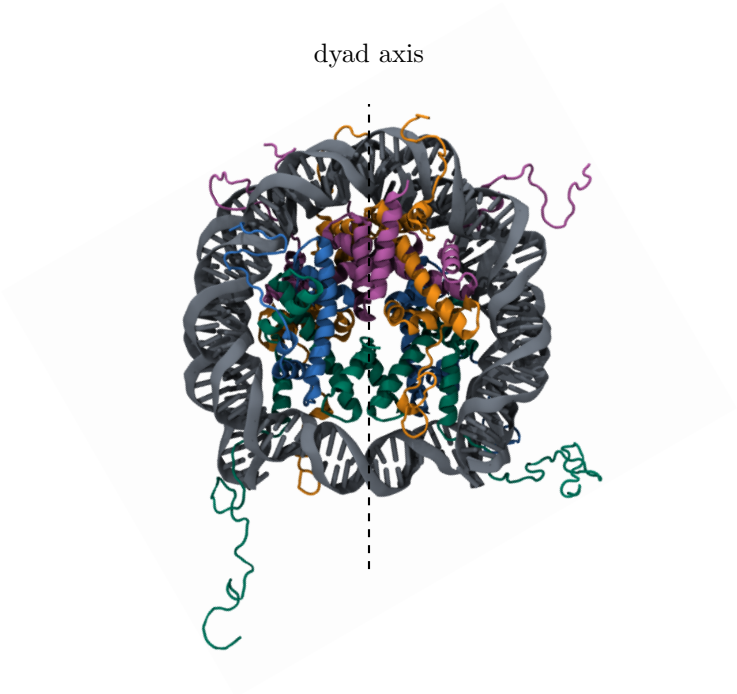


Figure 2.4: Atomistic structure of the nucleosome core particle from PDB ID 1KX5 [20]. DNA is shown in grey, while the core histones are coloured by type: H3 in green, H4 in blue, H2A in orange and H2B in purple. The molecular visualization was generated in this work with VMD [21] from the crystallographic coordinates.

The DNA completes approximately 1.7 left-handed turns around the histone octamer. Although the DNA double helix itself remains right-handed, its helical axis follows a larger left-handed path around the protein core. This curved path is referred to as the nucleosomal superhelix.

The global geometry of this path can be characterized approximately through an effective radius and pitch. The superhelix radius describes the perpendicular distance between the DNA helical axis and a reference axis passing through the centre of the nucleosome. The pitch corresponds to the displacement parallel to this reference axis over one complete superhelical turn. For the 146-bp nucleosome core particle resolved by Luger et al., the best ideal-superhelix fit to the wrapped region of the DNA axis excluded the approximately ten base pairs at each terminus that remained nearly straight and yielded an effective radius of 4.2 nm and a pitch of 2.4 nm per superhelical turn [22]. These values provide a structural reference for the analysis of the simulated nucleosomal configurations presented later in this thesis. Radius and pitch describe the global organization of DNA around the histone core and are distinct from the local helical parameters of the DNA double helix. In particular, the pitch of the nucleosomal superhelix should not be confused with the pitch of B-DNA introduced in Section 2.1.1. Because the DNA path around the histone core is not perfectly regular, a single radius and pitch cannot describe every local segment exactly. The fitted values should therefore be regarded as global descriptors that summarize the overall superhelical geometry.

The two ends of the wrapped fragment form the entry and exit regions, where nucleosomal DNA connects to the adjacent linker DNA. The central base pair lies close to the dyad axis, an approximate twofold symmetry axis passing through the centre of the nucleosome. Positions along the wrapped DNA are conventionally identified by their superhelical location (SHL), expressed as the approximate number of DNA helical turns from the dyad. The dyad defines SHL 0; for example, SHL +1 and SHL −1 identify positions approximately one helical turn, or about 10 base pairs, from the dyad on the two opposite sides of the nucleosome. The wrapped 147-bp fragment extends approximately from SHL −7 to SHL +7. This coordinate system provides a common reference for locating histone contacts and local DNA deformations along the nucleosomal path.

Histone–DNA interactions are organized around approximately 14 contact sites distributed along the nucleosomal superhelix. At these sites, which occur at intervals of roughly one DNA helical turn, the minor groove faces the histone octamer and histone residues interact primarily with the sugar–phosphate backbone. At several of these sites, arginine side chains approach or penetrate the inward-facing minor groove and constrain the local arrangement of the DNA backbone. These interactions involve electrostatic attraction, hydrogen bonding and water-mediated contacts. Since many contacts involve the backbone rather than the exposed edges of the bases, histones recognize DNA sequence mainly through shape readout rather than through direct base-specific contacts [20, 23].

The approximately helical spacing of these contacts also produces a characteristic organization along the two DNA strands. When the strands are considered separately, atomistic simulations show a periodicity of approximately ten nucleotides on each strand, with the contacts on the two strands displaced by approximately half a helical turn and therefore alternating along the duplex [24].

The arginines located in the inward-facing minor grooves also connect the DNA–histone interface with the local deformation of nucleosomal DNA. As discussed by Olson and Zhurkin [25], some of the largest coupled bending and lateral-displacement deformations occur close to these residues. These deformations, often described as kink-and-slide deformations, contribute to the construction of the overall superhelical path. Histone contacts therefore do not simply stabilize a pre-existing wrapped conformation, but also impose local geometrical constraints that different DNA sequences may accommodate in different ways. Because these constraints recur along the wrapped DNA path, the associated local deformations can also display a recurrent spatial organization. The exact amplitudes and positions of the corresponding changes need not be identical between sequences, since each sequence has different preferred local geometries and may accommodate the nucleosomal constraints in a different way.

The spatial organization of the structured histone core differs from that of the flexible terminal tails. Positively charged lysine and arginine residues in the tails can interact with different regions of nucleosomal and linker DNA while sampling multiple conformations. Their contacts are therefore not expected to reproduce the same regular spatial pattern as the contacts formed by the structured core [24, 26].

The nucleosome therefore imposes more than an overall bend on the DNA. It also constrains the position of the backbone, the rotational orientation of the grooves and the distribution of local deformations along the sequence. The energetic cost of satisfying these constraints, together with the stabilizing interactions formed with the histone core, is considered in the following subsection.

2.3.2 Mechanical and energetic cost of nucleosomal wrapping

The structure described in the previous subsection defines the geometry that DNA must adopt in the nucleosome. As discussed in Section 2.1.2, the contour length of approximately 147 base pairs is close to the persistence length of double-stranded DNA under typical solution conditions (50 nm). Wrapping this fragment around the histone octamer therefore represents a substantial deformation rather than a small fluctuation around the preferred conformation of free DNA.

This deformation cannot be described as a uniform bend of an otherwise unchanged double helix. Along the nucleosomal path, successive base-pair steps undergo different adjustments in roll, tilt and twist, while the widths and orientations of the grooves vary relative to the histone surface. The

DNA must therefore satisfy both the overall curvature of the nucleosomal superhelix and the local geometrical requirements imposed at the histone contact sites.

Nucleosome formation results from a balance between opposing free-energy contributions. DNA pays a mechanical cost to deviate from its preferred shape and loses part of the conformational freedom available in solution. At the same time, wrapping creates favourable interactions with the histone core, including electrostatic attraction between the negatively charged phosphate backbone and positively charged histone residues, hydrogen bonds and solvent-mediated contacts. These contributions are coupled, since the DNA must adopt the wrapped geometry to form the stabilizing contacts, and they should not be regarded as completely independent terms.

The strength of histone–DNA association is not uniform along the wrapped fragment. The entry and exit regions are comparatively dynamic and can transiently detach from the histone surface, while the inner region close to the dyad is more strongly constrained. Single-molecule experiments and molecular simulations accordingly describe nucleosome unwrapping as a multistage process, in which the outer DNA turn is released more readily than the inner turn. Furthermore, unwrapping does not necessarily proceed symmetrically from the two DNA ends: differences in local DNA mechanics can make one side detach more readily than the other [27]. Histone tails also contribute to nucleosome stability through electrostatic contacts with the negatively charged DNA backbone. These contacts can change as DNA progressively detaches from the histone surface and may also involve the entry, exit and linker regions [26]. This provides additional context for the comparison of genomic systems with and without linker DNA later in this thesis, since the added linker segments extend beyond the entry and exit regions and can interact with the histone tails.

Although this thesis focuses on DNA sequence, the properties of the histones can also modify this energetic balance. For example, simulations of nucleosomes containing the histone variant H2A.Z showed that its incorporation can reduce the free-energy barrier for DNA unwrapping and increase DNA accessibility [28]. This example is mentioned here to emphasize that nucleosome stability is a collective property of a complex system, to which several molecular variables contribute. In this thesis, the histone composition is kept fixed, allowing the analysis to focus specifically on one of these contributions: the effect of changing the DNA sequence.

A structural snapshot alone is therefore not sufficient to determine nucleosome stability. Geometrical descriptors can show how the DNA bends, twists or changes its groove profile, but stability is a thermodynamic property that also includes molecular interactions and conformational entropy. Free-energy calculations are required to quantify the resulting balance. The next subsection examines how this balance changes when the nucleotide sequence, and consequently the preferred shape and mechanical response of the DNA, is altered.

2.3.3 Sequence-dependent nucleosome preference

The previous subsection described nucleosome formation as a balance between the mechanical cost of DNA deformation and the interactions established with the histone core. When the histone composition is kept fixed, changing the DNA sequence can affect both contributions. Different sequences may require different amounts of energy to follow the nucleosomal path because their preferred local geometries and stiffnesses are not identical. This compatibility depends not only on how easily the DNA bends, but also on where the deformation occurs and in which direction. Sequence motifs that favour minor-groove compression, for example, can be advantageous when they occur at positions where the minor groove faces the histone core, but less favourable in a different rotational register.

Sequence can also influence the stabilizing interactions formed in the nucleosome. Variations in groove geometry alter the position of the phosphate groups and the local electrostatic environment presented to positively charged histone residues. Histones can therefore respond indirectly to nucleotide sequence through its shape and mechanical properties, even though most contacts do not involve direct recognition of the bases. Structural studies and molecular simulations indicate that intrinsic shape, groove geometry and electrostatic interactions act together in determining relative nucleosome affinity, while

local flexibility alone provides an incomplete description [2, 3].

The thermodynamic objective of this thesis is to determine how DNA sequence affects relative nucleosome-binding free energies. This quantity is obtained by combining the same target-to-reference alchemical transformation in the DNA-only and histone-associated systems. The DNA-only calculation provides the reference branch required to isolate the difference between the nucleosome-binding free energies of the two sequences. This construction provides the connection between sequence-dependent DNA mechanics and the free-energy framework introduced in Section 2.5.4.

2.3.4 From sequence preference to nucleosome occupancy

Sequence-dependent nucleosome preference can be investigated through different experimental observables. These observables are related, but they do not describe exactly the same physical quantity.

It is useful to distinguish three concepts. Nucleosome affinity describes the strength of association between the histone octamer and a DNA sequence and can be compared across different sequences. Nucleosome positioning refers to the preferred placement of a nucleosome along DNA. Nucleosome occupancy instead describes how frequently a genomic position is covered by a nucleosome within a population. Sequence-dependent energetic preferences therefore represent one of several factors that influence nucleosome positioning and occupancy.

Genome-wide occupancy measurements provide a way to investigate these effects in their genomic context. Kaplan et al. reconstituted nucleosomes on purified genomic DNA from *Saccharomyces cerevisiae* and used sequencing to construct an in vitro nucleosome-occupancy map. Because the experiment contained purified DNA and histones, the resulting organization reflected intrinsic nucleosome sequence preferences more directly than an in vivo map, where chromatin remodellers, transcription factors and transcription also contribute. Under equilibrium conditions, relative nucleosome occupancies are directly related to free-energy differences and can therefore be used to compare sequence-dependent nucleosome preferences.

The measurements also revealed sequence features associated with nucleosome enrichment or depletion. Among all five-base-pair sequences examined, AAAAA had the lowest average nucleosome occupancy, consistently with the reduced nucleosome affinity and depletion associated with poly(dA:dT) regions. Approximately ten-base-pair periodicities in the occurrence of specific dinucleotides were also observed, supporting the importance of rotationally phased sequence features for nucleosome positioning [9]. These trends do not imply that adenine content alone determines nucleosome preference, since the effect of an A-rich region also depends on its length, arrangement and surrounding sequence.

These experimental observations provide two complementary points of reference for this thesis. The known nucleosomal behaviour of A-rich motifs motivates the use of controlled sequence modifications, while the occupancy map provides an external reference for the comparison between naturally occurring genomic sequences. The precise sequences, their selection and the construction of the simulated systems are described in Chapter 3.

2.4 Coarse-grained models for nucleosome simulations

2.4.1 Modelling DNA at different resolutions

The previous sections have shown that the problem addressed in this thesis involves several length scales. Sequence dependence originates from local properties of base pairs and base-pair steps, whereas nucleosome formation involves the collective deformation of approximately 147 base pairs and their interaction with the histone core. A suitable computational description must therefore retain enough local information to distinguish different sequences while remaining efficient enough to sample the conformations of complete nucleosomal systems.

At atomistic resolution, every atom of DNA and proteins can be represented explicitly, together with solvent molecules and ions. This level of detail is particularly valuable when specific chemical

interactions, hydration effects or ion binding are central to the problem. However, its computational cost makes the systematic comparison of many sequences and thermodynamic states highly demanding. At the opposite limit, continuum models such as the worm-like chain describe DNA using a small number of effective mechanical parameters. They are efficient for studying global polymer behaviour, but do not explicitly represent the backbone, the bases or the local sequence-dependent geometry required here.

Coarse-grained models provide an intermediate level of description. Groups of atoms are replaced by effective interaction sites, commonly referred to as beads, and the effects of the omitted degrees of freedom are incorporated into effective interaction potentials. A coarse-grained model is therefore not simply an atomistic model with fewer particles. Its resolution and interactions are specifically chosen to reproduce a selected set of structural, mechanical or thermodynamic properties. This reduction makes broader conformational sampling possible, but it also limits the model to phenomena that can be described using the retained degrees of freedom. The reduction from an atomistic representation of DNA to the three-site-per-nucleotide description used by the Mechanically Accurate DNA (MADna) coarse-grained model is illustrated in Figure 2.5.

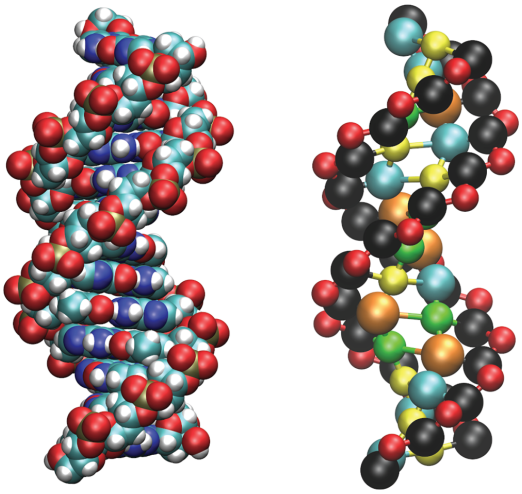


Figure 2.5: Atomistic representation of double-stranded DNA (left) and the corresponding coarse-grained representation used by MADna (right). In the coarse-grained model, each nucleotide is represented by three interaction sites located at the phosphate group, sugar and base. Source: Assenza and Perez (2022) [8].

Coarse-grained DNA models can themselves be constructed at different levels of resolution. Some describe each nucleotide or base pair as a rigid unit, providing a highly efficient representation of DNA conformation. Others retain several particles per nucleotide, associated with distinct molecular groups such as the base, sugar and phosphate. This second approach preserves more information about the spatial organization of the backbone and bases and is particularly useful when DNA must interact explicitly with another molecular component, such as the histone core.

The simulations performed in this thesis require a model that can distinguish the geometry and mechanical response of different sequences, describe complete nucleosomal-length DNA fragments and represent their interactions with the histones. MADna was selected because it meets these requirements while using only three interaction sites per nucleotide. Its molecular representation, effective interactions and sequence-dependent parametrization are described in the following subsection [8].

2.4.2 Representation and interactions in MADna

MADna (Mechanically Accurate DNA) represents each nucleotide using three interaction sites, or beads, positioned at the phosphate group, sugar and nitrogenous base. Since the four bases are represented by distinct bead types, the model contains six types of interaction sites: phosphate, sugar, adenine, cytosine, guanine and thymine. This resolution preserves the distinction between the backbone and the bases while considerably reducing the number of particles compared with an

atomistic representation, particularly because solvent and ions are described implicitly rather than represented by individual particles.

The double-stranded structure is maintained by bonded potentials defined over groups of two, three or four beads. Two-bead terms maintain connectivity and control selected distances, three-bead terms control angles, and four-bead terms control the relative orientation around a dihedral angle. These interactions are therefore referred to as bonds, angles and dihedrals, respectively. Together, they determine how the beads are arranged and how the DNA molecule responds when it is bent, twisted or stretched. They connect beads both within the same strand and between the two strands, representing sugar–phosphate connectivity, Watson–Crick base pairing and base stacking. The complete set of bonded interactions included in MADna is shown in Figure 2.6.

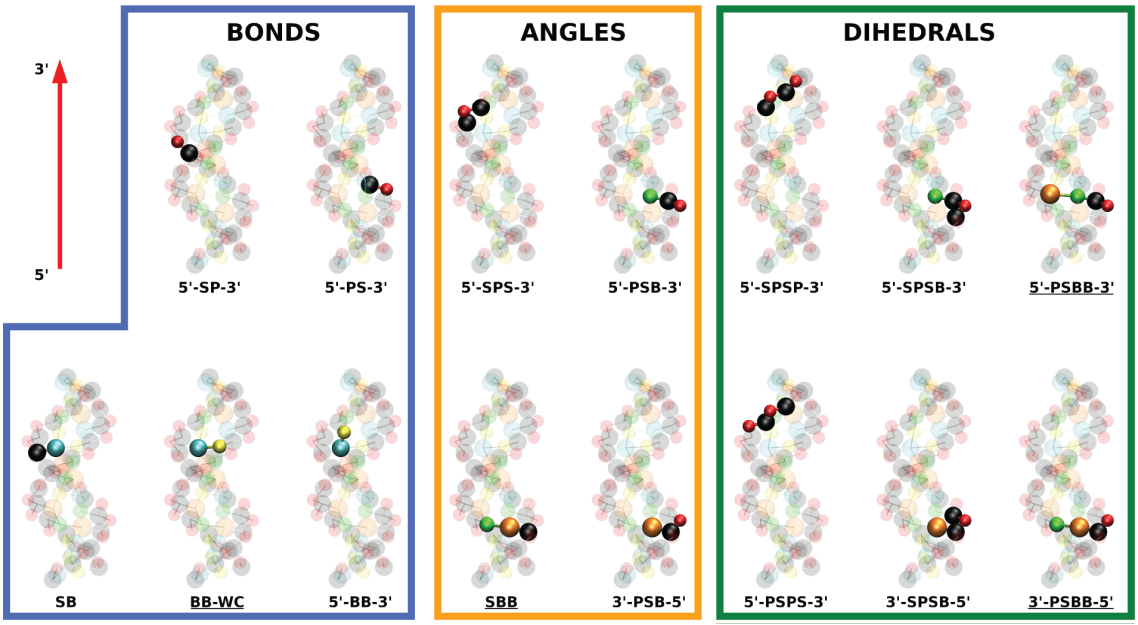


Figure 2.6: Bonded interactions included in the MADna model. Labels containing 5′ and 3′ identify step-dependent interactions, while underlined labels indicate interactions between the two DNA strands. P, S and B denote phosphate, sugar and base beads, respectively. Source: Assenza and Perez (2022), Figure 3 [8].

Bond stretching is described by a harmonic potential,

$$U_{\text{bond}} = \sum_i k_{b,i} \left(r_i - r_{i,0}\right)^2 , \tag{2.5}$$

where r_i is the instantaneous distance between the two beads involved in bond i, $r_{i,0}$ is its equilibrium value and $k_{b,i}$ is the corresponding force constant.

Similarly, angle bending is described by

$$U_{\text{angle}} = \sum_j k_{\theta,j} \left(\theta_j - \theta_{j,0}\right)^2 , \tag{2.6}$$

where θ_j is the angle formed by the three beads defining interaction j, $\theta_{j,0}$ is its equilibrium value and $k_{\theta,j}$ is the corresponding angular force constant.

The torsional response is described through four-body dihedral interactions,

$$U_{\text{dihedral}} = \sum_k k_{\phi,k} \left[1 + \cos\left(\phi_k - \phi_{k,0}\right)\right] . \tag{2.7}$$

Here, ϕ_k is the dihedral angle defined by four beads and describes the relative orientation of the planes formed by the first and last three beads. The parameter $\phi_{k,0}$ is related to the equilibrium dihedral angle $\phi_{k,\min}$ by $\phi_{k,0} = \phi_{k,\min} - \pi$, while $k_{\phi,k}$ is the corresponding force constant.

The parameters of the bonded interactions depend on the identities of the bases involved. Depending on the interaction, this dependence is defined at the level of an individual base pair or of a base-pair step. Changing the DNA sequence therefore modifies both the preferred local geometry and the energetic cost of deforming the molecule. The complete list of interactions and parameters is provided in the original MADna publication [8].

The selection of bonded interactions and their parameters was based on information extracted from atomistic simulations. Atomistic trajectories containing the 10 non-redundant base-pair-step types were first mapped onto the three-site-per-nucleotide representation. This number accounts for the equivalence of reverse-complement steps in the antiparallel duplex; for example, a CA step on one strand corresponds to TG on the complementary strand. These 10 base-pair-step types should not be confused with the 16 distinct nucleotide-step combinations defined along an individual strand and used in the sequence-dependent parametrization of MADna. The distributions of bond distances, angles and dihedral angles provided initial estimates of the corresponding equilibrium values and force constants through Boltzmann inversion, which relates the probability distribution of a geometrical coordinate to an effective potential. The authors then refined the model iteratively, adding or removing bonded interactions according to how well the coarse-grained simulations reproduced the structural properties observed in the atomistic trajectories. Finally, a limited subset of parameters was adjusted to improve the description of the elastic response of DNA under stretching and torsional deformation. The model was subsequently evaluated using additional sequences that had not been included in the parametrization [8].

Since every sequence-dependent bonded interaction involves at most one base-pair step, structural properties extending over longer length scales arise from the propagation of these local effects along the molecule, together with the electrostatic repulsion between phosphate groups. In this way, the model can produce sequence-dependent properties at scales larger than an individual step without introducing a separate interaction specifically for each global structural feature.

In addition to the bonded terms, excluded-volume interactions prevent different beads from overlapping. These interactions are described using the Weeks–Chandler–Andersen potential,

$$U_{\mathrm{WCA}}(r) = \begin{cases} \epsilon \left[\left(\dfrac{\sigma_{ij}}{r} \right)^{12} - 2 \left(\dfrac{\sigma_{ij}}{r} \right)^{6} + 1 \right], & r \leq \sigma_{ij}, \\ 0, & r > \sigma_{ij}. \end{cases} \tag{2.8}$$

Here, r is the distance between the two beads, $\epsilon = 1\,\mathrm{kcal\,mol^{-1}}$ sets the energy scale of the repulsion and $\sigma_{ij} = R_i + R_j$ is the sum of their bead radii. Six bead radii are defined, corresponding to phosphate, sugar and the four possible bases. In this convention, σ_{ij} is both the position of the minimum of the underlying Lennard–Jones interaction and the cutoff distance of the repulsive potential.

Electrostatic interactions are included through the Debye–Hückel potential,

$$U_{\mathrm{elec}} = \sum_{i<j} \frac{q_i q_j e_0^2}{4\pi\epsilon_0\epsilon_r r_{ij}} \exp\left(-\frac{r_{ij}}{\lambda_D} \right), \tag{2.9}$$

where q_i and q_j are the effective charges of the interacting beads, e_0 is the elementary charge, ϵ_0 is the vacuum permittivity, ϵ_r is the relative dielectric constant and r_{ij} is the distance between the two beads. The Debye screening length for a solution containing monovalent ions is

$$\lambda_D = \sqrt{\frac{\epsilon_0\epsilon_r k_B T}{2 e_0^2 \rho_0}}, \tag{2.10}$$

where k_B is the Boltzmann constant, T is the temperature and ρ_0 is the bulk number density of salt ions. Electrostatic interactions are therefore progressively screened at distances larger than λ_D. In MADna, the charged sites are the phosphate beads, which are assigned an effective charge $q = -0.6$ for DNA–DNA interactions to account implicitly for counterion condensation.

The resulting model combines a sequence-dependent bonded description with steric and screened electrostatic interactions. Its ability to reproduce different structural and mechanical properties of double-stranded DNA, together with the limitations introduced by coarse graining, is discussed in the following subsection.

2.4.3 Capabilities and limitations of MADna

The usefulness of a coarse-grained model depends not only on its computational efficiency, but also on which properties remain accurately represented after reducing the molecular resolution. Assenza and Perez evaluated MADna by comparing its predictions with atomistic simulations and experimental measurements. Although its sequence-dependent parameters are defined locally, at most at the level of a base-pair step, the model reproduces several properties that extend over longer length scales. These include helical rise and twist, major- and minor-groove geometry, intrinsic shape and the mechanical response of double-stranded DNA. Importantly, good agreement with atomistic simulations was also obtained for testing sequences that had not been used to parametrize the model [8].

MADna was also designed to reproduce the response of DNA to different types of mechanical deformation. The model provides a consistent description of stretching and torsional rigidity, persistence length and couplings between different deformation modes. In particular, it reproduces the negative twist–stretch coupling of double-stranded DNA, meaning that DNA initially overwinds when it is stretched. Its ability to connect local sequence-dependent interactions with the mechanical behaviour of the complete molecule makes it suitable for comparing DNA sequences that must adapt to strongly deformed configurations, such as the path followed by DNA in a nucleosome.

The three-site-per-nucleotide representation also retains an explicit geometrical description of the backbone and bases. This allows DNA to interact with other coarse-grained molecular components, including the histone octamer, while remaining substantially less expensive to simulate than an atomistic system. For the present thesis, this balance is particularly useful because complete nucleosomal-length fragments must be simulated repeatedly for different sequences and alchemical states.

These advantages are accompanied by important limitations. MADna was developed to describe intact double-stranded DNA predominantly in the B form. The two strands are held together by permanent bonded interactions, including those representing Watson–Crick base pairing. The model therefore cannot describe the disruption of Watson–Crick base pairing. Its parameters were obtained and tested within a particular range of structures and mechanical perturbations, so predictions involving conformations far outside this regime should be interpreted cautiously. Moreover, because the parametrization was derived from atomistic trajectories, MADna necessarily retains some dependence on the atomistic force field and simulation conditions used to generate those trajectories.

The reduced resolution also means that individual solvent molecules, ions and atom-specific interactions are not represented explicitly. Solvent and ionic screening are incorporated through effective interactions, while chemical details such as hydration patterns and specific directional hydrogen bonds cannot be resolved at the same level as in an atomistic simulation. Since MADna describes only the DNA component of the nucleosome, representing the complete system requires an additional coarse-grained protein model and effective DNA–protein interaction terms. These interactions include excluded-volume and electrostatic contributions, but do not resolve individual directional contacts, such as specific hydrogen bonds, at atomistic resolution. The representation of the histone octamer and its coupling to DNA are introduced in the following sections.

Finally, the agreement with the atomistic reference is not equally accurate for every observable. The original validation reported small systematic deviations for some geometrical properties and a less accurate description of the detailed sequence dependence of the torsional modulus. Results obtained

with MADna should therefore be interpreted as predictions within an effective coarse-grained description rather than as an atomically complete representation of DNA. Within these limits, the model provides the combination of sequence sensitivity, mechanical accuracy and computational efficiency required for the comparisons performed in this thesis.

2.4.4 The CALVADOS 3 protein model

The DNA model described above provides the sequence-dependent representation of the double helix, but a complete nucleosomal model also requires a description of the histone proteins. In this work, the protein component is represented using CALVADOS 3, a coarse-grained model developed to describe proteins containing both folded domains and intrinsically disordered regions [29]. This combination is particularly relevant for histones, which contain a structured globular core together with flexible terminal tails.

CALVADOS 3 represents each amino acid by a single interaction site. In intrinsically disordered regions, the bead is placed at the C_α position. In folded domains, it is instead placed at the centre of mass of the residue. The centre-of-mass representation provides a more realistic spatial distribution of the interaction sites within a folded structure and reduces the excessive inter-domain compaction that can arise when all protein beads are placed at the C_α positions [29].

Consecutive residues are connected by harmonic bonds, while non-bonded interactions are described through short-range residue-dependent interactions and screened electrostatics. The short-range contribution is represented by an Ashbaugh–Hatch potential, whose strength depends on the identities and hydrophobic character of the interacting amino acids. This term provides an effective description of both repulsive and attractive residue–residue interactions. Electrostatic interactions between charged residues are represented by a Debye–Hückel potential, so that their magnitude depends on residue charge and decreases with distance according to the ionic screening of the implicit solvent.

A one-bead-per-residue model does not, by itself, preserve the tertiary structure of a folded protein. CALVADOS 3 therefore combines the non-bonded force field with an elastic network model for folded domains. In this network, residue pairs that are sufficiently close in a reference structure are connected by harmonic restraints. The network maintains the organization of folded regions, while residues outside it remain free to explore different conformations. This separation makes it possible to preserve the structured histone core without imposing the same restraints on the flexible histone tails.

As in other coarse-grained descriptions, the reduction in molecular resolution limits the chemical detail that can be represented. Solvent molecules and ions are treated implicitly, and atom-specific interactions such as directional hydrogen bonds are not resolved individually. CALVADOS 3 instead provides an effective description of protein conformations and residue-dependent interactions at a computational cost compatible with simulations of the complete histone octamer. The construction of the specific histone model used in this thesis, including its mapping from the 1KX5 structure and the elastic restraints applied to the octamer, is described in Section 3.3.3.

2.4.5 Coupling MADna and CALVADOS 3

MADna and CALVADOS 3 define the internal interactions of DNA and proteins, respectively. To simulate a complete nucleosome, additional cross-interactions are required between beads belonging to the two molecular components. In the framework used here, the models are coupled through excluded-volume repulsion and screened electrostatic interactions.

Excluded-volume interactions prevent DNA and protein beads from overlapping and are described using the Weeks–Chandler–Andersen potential introduced in Eq. 2.8. The interaction distance depends on the sizes of the two beads involved, allowing the steric organization of the DNA backbone, bases and histone residues to be retained in the combined coarse-grained representation.

The main attractive contribution between DNA and histones is electrostatic. The negatively charged phosphate beads of the DNA backbone interact with positively charged histone residues, particularly

lysine and arginine, through the Debye–Hückel potential introduced in Eq. 2.9. The same formulation also accounts for repulsion between particles carrying charges of the same sign, while ionic screening limits the range of the interaction. The charge convention and the parameters used for these cross-interactions are reported in Section 3.3.4.

No permanent contacts are imposed between DNA and the histone octamer. The interface can therefore reorganize during the simulations as DNA changes conformation and the histone tails explore different regions around the nucleosome. At the same time, the coupling does not include attractive terms specifically parametrized for particular combinations of DNA bases and amino acids, nor does it explicitly represent directional hydrogen bonds. Sequence dependence enters the DNA–histone interface mainly through the MADna description: changing the nucleotide sequence changes the preferred DNA geometry, its mechanical response and the dimensions of the base interaction sites, which can alter the spatial arrangement of the backbone relative to the histone surface.

The framework employed in this thesis therefore combines two established coarse-grained models with complementary roles: MADna provides a sequence-dependent description of DNA mechanics, whereas CALVADOS 3 represents both the structured histone domains and the flexible terminal tails. To the best of our knowledge, this specific combination has not previously been reported for nucleosome simulations. The methodological novelty lies in coupling the two models within a common nucleosomal representation rather than in reparametrizing either individual force field.

2.5 Molecular simulations and free-energy calculations

2.5.1 Molecular dynamics and statistical ensembles

Once the interaction potential of a molecular model has been defined, molecular dynamics provides a way to explore the configurations that the system can adopt at finite temperature. A configuration is simply the set of positions of all particles at a particular instant. Starting from an initial configuration, the forces acting on the particles are calculated from the potential energy and used to update their positions and velocities over a short time interval. Repeating this procedure produces a molecular dynamics trajectory.

For a particle i with mass m_i and position $\mathbf{r}_i$, the classical equation of motion is

$$m_i \frac{d^2\mathbf{r}_i}{dt^2} = \mathbf{F}_i = -\nabla_i U(\mathbf{r}), \tag{2.11}$$

where

$$\mathbf{r} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) \tag{2.12}$$

is a compact notation for the positions of all N particles in the system. The force is calculated separately for each particle, but generally depends on the positions of the other particles with which it interacts. At each integration step, the forces are evaluated and used to update the positions and velocities of all particles, producing the next configuration of the trajectory.

At non-zero temperature, the molecule evolves along the trajectory established by Newton's laws. Thermal motion causes the particles to fluctuate, allowing the DNA and the histone complex to explore different conformations. The simulations considered in this thesis are performed in the canonical, or NVT, ensemble, in which the number of particles, simulation volume and temperature remain constant.

To sample this ensemble, the temperature is controlled through a thermostat that allows the simulated system to exchange energy with an effective thermal reservoir. One common approach is Langevin dynamics, in which the conservative force derived from the interaction potential is supplemented by a friction term and a random force,

$$m_i \frac{d^2\mathbf{r}_i}{dt^2} = -\nabla_i U(\mathbf{r}) - m_i \gamma_i \frac{d\mathbf{r}_i}{dt} + \mathbf{R}_i(t), \tag{2.13}$$

Here, γ_i is the friction coefficient and $\mathbf{R}_i(t)$ is a stochastic force representing the effect of degrees of freedom that are not described explicitly, such as collisions with solvent molecules. The amplitude of the stochastic force is related to the friction coefficient and temperature through the fluctuation–dissipation theorem, ensuring that the system samples the canonical ensemble at the chosen temperature [30]. The specific numerical integrator, timestep and friction coefficient used in the simulations will be reported in Chapter 3.

At thermal equilibrium, the system does not visit all configurations with the same frequency. The probability of observing a configuration $\mathbf{r}$ is proportional to its Boltzmann weight,

$$P(\mathbf{r}) \propto \exp\left[-\beta U(\mathbf{r})\right], \qquad \beta = \frac{1}{k_{\mathrm{B}} T}. \tag{2.14}$$

Configurations with lower potential energy generally have a larger statistical weight, although thermal fluctuations allow the system to explore higher-energy configurations as well. The complete distribution of possible configurations and their probabilities defines the canonical ensemble.

An equilibrium quantity is obtained by averaging its value over the configurations of the canonical ensemble. For an observable $A(\mathbf{r})$, this average can be written as

$$\langle A \rangle = \frac{\displaystyle\int d\mathbf{r}\, A(\mathbf{r}) e^{-\beta U(\mathbf{r})}}{\displaystyle\int d\mathbf{r}\, e^{-\beta U(\mathbf{r})}}. \tag{2.15}$$

In practice, this ensemble average is estimated from the configurations sampled along the molecular dynamics trajectory,

$$\langle A \rangle \simeq \frac{1}{N_{\mathrm{frames}}} \sum_{j=1}^{N_{\mathrm{frames}}} A(\mathbf{r}_j). \tag{2.16}$$

For this estimate to be representative, the trajectory must be analysed only after the system has reached equilibrium and must be sufficiently long to include the relevant molecular fluctuations. The initial part of the simulation is therefore excluded because the system may still be relaxing from its starting structure. Moreover, consecutive frames are not fully independent, since the molecular configuration changes continuously and configurations saved close together in time tend to be similar. Multiple independent realizations, equilibration checks and an appropriate treatment of statistical correlations are consequently required to obtain reliable averages and uncertainties. The specific procedures adopted in this work will be described in Chapter 3.

In coarse-grained molecular dynamics, the simulated trajectory should primarily be interpreted as a method for sampling the effective model. Because many microscopic degrees of freedom have been removed and solvent effects are represented implicitly, the simulation time cannot be mapped directly onto the physical time of the real system. Coarse-graining generally smooths the free-energy landscape and therefore accelerates the overall dynamics, although the major conformational steps sampled along kinetic pathways are expected to remain qualitatively representative. In this thesis, molecular dynamics is used mainly to obtain equilibrium averages, including the energy derivatives required for the free-energy calculations, rather than to make quantitative predictions about kinetic rates.

Molecular dynamics thus provides the statistical sampling needed to connect the MADna potential with thermodynamic observables. The next section introduces the partition function and the free energy that characterize the equilibrium distribution sampled by these simulations.

2.5.2 Free-energy differences

The molecular dynamics trajectory described above provides a representation of the configurations sampled by the system at equilibrium. The next question is how this information can be used to compare the thermodynamic stability of two systems containing different DNA sequences. Looking only at their minimum-energy structures or average potential energies is not sufficient, because stability depends on the complete distribution of accessible configurations. The relevant quantity for this comparison is the free energy.

A system can be favoured because its configurations have a low energy, but also because it can access a larger number of configurations. These two contributions are summarized by the Helmholtz free energy,

$$F = U_{\mathrm{int}} - TS, \tag{2.17}$$

where U_{int} is the mean internal energy, S is the entropy and T is the temperature. A lower internal energy reduces the free energy, while a larger number of accessible configurations generally increases the entropy and therefore also lowers F. Thermodynamic stability consequently depends on both the energies of the configurations and the way in which the system can explore them.

For the configurational degrees of freedom sampled by the molecular dynamics simulations, the corresponding partition function is

$$Z_{\mathrm{conf}} = \int d\mathbf{r}\, e^{-\beta U(\mathbf{r})}, \tag{2.18}$$

and the configurational contribution to the Helmholtz free energy is

$$F_{\mathrm{conf}} = -k_{\mathrm{B}} T \ln Z_{\mathrm{conf}}. \tag{2.19}$$

Here, the integral extends over all possible configurations of the system and each configuration contributes according to its Boltzmann weight. The Thermodynamic Integration calculations performed in this work determine differences in F_{conf}. Although the alchemical transformations also change the masses assigned to the base beads, the corresponding kinetic contribution is identical in the DNA-only and histone-associated branches and therefore cancels in the calculation of $\Delta\Delta F_{\mathrm{bind}}$. For simplicity, these configurational free-energy differences are denoted by ΔF throughout this thesis.

In practice, the quantity of interest is generally not the absolute free energy of a single system, but the difference between two states A and B,

$$\Delta F_{A\to B} = F_{\mathrm{conf},B} - F_{\mathrm{conf},A}. \tag{2.20}$$

With this definition, $\Delta F < 0$ indicates that endpoint B has a lower configurational free energy than endpoint A, whereas $\Delta F > 0$ indicates the opposite.

In this thesis, the two endpoint states correspond to systems containing different DNA sequences under the same physical conditions. Changing the sequence modifies both the preferred geometry and the flexibility of DNA, and therefore affects the configurational free-energy difference between the alchemical endpoints. The corresponding transformation free energies provide the quantities required to construct the thermodynamic cycle and calculate the relative nucleosome-binding free energy.

Calculating the absolute free energy directly from the partition function would require evaluating an integral over the very large number of configurations available to the molecular system. This is not practical for the systems considered here. A free-energy difference can instead be obtained by connecting the two endpoint systems through a continuous thermodynamic path. The method used

in this thesis to perform this calculation is Thermodynamic Integration, discussed in the following section [30].

2.5.3 Thermodynamic integration

Thermodynamic Integration is a computational strategy used to avoid the direct calculation of the partition functions of two complex molecular systems. The key idea is to replace this difficult calculation with a continuous transformation between the systems, constructed so that the corresponding free-energy difference can be obtained from equilibrium averages.

Consider two systems simulated in the same molecular environment, either DNA-only or histone-associated, that differ in the DNA sequence they contain. One sequence is denoted as the target and the other as the reference.

A continuous path between the two systems is introduced through an alchemical parameter

$$\lambda \in [0, 1], \tag{2.21}$$

with

$$U(\mathbf{r}; 0) = U_{\text{target}}(\mathbf{r}), \qquad U(\mathbf{r}; 1) = U_{\text{reference}}(\mathbf{r}). \tag{2.22}$$

Each intermediate value of λ defines a different potential energy $U(\mathbf{r}; \lambda)$. In this way, the calculation does not pass directly from the target system to the reference system, but through a series of intermediate models that connect the two endpoints smoothly.

These intermediate models do not correspond to physical DNA sequences composed of ordinary nucleotides. They are fictitious alchemical systems in which the sequence-dependent interaction parameters are mixed between their target and reference values. For a generic parameter x, this interpolation is written as

$$x(\lambda) = (1 - \lambda)x_{\text{target}} + \lambda x_{\text{reference}}. \tag{2.23}$$

At $\lambda = 0$, all interpolated parameters have their target values; at $\lambda = 1$, they have their reference values. Intermediate values of λ provide the mathematical path needed for the free-energy calculation.

Since the potential energy varies continuously with λ, the corresponding free energy $F(\lambda)$ also changes along this path. Differentiating $F(\lambda) = -k_{\text{B}}T \ln Z(\lambda)$ gives the central relation of Thermodynamic Integration,

$$\frac{\partial F}{\partial \lambda} = \left\langle \frac{\partial U}{\partial \lambda} \right\rangle_{\lambda}, \tag{2.24}$$

where the brackets indicate an equilibrium average over the configurations sampled at a fixed value of λ. Integrating this relation from zero to one gives

$$\Delta F_{\text{target}\to\text{reference}} = F_{\text{reference}} - F_{\text{target}} = \int_0^1 \left\langle \frac{\partial U}{\partial \lambda} \right\rangle_{\lambda} d\lambda. \tag{2.25}$$

Since free energy is a state function, the exact free-energy difference depends only on the two endpoints and not on the particular intermediate path used to connect them. The choice of path can nevertheless affect the numerical efficiency of the calculation and the sampling required to obtain a reliable estimate.

This is the practical advantage of Thermodynamic Integration: the partition functions and absolute free energies of the two endpoints do not need to be calculated separately. Instead, molecular dynamics

simulations are performed for several values of λ. For each value, $\partial U/\partial\lambda$ is calculated for the sampled configurations and averaged over the equilibrium trajectories and independent realizations. Integrating the resulting averages over λ then provides the free-energy difference between the target and reference systems.

Only interaction parameters that change along the alchemical path contribute directly to $\partial U/\partial\lambda$. Interactions whose parameters remain fixed can nevertheless influence the configurations sampled at each value of λ, and therefore affect the resulting equilibrium averages. The specific contributions included in the derivative, together with the numerical integration and uncertainty-estimation procedures, will be described in Chapter 3.

In this thesis, the same target-to-reference transformation is performed in the DNA-only and histone-associated environments. Combining the two resulting free-energy differences provides the difference between the nucleosome-binding free energies of the target and reference sequences.

2.5.4 Thermodynamic cycle for relative nucleosome-binding free energy

An alchemical free-energy difference describes the thermodynamic effect of changing the DNA sequence within a particular molecular environment. On its own, however, it does not measure whether one sequence is more favourable than another for nucleosome formation. To isolate this relative nucleosome preference, the same target-to-reference transformation must be compared in the DNA-only and histone-associated environments.

For each sequence pair, Thermodynamic Integration provides two free-energy differences,

$$\Delta F_{\mathrm{DNA}} = F_{\mathrm{DNA}}^{\mathrm{reference}} - F_{\mathrm{DNA}}^{\mathrm{target}}, \tag{2.26}$$

and

$$\Delta F_{\mathrm{nucleosome}} = F_{\mathrm{nucleosome}}^{\mathrm{reference}} - F_{\mathrm{nucleosome}}^{\mathrm{target}}. \tag{2.27}$$

The first describes the free-energy effect of changing the sequence in isolated DNA. The second describes the same transformation when the DNA is associated with the histone octamer. Their difference therefore provides the relative nucleosome-binding free energy of the target sequence with respect to the reference.

Four thermodynamic states are involved in this comparison: the target and reference sequences are each considered as isolated DNA and in the nucleosomal environment. These states form the thermodynamic cycle shown in Figure 2.7. The two horizontal paths change the DNA sequence while keeping the molecular environment fixed and correspond to the transformations calculated by Thermodynamic Integration. The two vertical paths instead represent nucleosome formation without changing the sequence. Their absolute binding free energies are not calculated directly in the simulations.

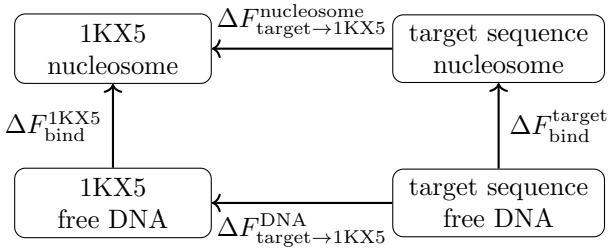


Figure 2.7: Thermodynamic cycle used to obtain the relative nucleosome-binding free energy of a target sequence with respect to the 1KX5 reference. Horizontal arrows represent the alchemical transformation from the target sequence ($\lambda = 0$) to 1KX5 ($\lambda = 1$) in the DNA-only and nucleosomal environments. Vertical arrows represent nucleosome formation at fixed sequence and are not calculated directly. With the convention used in this thesis, the relative binding free energy is $\Delta\Delta F_{\mathrm{bind}} = \Delta F_{\mathrm{DNA}} - \Delta F_{\mathrm{nucleosome}}$.

For a sequence s, the vertical binding process can be formally associated with the free-energy difference

$$\Delta F^{s}_{\mathrm{bind}} = F^{s}_{\mathrm{nucleosome}} - F^{s}_{\mathrm{DNA}} - F_{\mathrm{histone}}, \tag{2.28}$$

where s denotes either the target or the reference sequence and F_{histone} is the free energy of the isolated histone octamer. The same histone model and composition are used for both DNA sequences, so this term is identical for the target and reference states and cancels when their binding free energies are subtracted. Since free energy is a state function, the total free-energy change around the closed thermodynamic cycle must be zero. Following the cycle in the direction of the arrows gives

$$\Delta F_{\mathrm{DNA}} + \Delta F^{\mathrm{reference}}_{\mathrm{bind}} - \Delta F_{\mathrm{nucleosome}} - \Delta F^{\mathrm{target}}_{\mathrm{bind}} = 0. \tag{2.29}$$

Rearranging this expression gives the relative nucleosome-binding free energy,

$$\Delta\Delta F_{\mathrm{bind}} = \Delta F^{\mathrm{target}}_{\mathrm{bind}} - \Delta F^{\mathrm{reference}}_{\mathrm{bind}} = \Delta F_{\mathrm{DNA}} - \Delta F_{\mathrm{nucleosome}}. \tag{2.30}$$

The thermodynamic cycle therefore makes the difference between the two vertical binding processes accessible from the horizontal transformations. Only ΔF_{DNA} and $\Delta F_{\mathrm{nucleosome}}$ are calculated directly through Thermodynamic Integration; their difference yields the relative nucleosome-binding free energy.

With the convention used in this thesis, a positive value of $\Delta\Delta F_{\mathrm{bind}}$ means that the target sequence has a higher, and therefore less favourable, nucleosome-binding free energy than the reference sequence. A negative value means that the target is relatively favoured, while a value close to zero indicates that no difference between the nucleosome-binding free energies of the two sequences is resolved.

This quantity should not be interpreted as an absolute nucleosome-binding free energy. It represents the difference between the nucleosome-binding free energies of the target and reference sequences within the computational model. Figure 2.7 illustrates the cycle for a comparison with the 1KX5 reference, but the same construction can be applied to any pair of endpoint sequences.

At this point, the physical description of sequence-dependent DNA mechanics has been connected to the thermodynamic quantity used in this work. The following chapter describes how the simulated systems and alchemical transformations were constructed and how the corresponding free-energy differences were calculated.

Chapter 3

Computational Methods

3.1 Overview of the computational workflow

This chapter describes the practical implementation of the free-energy calculations introduced in Chapter 2. It also presents the validation procedures and structural analyses used to interpret the resulting thermodynamic trends. Starting from two endpoint DNA sequences, a series of mixed-λ systems was generated in both DNA-only and histone-associated environments. The complete workflow combines coarse-grained molecular dynamics simulations, Thermodynamic Integration, post-simulation trajectory analysis and a sequence-based prediction of DNA crookedness.

The investigated systems were organized into three groups: A-rich variants derived from the 1KX5 nucleosomal DNA sequence; five random control sequences and Widom 601, also compared with 1KX5; and two genomic fragments from chromosome XIII. For every comparison, a target sequence and a reference sequence of equal length were connected through an alchemical transformation controlled by the coupling parameter λ.

The same alchemical transformation was examined in two molecular environments: isolated DNA and DNA associated with the histone octamer. For the chromosome XIII comparison, additional constructs containing terminal linker DNA were generated to assess whether the finite boundaries of the 147-bp systems affected the calculated relative nucleosome-binding free energy.

The simulation outputs were processed to calculate and average $\partial U/\partial\lambda$ at each value of the coupling parameter. These averages were numerically integrated to obtain the free-energy difference associated with each sequence transformation. Finally, the DNA-only and histone-associated results were combined through the thermodynamic cycle described in Section 2.5.4. The reliability of the resulting estimates was examined through equilibration and convergence checks, comparisons between independent realizations, sensitivity to the discretization of the alchemical path and a thermodynamic-cycle consistency test.

Selected simulation trajectories were also examined through structural post-processing to support the interpretation of the thermodynamic results. The structural analysis considered the six base-pair-step parameters introduced in Section 2.1.1, minor-groove width, the global path followed by DNA around the histone core and DNA–histone contacts. Comparisons between chromosome XIII constructs with and without linker DNA were performed over equivalent central DNA regions to prevent the additional terminal base pairs from changing the region included in the analysis.

Finally, the crookedness parameter introduced in Section 2.1.3 was predicted directly from each nucleotide sequence using the model described by Luengo Márquez [17]. Differences in crookedness between the reference and target sequences were compared with the corresponding relative nucleosome-binding free energies to test whether this compact descriptor of intrinsic DNA geometry captures part

of the observed thermodynamic behaviour.

Part of the simulation campaign was conducted in collaboration with Antonio Estacio Mateos, an MSc student working on a related thesis project within the same research group. The two projects used the same computational framework, and some simulations and their resulting data were shared between them [31].

3.2 DNA sequences

3.2.1 The 1KX5 reference and A-rich variants

The first set of sequence comparisons was designed to examine the effect of progressively increasing the length of a central adenine tract while preserving a common reference system. The reference was the 147-bp DNA sequence associated with the experimentally resolved nucleosome core particle structure 1KX5 [20]. The 1KX5 structure served two distinct purposes: its DNA sequence defined the common reference endpoint for the alchemical transformations, while the experimentally resolved coordinates provided the initial wrapped configuration from which the coarse-grained nucleosomal systems were constructed.

The choice of adenine-rich sequences was motivated by the characteristic structural and mechanical properties of A-tracts discussed in Section 2.2.3, including their effects on groove geometry, intrinsic curvature and deformation response. Systematically extending the central adenine tract provides a controlled way of introducing an increasingly long and comparatively rigid segment into the 1KX5 sequence, making it possible to examine how this localized mechanical perturbation affects relative nucleosome-binding free energies.

The A-rich variants were obtained by replacing a central region of the 1KX5 sequence with a continuous tract of adenines, while keeping the total DNA length fixed at 147 bp. The remaining portions of the original 1KX5 sequence were retained symmetrically on the two sides of the central tract. The set comprised sequences containing 25, 50, 75, 100 and 125 central adenines, together with A147, in which the entire DNA sequence consists of adenines. This construction provides a controlled series in which the size and position of the A-rich region change systematically, while the overall molecular length remains constant. The complete nucleotide sequences of 1KX5 and the A-rich variants are reported in Appendix 6.2, Table 6.1.

3.2.2 Random control sequences and Widom 601

A second set consisted of five randomized DNA sequences, denoted R1, R2, R3, R4 and R5. The sequences were generated using an online random DNA sequence generator (`https://faculty.ucr.edu/~mmaduro/random.htm`), setting the length to 147 bp and the target GC content to 50%. They were included as non-specific controls without a deliberately introduced central motif, allowing the A-rich variants to be compared with sequences of the same length and broadly similar overall composition but with a different arrangement of the bases.

The Widom 601 sequence was included as an additional benchmark. This 147-bp sequence is widely used in experimental studies because of its strong nucleosome-positioning properties [2]. It therefore provides a useful comparison with both the structurally characterized 1KX5 reference and the artificial sequences considered in this work.

All sequences in this set had the same total length. This is required by the mixed-λ construction, in which the two endpoints are mapped nucleotide by nucleotide and must contain the same number of coarse-grained particles. The complete nucleotide sequences of R1, R2, R3, R4, R5 and Widom 601 are reported in Appendix 6.2, Table 6.1.

3.2.3 Genomic chromosome XIII sequences

An additional pair of DNA sequences was introduced to test whether the relative nucleosome preferences predicted by the simulations were consistent with an experimentally observed difference in nucleosome occupancy. The starting data were the genome-wide *in vitro* nucleosome-occupancy measurements obtained for *Saccharomyces cerevisiae* by Kaplan et al. [9]. Since these measurements were performed by assembling purified histone octamers on genomic DNA, they provide information about sequence-dependent nucleosome formation under controlled conditions, without the contribution of chromatin remodellers, transcription factors or other cellular components.

The occupancy data used in this work were collected within the host research group from the literature and combined with the genomic coordinates of transcription start sites (TSSs) to obtain the processed profiles. The resulting dataset contained profiles for more than 5000 TSSs, each covering a genomic region of approximately 2000 bp centred on the corresponding start site. Averaging the profiles over all TSSs produced the expected depletion in nucleosome occupancy around the transcription start site. This average provided a qualitative check of the processed dataset, whereas the sequences used in the simulations were selected from individual profiles.

The individual profiles were screened for pairs of positions displaying a large difference in experimental occupancy. The pair selected for this thesis belongs to chromosome XIII and comprises positions -397 and $+920$ relative to the same TSS. The -397 position lies in a region of comparatively low occupancy, whereas the $+920$ position has higher occupancy. A 147-bp DNA fragment centred on each position was extracted for the simulations. Selecting both fragments from the same TSS-centred genomic profile provides a comparison between two regions associated with the same broader genomic locus.

Experimental occupancy was used here as a proxy for relative nucleosome stability. Under an equilibrium interpretation, the relative probability of nucleosome formation on two sequences can be related to their effective free-energy difference through Boltzmann statistics:

$$\Delta F_A - \Delta F_B \simeq -k_{\mathrm{B}}T \ln\left(\frac{O_A}{O_B}\right), \tag{3.1}$$

where O_A and O_B denote the corresponding occupancy measures. This relation does not convert occupancy into an absolute nucleosome-binding free energy. It instead provides an effective relative benchmark when the two sequences are measured under the same experimental conditions. In the selection analysis, the contrast between the two chromosome XIII regions was estimated as approximately $3.45\,k_{\mathrm{B}}T$, corresponding to about 2.1 kcal mol^{-1} at room temperature, in favour of the $+920$ sequence.

In the alchemical transformations, the -397 sequence was assigned to $\lambda = 0$, whereas the $+920$ sequence was assigned to $\lambda = 1$. Both DNA-only and histone-associated systems were simulated, allowing the resulting relative nucleosome-binding free energy to be compared with the direction and approximate magnitude inferred from the experimental occupancy data. The complete nucleotide sequences are reported in Appendix 6.2, Table 6.2.

3.2.4 Summary of the simulated sequence pairs

Table 3.1 summarizes the endpoint sequences, DNA lengths and molecular environments considered in the final thermodynamic-integration calculations.

Sequence set	$\lambda = 0$ endpoint	$\lambda = 1$ endpoint	Length	Simulated environments
A-rich variants	A25, A50, A75, A100, A125, A147	1KX5	147 bp	DNA-only; histone-associated
Random controls	R1, R2, R3, R4, R5	1KX5	147 bp	DNA-only; histone-associated
Positioning sequence	Widom 601	1KX5	147 bp	DNA-only; histone-associated
chrXIII, no linkers	−397	+920	147 bp	DNA-only; histone-associated
chrXIII, with linkers	−397 with linkers	+920 with linkers	187 bp	DNA-only; histone-associated

Table 3.1: Summary of the sequence pairs considered in the final thermodynamic-integration calculations. The endpoint columns follow the λ convention used throughout this thesis. The genomic constructs with linker DNA contain the same central 147-bp chromosome XIII fragments as the corresponding constructs without linkers, extended by 20 bp at each end.

3.3 Construction of the simulated systems

3.3.1 General construction strategy

Each sequence transformation was simulated in two molecular environments, using the same endpoint sequences and the same λ convention: a DNA-only system and a histone-associated system. These systems correspond to the two horizontal transformations of the thermodynamic cycle introduced in Section 2.5.4.

DNA was represented using the thermodynamic-integration implementation of MADna, while the histone-associated systems combined this DNA description with a coarse-grained representation of the histone octamer. The following subsections describe the composition and initial construction of the two environments. The generation of the mixed-λ parameters and the corresponding simulation inputs is discussed separately in Section 3.4.

3.3.2 DNA-only systems

The DNA-only systems contained a single double-stranded DNA molecule, without histone proteins or external restraining potentials. DNA was described using a version of the MADna model adapted for Thermodynamic Integration. This implementation retains the coarse-grained representation and interaction terms of MADna, but can use modified parameter files containing the additional particle types required to represent intermediate values of λ. The construction of these mixed models and the interpolation of their sequence-dependent parameters are described in Section 3.4.

Initial particle positions were read from previously prepared coordinate files rather than generated independently for each thermodynamic window. The original reference geometry was derived from the experimentally determined nucleosome structure deposited in the Protein Data Bank (PDB ID: 1KX5), which was subsequently mapped onto the coarse-grained representation used in the simulations. The resulting coarse-grained nucleosomal configuration was then used as the structural reference for preparing the initial coordinates. For the DNA-only systems, the histone particles were removed from this reference configuration and only the DNA coordinates were retained.

Using geometrically consistent starting configurations reduces severe particle overlaps and highly distorted bonded interactions, which could otherwise produce very large forces during the initial integration steps. It also ensures that the different thermodynamic windows are initialized from comparable molecular conformations. The DNA was subsequently allowed to relax and equilibrate under its own interaction potential. These simulations provide the DNA-only free-energy branch, ΔF_{DNA}, of the

thermodynamic cycle.

3.3.3 Construction of the coarse-grained histone octamer

The histone octamer was represented using the CALVADOS 3 model introduced in Section 2.4.4. This section describes how the model was applied to construct the histone component of the nucleosomal systems used in the simulations. The octamer contained two copies of each core histone, H2A, H2B, H3 and H4, and its initial structure was obtained from PDB ID 1KX5.

The atomistic histone structure was mapped onto the CALVADOS 3 representation. In the disordered regions, including the flexible histone tails, each interaction site was placed at the corresponding C_α position. Within folded domains, the bead was instead placed at the centre of mass of the residue. The resulting coarse-grained representation of the histone octamer is shown in Figure 3.1.

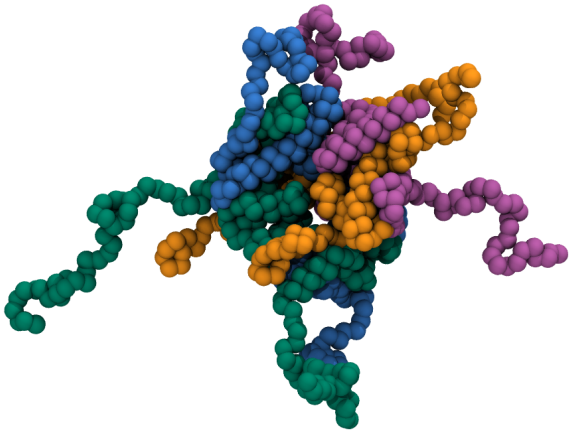


Figure 3.1: Coarse-grained representation of the histone octamer obtained by mapping PDB ID 1KX5 [20] onto the one-bead-per-residue CALVADOS 3 model [29]. Each sphere represents one amino acid and is displayed with the protein-bead radius used in the simulations, 3.0 Å. The colour convention is the same as in Figure 2.4: H3 is shown in green, H4 in blue, H2A in orange and H2B in purple. The molecular visualization was generated in this work with VMD [21].

Protein connectivity was maintained through harmonic bonds between consecutive residues, while residue-dependent non-bonded and screened electrostatic interactions followed the CALVADOS 3 parametrization. To preserve the tertiary organization of the structured histone regions, an elastic network model (ENM) was applied to the folded domains.

Pairs of residues belonging to the same folded domain were connected by harmonic restraints when their distance in the 1KX5 reference structure was smaller than 0.9 nm. The elastic-network energy was

$$U_{\mathrm{ENM}} = \sum_{(i,j)\in\mathcal{R}} \frac{1}{2} k_{\mathrm{ENM}} \left(r_{ij} - r_{ij,0}\right)^2 , \tag{3.2}$$

where $\mathcal{R}$ denotes the set of restrained residue pairs, r_{ij} is their instantaneous distance and $r_{ij,0}$ is the corresponding distance in the reference structure. The force constant used in the simulations was

$$k_{\mathrm{ENM}} = 1.67 \text{ kcal mol}^{-1} \text{ Å}^{-2}, \tag{3.3}$$

equivalent to the value of 700 kJ mol^{-1} nm^{-2} specified for CALVADOS 3.

In addition to the restraints within individual folded domains, selected residue pairs at the interfaces between histone chains were connected by reference-based harmonic restraints to preserve the assembled organization of the octamer. Residues outside the restrained regions, particularly those in the terminal tails, remained flexible during the simulations. The complete nucleosomal systems were obtained by combining this histone representation with MADna DNA through the cross-interactions described in the following subsection.

3.3.4 Implementation of DNA–histone interactions

The general coupling between MADna and CALVADOS 3 was introduced in Section 2.4.5. In the simulated systems, the interaction between the DNA and protein components consisted of steric repulsion and screened electrostatics. These cross-interactions were evaluated between beads belonging to the two different molecular components, without introducing permanent DNA–histone contacts. Steric interactions between DNA and protein beads were described using the Weeks–Chandler–Andersen potential defined in Eq. 2.8. For a pair formed by a DNA bead i and a protein bead j, the effective interaction distance was calculated as $\sigma_{ij} = R_i + R_j$, where R_i and R_j are the corresponding bead radii. This repulsive contribution prevents unphysical overlap between DNA and the histone octamer.

Electrostatic interactions between DNA phosphate beads and charged histone residues were described using the Debye–Hückel potential introduced in Eq. 2.9. For DNA–protein interactions, phosphate beads were assigned a charge of $q = -1$, rather than the reduced effective charge $q = -0.6$ used for DNA–DNA electrostatics in MADna. As discussed in Section 2.4.2, the reduced MADna charge accounts implicitly for counterion condensation between DNA phosphates and was not applied to direct DNA–protein interactions. Histone residues carried the charges defined by CALVADOS 3: $+1$ for arginine and lysine, -1 for aspartate and glutamate, and $q = +0.09$ for histidine at pH 7. The remaining amino-acid beads were neutral. This treatment represents the screened electrostatic attraction between the negatively charged DNA backbone and positively charged histone residues.

No additional attractive potential was introduced for specific combinations of DNA bases and amino acids, and no permanent contacts were imposed between DNA and the histone octamer. Sequence dependence enters the DNA–protein interface through the MADna description: changing the DNA sequence modifies its preferred geometry and the radii of the base beads, thereby affecting the spatial arrangement of the interaction sites and the steric component of the cross-interactions. Directional hydrogen bonds and atom-specific contacts are not included in the model.

3.3.5 Nucleosome systems without linker DNA

The nucleosome systems without linker DNA combined the coarse-grained histone octamer described above with a 147-bp double-stranded DNA molecule. In the MADna representation, a duplex of N base pairs contains $6N - 2$ DNA beads: each strand contains N sugar beads and N base beads, but only $N - 1$ phosphate beads associated with the phosphodiester connections between consecutive nucleotides. The 147-bp constructs therefore contained 880 DNA beads. The A-rich variants, the random control sequences, Widom 601 and the chromosome XIII fragments without linkers were all represented using this same nucleosomal architecture.

Consistently with the 1KX5-based construction described above, the initial coordinates placed the histone octamer in its reference organization and the DNA along the wrapped path observed in the experimental structure. The particle arrangement and initial nucleosomal geometry were therefore common to the two alchemical endpoints, while the sequence-dependent MADna parameters varied between them. For the A-rich variants, the random controls and Widom 601, the ten realizations at each thermodynamic window were initialized from the same 1KX5-derived coordinate file. For the chromosome XIII systems, ten separate starting-coordinate files were instead assigned to the ten realizations.

After initialization, the DNA was free to move and change conformation under the MADna potential and the DNA–histone interactions described in Section 3.3.4. The elastic network introduced in

Section 3.3.3 acted only within the selected structured regions of the histone octamer, preserving the organization of its folded domains while leaving the histone tails flexible. No restraint or permanent DNA–histone contact was imposed to keep the DNA on its initial path or force it to remain wrapped around the octamer. Its association with the histones was determined solely by the non-bonded DNA–protein interactions, with electrostatics providing the attractive contribution.

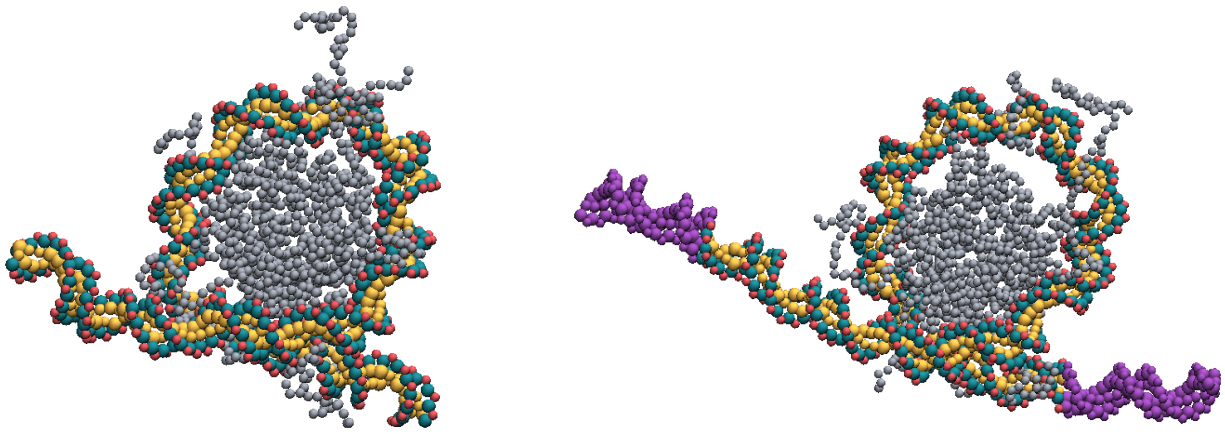
3.3.6 Genomic systems with linker DNA

The genomic systems with linker DNA were constructed from the same chromosome XIII sequences used in the 147-bp comparison. Each central fragment was extended by adding 20 adenines to both ends of the sequence-defining strand, together with the corresponding complementary thymine segments on the opposite strand. The resulting constructs had a total length of 187 bp and contained 1120 DNA beads in the MADna representation. The linker sequences were identical in the two alchemical endpoints and remained unchanged along the transformation; only the central 147-bp genomic region differed between the −397 and +920 systems.

Poly(dA:dT) terminal segments were selected because extended A-rich tracts generally have a low propensity for nucleosome formation, as discussed in Section 2.2.3. They were therefore expected to remain preferentially outside the wrapped region, providing additional flanking DNA without introducing strongly nucleosome-favouring terminal sequences. Since the same linker sequence was present in both alchemical endpoints, the added segments did not contribute directly to the sequence transformation.

A VMD visualization of the constructs with and without linker DNA is shown in Figure 3.2.

As in the systems described above, a physically reasonable initial configuration had to be prepared before starting the simulations. Once again, the experimentally determined 1KX5 structure was used as the structural reference. Its atomistic coordinates had first been mapped onto the coarse-grained representation employed in this work. The resulting nucleosomal configuration provided the initial arrangement of the histone octamer and the central wrapped DNA fragment, while the additional terminal segments were constructed to extend the DNA beyond the entry and exit regions.



(a) 147-bp construct without linker DNA. **(b)** 187-bp construct with linker DNA.

Figure 3.2: VMD representations of the histone-associated chromosome XIII +920 endpoint ($\lambda = 1$) at frame 0: (a) the 147-bp construct without linker DNA and (b) the 187-bp construct containing 20-bp linker regions at both DNA ends. Histones are shown in grey; phosphates, sugars and bases in the central DNA fragment are shown in coral, teal and gold, respectively; the terminal linker regions are highlighted in purple. The configurations were visualized in this work using VMD [21].

Because this procedure produced a newly assembled molecular configuration, a preliminary relaxation test was performed before the production simulations. The histone-associated system was simulated

at $\lambda = 0.5$ for 500 ps using a timestep of 0.001 ps, ten times smaller than the timestep used in the production simulations and described in Section 3.5.2. The smaller timestep allowed close contacts and local distortions introduced during the construction process to relax more gradually, reducing the risk of very large initial forces and numerical instabilities. This short simulation was used as a preparation and stability check and was not included in the Thermodynamic Integration sampling.

For the corresponding DNA-only systems, the initial DNA configurations were obtained from the prepared full-system coordinates by removing the histone beads. The DNA-only and histone-associated branches therefore started from geometrically corresponding DNA conformations. The assignment of the prepared coordinate files to the ten production realizations is described in Section 3.5.3.

The comparison with the systems without linkers was designed to determine whether the presence of flanking DNA modifies the calculated relative nucleosome-binding free energy. In particular, the added segments extend beyond the nucleosomal entry and exit regions and can affect the conformational freedom of DNA and its possible interactions with the histone tails, while leaving the central genomic sequence comparison unchanged.

3.4 Construction of mixed-λ systems

3.4.1 Alchemical mapping and endpoint convention

The theoretical basis of Thermodynamic Integration and the target-to-reference endpoint convention were introduced in Section 2.5.3. They are recalled only briefly here to describe the sequence-encoding procedure used to construct the mixed MADna systems.

For each comparison, the endpoint DNA sequences were denoted as the target and reference sequences. Artificial intermediate systems were generated by interpolating the sequence-dependent MADna parameters through the coupling parameter λ. For a generic sequence-dependent parameter x, the interpolation was defined as

$$x(\lambda) = \lambda x_{\text{reference}} + (1 - \lambda) x_{\text{target}}. \tag{3.4}$$

Therefore, $\lambda = 0$ corresponds to the target sequence and $\lambda = 1$ to the reference sequence. Intermediate values do not represent physical nucleotide sequences, but DNA models whose sequence-dependent parameters combine the corresponding target and reference values.

The endpoint sequences had the same length and were aligned position by position. Consequently, the number of particles and their connectivity remained unchanged along the alchemical path, while the sequence-dependent interaction parameters varied continuously with λ.

At each sequence position, the nucleotide in the reference was associated with the nucleotide occupying the same position in the target. The resulting ordered pair represents the two alternative nucleotide identities assigned to that position at the alchemical endpoints; it does not describe two bases paired across the complementary DNA strands. Each of the sixteen possible combinations was assigned an artificial mixed base type:

AA → a	AC → b	AG → c	AT → d
CA → e	CC → f	CG → g	CT → h
GA → i	GC → j	GG → k	GT → l
TA → m	TC → n	TG → o	TT → p

The first letter denotes the nucleotide in the reference sequence, whereas the second denotes the nucleotide in the target sequence. The order is therefore important. For example, the mixed type `d` represents adenine in the reference sequence and thymine in the target sequence. At $\lambda = 0$, its parameters are those of thymine; at $\lambda = 1$, they are those of adenine; and at $\lambda = 0.5$, each interpolated parameter is the equally weighted average of the corresponding thymine and adenine values.

The complete mixed sequence supplied to the molecular builder was written using the artificial alphabet `a` to `p`. Positions containing the same nucleotide in both endpoints were also assigned their corresponding mixed type, such as `AA`. At these positions, the two endpoint parameter sets are identical and no effective change occurs along λ. This uniform representation allowed the same construction procedure to be applied to the entire sequence. The corresponding mixed types on the complementary strand were assigned consistently with Watson–Crick pairing. The endpoint assignments for the different sequence comparisons are summarized in Section 3.2.4.

The same positional mapping and endpoint convention were used for the DNA-only and histone-associated systems. The Thermodynamic Integration calculation therefore gives

$$\Delta F = F_{\text{reference}} - F_{\text{target}} \tag{3.5}$$

in both environments.

The alchemical path was sampled at eleven uniformly spaced values,

$$\lambda = 0.0,\ 0.1,\ 0.2,\ \ldots,\ 1.0, \tag{3.6}$$

corresponding to a spacing of $\Delta\lambda = 0.1$. These windows provided the discrete values required for the numerical integration described in Section 3.7.3. The sensitivity of the resulting free-energy estimate to the λ spacing is examined in Section 4.1.1.

3.4.2 Interpolation of sequence-dependent parameters

As described in the previous subsection, the first step in constructing an intermediate system was to express the DNA sequence using the artificial alphabet. Before generating the simulation input for a given value of λ, the artificial base types also had to be assigned the numerical parameters required by MADna.

For this purpose, the standard MADna parameter set was extended by introducing the sixteen mixed base types defined in the previous subsection. For each of the eleven λ values, their particle properties and sequence-dependent interaction parameters were obtained using the interpolation equation introduced above. A separate parameter file was therefore generated for every λ value and used together with the artificially encoded sequence to construct the corresponding molecular system.

For example, the artificial type `d` represents thymine at the target endpoint and adenine at the reference endpoint. At $\lambda = 0.3$, each of its interpolated particle parameters contains a contribution of 70% from thymine and 30% from adenine. For its radius, this gives

$$r_{\mathtt{d}}(0.3) = 0.7 r_{\mathrm{T}} + 0.3 r_{\mathrm{A}}. \tag{3.7}$$

The same procedure was applied to all sequence-dependent quantities required by the MADna model.

At the level of individual particles, the interpolation was applied to the mass and radius associated with each base type. The four standard base beads are electrically neutral in MADna, so their interpolated charges remain zero. Their masses and radii, however, depend on base identity, with the interpolated radii determining the excluded-volume interactions involving the mixed base beads.

The interpolation was also applied to all sequence-dependent bonded interactions introduced in Section 2.4.2. Depending on the interaction, the corresponding parameters are determined by the identity of a single nucleotide, a complementary base pair or two consecutive base pairs. These terms include the sugar–base and base–base bonds, the sequence-dependent angles and the interactions associated with neighbouring base-pair steps. Both their equilibrium values and force constants were interpolated between the target and reference parameter sets.

The properties of the sugar and phosphate beads and the parameters controlling electrostatic screening were not modified by the alchemical transformation. The histone model and the elastic network preserving its structured regions also remained fixed. In the histone-associated systems, the interpolated radii of the DNA base beads modified the sequence-dependent DNA–protein excluded-volume interactions, whereas the electrostatic interactions involving the phosphate beads remained unchanged throughout the transformation.

3.4.3 Generation of the simulation inputs

For each of the eleven λ values, the artificially encoded sequence and the corresponding interpolated MADna parameter set were used to construct a separate molecular system. A generation script performed the sequence conversion and parameter interpolation described in the previous subsections and then supplied the resulting model to VLMP.

VLMP (Virtual Lab Modeling Platform) is a Python framework that combines molecular models, interaction parameters, coordinates and simulation protocols to generate the input required by molecular-simulation programs. In the present workflow, it was used to assemble the mixed DNA model and prepare the complete description of each system.

For the DNA-only systems, a custom VLMP component interpreted the artificial base types and constructed the DNA particles, connectivity and interactions. For the histone-associated systems, the same mixed DNA representation was combined with the CALVADOS 3 model of the histone octamer and with the DNA–protein interactions described in Section 3.3.4.

After constructing the particles and interactions, VLMP assigned the previously prepared starting coordinates to the corresponding system. The intermediate systems therefore started from the intended molecular configuration while differing in their sequence-dependent parameter values. The use of different starting configurations for the chromosome XIII simulations is described in Section 3.5.3.

For each λ value, VLMP generated ten simulation inputs corresponding to ten separate realizations. Each input contained the particles, molecular interactions, initial coordinates, thermodynamic ensemble, integration protocol and requested outputs required to execute the simulation.

The resulting inputs were executed with UAMMD-structured, as described in Section 3.5.

3.5 Molecular dynamics simulations

3.5.1 Software and computational resources

The simulation inputs generated with VLMP were executed using UAMMD-structured, a GPU-accelerated molecular-simulation framework based on the UAMMD platform [32] that evaluated the forces associated with the interactions defined in the inputs, propagated the molecular dynamics trajectories and recorded the configurations and thermodynamic quantities required for the subsequent analyses.

The production simulations were performed on a GPU computing cluster managed through the SLURM workload manager. The software documentation for VLMP and UAMMD-structured is provided in Refs. [33, 34].

3.5.2 Simulation protocol

All production simulations used the same thermodynamic conditions and numerical integration scheme. The systems were simulated in the canonical NVT ensemble using the Langevin dynamics introduced in Section 2.5.1, integrated with the Brünger–Brooks–Karplus algorithm [35].

Solvent molecules and ions were represented implicitly. The ionic strength reported in Table 3.2 determined the electrostatic screening in both the DNA-only and histone-associated systems, while

the large simulation box prevented interactions between periodic copies of the molecular system. The principal simulation parameters are summarized in Table 3.2.

Table 3.2: Principal parameters used in the production molecular dynamics simulations.

Parameter	Value
Temperature	300 K
Simulation box	cubic, side length 10000 Å
Ionic strength	150 mM
Histone-model pH	7.0
Integration timestep	0.01 ps
Langevin friction coefficient	0.02 ps^{-1}
Production time: A-rich, random and Widom 601	500 ns per realization
Production time: chromosome XIII	1500 ns per realization
Configuration and thermodynamic-output interval	100 ps
Restart interval	1000 ps

The timestep and friction coefficient were converted to the internal time units associated with the `KcalMol_A` unit system before the simulation inputs were generated.

3.5.3 Independent realizations and simulation outputs

As described above, ten independent realizations were performed for each value of λ. These repeated simulations were used to obtain more representative averages and to estimate the statistical variability of the calculated quantities, as discussed in Section 3.7.

The starting coordinates assigned to the ten realizations depended on the group of systems considered. For the A-rich, random and Widom 601 transformations, all realizations were initialized from the same prepared molecular configuration. For the chromosome XIII transformations, the ten realizations were initialized from ten configurations extracted as snapshots at different times from a shorter preparatory simulation, so that they did not all start from identical molecular coordinates.

During each simulation, particle coordinates and thermodynamic quantities were recorded at the intervals reported in Table 3.2. The saved coordinate trajectories were subsequently used to calculate the energy derivative required for Thermodynamic Integration and to perform the structural analyses described later in this chapter.

The potential-energy derivative with respect to λ was not recorded directly during the production simulations. It was evaluated afterwards by applying the procedure described in Section 3.6 to the saved molecular configurations.

3.6 Calculation of $\partial U/\partial\lambda$

3.6.1 Preparation of the derivative inputs

Before calculating the energy derivative from the saved trajectories, a Python preprocessing script prepared the parameters required for each λ window. The script read the force-field parameter sets used at the two endpoints of the transformation, $\lambda = 0$ for the target sequence and $\lambda = 1$ for the reference sequence, together with the parameter set used to simulate the intermediate system at the λ value being analysed.

The endpoint parameter sets were compared to determine how every sequence-dependent parameter changes during the alchemical transformation. Because the interpolation was linear, the derivative of a generic parameter x was constant along the entire transformation and was given by

$$\frac{dx}{d\lambda} = x_{\lambda=1} - x_{\lambda=0} = x_{\text{reference}} - x_{\text{target}}. \tag{3.8}$$

The parameter set associated with the current λ value was required for a different purpose: it provided the actual force-field parameters with which the corresponding trajectory had been generated. Therefore, for each interaction, the preprocessing step stored both its parameter values at the current λ and their derivatives with respect to λ.

For example, consider a bond whose force constant and equilibrium distance depend on the DNA sequence. The target parameter set provides the values k_{target} and $r_{0,\text{target}}$, whereas the reference parameter set provides $k_{\text{reference}}$ and $r_{0,\text{reference}}$. Using the linear interpolation defined in Eq. 3.4, these parameters are

$$\begin{aligned} k(\lambda) &= (1-\lambda)k_{\text{target}} + \lambda k_{\text{reference}}, \\ r_0(\lambda) &= (1-\lambda)r_{0,\text{target}} + \lambda r_{0,\text{reference}}. \end{aligned} \tag{3.9}$$

Their derivatives are consequently

$$\frac{dk}{d\lambda} = k_{\text{reference}} - k_{\text{target}}, \qquad \frac{dr_0}{d\lambda} = r_{0,\text{reference}} - r_{0,\text{target}}. \tag{3.10}$$

For the λ window being analysed, the script also retrieved the interpolated values $k(\lambda)$ and $r_0(\lambda)$ that had been used during the simulation. The saved molecular configuration then supplied the instantaneous bond distance r. These quantities—$k(\lambda)$, $r_0(\lambda)$, $dk/d\lambda$, $dr_0/d\lambda$ and r—were subsequently combined by the C++ program to calculate the contribution of that bond to $\partial U/\partial\lambda$.

The same preparation was performed for the sequence-dependent angle and dihedral interactions. For each bonded interaction, the prepared data identified the particles involved and contained both the parameters at the current λ value and their derivatives. For excluded-volume interactions, the script recorded the bead radii at the current λ value and their derivatives between the two endpoints. It also identified the particle pairs excluded from the non-bonded calculation because their interactions were already represented by bonded terms.

At a given λ value, all ten realizations shared the same molecular topology and force-field parameters. The parameter information was therefore prepared once for each λ window and then combined separately with the coordinates saved for each of the ten trajectories. The standard thermodynamic output was not used for this calculation because it did not contain $\partial U/\partial\lambda$; the derivative was reconstructed directly from the saved molecular configurations using the procedure described in the following subsection.

3.6.2 Frame-by-frame energy derivative

The information prepared in the previous subsection was combined with the coordinates saved during each simulation. A C++ program read each trajectory frame by frame and calculated the instantaneous geometrical quantities required by the force field: bond distances, bond angles, dihedral angles and distances between particles involved in excluded-volume interactions.

The dependence of the potential energy on λ arises from the sequence-dependent parameters interpolated between the target and reference systems. In the implemented model, the explicit contributions to the derivative are those associated with bonds, angles, dihedrals and excluded-volume interactions:

$$\frac{\partial U}{\partial \lambda} = \frac{\partial U_{\text{bond}}}{\partial \lambda} + \frac{\partial U_{\text{angle}}}{\partial \lambda} + \frac{\partial U_{\text{dihedral}}}{\partial \lambda} + \frac{\partial U_{\text{WCA}}}{\partial \lambda}. \tag{3.11}$$

Other interactions still influence the molecular configurations sampled during the simulation. However, the electrostatic parameters, the histone model and the elastic restraints applied to the histone core

do not change along the alchemical transformation. Their explicit derivatives with respect to λ are therefore zero, and they do not appear in Eq. 3.11.

For a harmonic interaction involving an instantaneous coordinate q, an equilibrium value q_0 and a force constant k, the convention used in the simulations and reproduced in the post-processing program is

$$U_{\text{harmonic}} = k(q - q_0)^2. \tag{3.12}$$

Since both k and q_0 may vary with λ, its derivative is

$$\frac{\partial U_{\text{harmonic}}}{\partial \lambda} = \frac{dk}{d\lambda}(q - q_0)^2 - 2k(q - q_0)\frac{dq_0}{d\lambda}. \tag{3.13}$$

This expression was applied to bonds by setting $q = r$ and $q_0 = r_0$, and to angles by setting $q = \theta$ and $q_0 = \theta_0$. In each frame, the instantaneous value of q was calculated from the particle coordinates, while the current values of k and q_0, together with their λ derivatives, were taken from the data prepared in Section 3.6.1.

The sequence-dependent dihedral potential was represented as

$$U_{\text{dihedral}} = k\left[1 + \cos(\phi - \phi_0)\right], \tag{3.14}$$

and its contribution was calculated as

$$\frac{\partial U_{\text{dihedral}}}{\partial \lambda} = \frac{dk}{d\lambda}\left[1 + \cos(\phi - \phi_0)\right] + k\sin(\phi - \phi_0)\frac{d\phi_0}{d\lambda}. \tag{3.15}$$

The excluded-volume contribution was evaluated for the non-excluded particle pairs involving DNA beads. For each pair, the interaction distance was determined by the sum of the two bead radii. When the instantaneous particle separation was within the repulsive interaction range, the program calculated the energy derivative associated with the variation of these radii along λ. In the histone-associated systems, this term also included changes in DNA–protein steric interactions resulting from the sequence-dependent radii of the DNA base beads.

For every saved frame, the program reported separately the bond, angle, dihedral and excluded-volume contributions, together with their sum. The resulting time series of the total $\partial U/\partial\lambda$ was calculated independently for each of the ten realizations and was subsequently used for equilibration removal, statistical averaging and numerical integration, as described in Section 3.7.

3.7 Free-energy analysis and validation

3.7.1 Equilibration and convergence assessment

For each realization, the frame-by-frame calculation described in Section 3.6 produced a time series of the total potential-energy derivative. Before calculating its mean, the first 100 saved values were discarded to remove the initial relaxation interval. Since configurations were recorded every 100 ps, this corresponds to the exclusion of the first 10 ns of each trajectory. The averages were therefore calculated over the remaining 490 ns for the A-rich, random and Widom 601 systems and over the remaining 1490 ns for the chromosome XIII systems.

Temporal convergence was assessed by inspecting representative derivative time series together with their moving averages. The moving average was used only to reveal possible long-term drifts beneath the instantaneous fluctuations and did not enter the free-energy calculation. After averaging the trajectories at each λ value, the resulting Thermodynamic Integration profile was also inspected to identify possible discontinuities or anomalous windows. The outcome of these checks is presented in Section 4.1.1.

3.7.2 Statistical averaging and uncertainty estimation

After removal of the equilibration interval, the derivative values were first averaged separately within each realization. This produced ten realization-specific estimates for every λ window. These ten values were subsequently combined to obtain the final mean derivative

$$\left\langle \frac{\partial U}{\partial \lambda} \right\rangle_{\lambda} \tag{3.16}$$

together with its statistical uncertainty. Thus, one time-averaged derivative was first obtained from each realization, and the dispersion among the ten trajectory means could be evaluated separately from the instantaneous fluctuations occurring within an individual trajectory.

Consecutive configurations belonging to the same realization are temporally correlated and therefore do not constitute independent samples. The analysis workflow accounts for the trajectories through the two-stage averaging described above. However, the precise estimator used to obtain the uncertainty from the realization-specific values must still be verified before it is identified as a standard deviation, standard error or bootstrap estimate.

The final mean and uncertainty obtained at each of the eleven λ values were then used for numerical integration. The limitations associated with finite trajectory length and incomplete sampling are discussed together with the results in Section 4.1.2.

3.7.3 Numerical integration and λ discretization

For each sequence transformation and molecular environment, the procedure described above produced eleven estimates of the mean energy derivative at

$$\lambda_i = 0.0, 0.1, \ldots, 1.0. \tag{3.17}$$

These values define the discrete Thermodynamic Integration profile. The free-energy difference was calculated using the trapezoidal rule with a uniform spacing $\Delta\lambda = 0.1$:

$$\Delta F \simeq \Delta\lambda \left[\frac{1}{2} g_0 + \sum_{i=1}^{9} g_i + \frac{1}{2} g_{10} \right], \tag{3.18}$$

where

$$g_i = \left\langle \frac{\partial U}{\partial \lambda} \right\rangle_{\lambda_i}. \tag{3.19}$$

The endpoint values receive half weight because each belongs to only one integration interval, whereas every intermediate value contributes to the intervals on both sides. With the endpoint convention adopted in this work, the result corresponds to

$$\Delta F = F_{\text{reference}} - F_{\text{target}}. \tag{3.20}$$

The uncertainty associated with the numerical integral was obtained by propagating the uncertainties of the eleven derivative estimates in quadrature, using the same trapezoidal weights:

$$\sigma_{\Delta F} = \Delta\lambda \sqrt{\frac{\sigma_0^2}{4} + \sum_{i=1}^{9} \sigma_i^2 + \frac{\sigma_{10}^2}{4}}. \tag{3.21}$$

This calculation was performed independently for the DNA-only and histone-associated branches of every transformation.

The sensitivity to the discretization of the alchemical path was examined separately for the A147 transformation in the nucleosomal environment. The integral obtained from all eleven windows was compared with values recalculated using progressively coarser subsets of the available λ points. This test was used to assess whether the selected spacing provided sufficient numerical resolution; its results are reported in Section 4.1.1.

3.7.4 Relative nucleosome-binding preference

The DNA-only and histone-associated calculations describe the same target-to-reference transformation in two different molecular environments:

$$\Delta F_{\mathrm{DNA}} = F_{\mathrm{DNA}}^{\mathrm{reference}} - F_{\mathrm{DNA}}^{\mathrm{target}}, \tag{3.22}$$

and

$$\Delta F_{\mathrm{nucleosome}} = F_{\mathrm{nucleosome}}^{\mathrm{reference}} - F_{\mathrm{nucleosome}}^{\mathrm{target}}. \tag{3.23}$$

Following the thermodynamic cycle introduced in Section 2.5.4, the relative nucleosome-binding preference was calculated as

$$\Delta\Delta F_{\mathrm{bind}} = \Delta F_{\mathrm{DNA}} - \Delta F_{\mathrm{nucleosome}}. \tag{3.24}$$

This quantity is the difference between the nucleosome-binding free energies of the target and reference sequences, rather than an absolute binding free energy. With the sign convention adopted here, a positive value indicates more favourable nucleosome binding for the reference sequence, whereas a negative value indicates more favourable binding for the target. A value compatible with zero indicates that no difference in their relative nucleosome-binding free energies is resolved by the calculation.

Assuming that the DNA-only and histone-associated calculations are statistically independent, their uncertainties were combined in quadrature:

$$\sigma_{\Delta\Delta F} = \sqrt{\sigma_{\Delta F_{\mathrm{DNA}}}^2 + \sigma_{\Delta F_{\mathrm{nucleosome}}}^2}. \tag{3.25}$$

For chromosome XIII, the same calculation was performed separately for the constructs with and without linker DNA.

3.8 Post-simulation trajectory analysis

3.8.1 Structural visualization in VMD

Before performing any quantitative calculation on the trajectories, the simulated systems were inspected using Visual Molecular Dynamics (VMD), a program for the visualization and analysis of molecular structures and trajectories [21]. This preliminary check was used to follow the systems over time, verify their structural integrity and confirm that the DNA remained associated with the histone core in the nucleosomal simulations.

The visualization also provided a qualitative comparison of the conformations sampled by the different sequences, including changes in overall DNA bending, nucleosomal wrapping and opening of the entry and exit regions. These observations were used only as an initial structural assessment and not as quantitative evidence of equilibrium behaviour. Representative VMD configurations from selected systems are reported in Appendix 6.1, Figures 6.1–6.3.

3.8.2 Trajectory processing with cLibrary

Following the preliminary visualization, the trajectories were analysed quantitatively using cLibrary, a collection of C++ programs developed within the research group for the geometrical analysis of coarse-grained DNA trajectories. Only the two physical endpoints of each alchemical transformation were considered: $\lambda = 0$ corresponds to the complete target sequence and $\lambda = 1$ to the complete reference sequence, whereas the intermediate-λ systems contain artificial mixed nucleotide types introduced exclusively for Thermodynamic Integration.

The selected trajectories were converted into the input format required by the analysis programs without modifying the DNA coordinates or the order of the saved configurations. The C++ program then processed the trajectory sequentially, reading one frame at a time and reconstructing the base-pair centres and local helical reference frames from the positions of the DNA beads. These geometrical quantities were used to calculate groove dimensions and the translational and rotational parameters describing consecutive base pairs. The resulting position- and time-dependent data were subsequently averaged and plotted using dedicated Python scripts.

The first 100 saved configurations were excluded before calculating temporal averages, consistently with the equilibration interval adopted for the free-energy analysis. Before selecting a single trajectory for the structural analysis, minor-groove-width profiles were compared across the available realizations. Spatial Pearson correlations were calculated along the base-pair index between the temporal mean profile of each realization and the corresponding mean over the available realizations. The high correlations indicated similar spatial patterns across the tested realizations for this observable. For a DNA-only test system, a profile calculated from one realization differed from the ten-realization mean by a median RMSD of approximately 0.001 nm, corresponding to about 0.08% of the mean groove width, and retained a correlation greater than 0.999. For chromosome XIII at $\lambda = 0$, the DNA-only profiles obtained from the available realizations showed correlations of approximately 0.9997–0.9998 with their mean profile. The nucleosomal profiles showed larger differences in amplitude than the DNA-only profiles but retained nearly identical spatial patterns, with correlations between approximately 0.997 and 0.998.

On the basis of these preliminary comparisons, the principal structural analyses were performed using the first of the ten production simulations, labelled R0. This choice substantially reduced the amount of trajectory processing while retaining the main spatial features observed in the tested mean profiles. However, because the comparison between realizations was performed using minor-groove width, it does not establish convergence for every structural observable. The resulting profiles are therefore interpreted as descriptive analyses of a representative trajectory rather than as ensemble averages over all ten realizations.

3.8.3 Local DNA geometry

The local DNA geometry was characterized using the six base-pair-step parameters introduced in Section 2.1.1. Shift, slide and rise describe translations between consecutive base pairs and are reported in nanometres, whereas tilt, roll and twist describe relative rotations and are reported in degrees. Minor-groove width, calculated for individual base pairs, was used as an additional descriptor and for the preliminary comparison between realizations described in Section 3.8.2.

For each retained trajectory frame, these quantities were calculated as a function of position along the DNA. To reduce the influence of the terminal regions, the analysis was restricted to base-pair steps 11–136 for the six step parameters and to base pairs 11–137 for minor-groove width. No spatial smoothing was applied to the resulting profiles.

For a generic base-pair-step parameter p, the local change associated with the histone environment was defined as

$$\Delta p(i) = \overline{p}_{\mathrm{nucleosome}}(i) - \overline{p}_{\mathrm{DNA}}(i), \tag{3.26}$$

where the overbars denote temporal averages after exclusion of the first 100 saved configurations. A positive value indicates that the parameter is larger in the nucleosomal trajectory than in the corresponding DNA-only trajectory at the same position, whereas a negative value indicates the opposite change. This difference describes a local geometrical response to the histone-associated environment and must not be interpreted as a free-energy difference.

The same definition was applied independently to shift, slide, rise, tilt, roll and twist. The resulting profiles were used to compare 1KX5 with Widom 601 and the two chromosome XIII sequences while keeping the colour assigned to each sequence consistent across all six parameters.

For the comparison of the chromosome XIII constructs with and without linker DNA, the analysis was restricted to equivalent regions of the two systems. Base-pair steps 31–156 of the 187-bp constructs were mapped onto positions 11–136 of the 147-bp constructs; for base-pair-centred observables, base pairs 31–157 were correspondingly mapped onto positions 11–137.

3.8.4 Global geometry of nucleosomal DNA

3.8.4.1 Nucleosome-fixed reference frame

During the simulations, the complete nucleosome can translate and rotate within the simulation box. Coordinates measured directly in the simulation reference frame would therefore combine the overall motion of the complex with changes in the position of the DNA relative to the histone core. To separate these contributions, the base-pair centres were expressed in a reference frame that follows the nucleosome in each saved configuration.

The construction of this reference frame followed the approach previously developed to analyse nucleosomal DNA dynamics within the same MADna-based coarse-grained framework [36]. Its origin, $\mathbf{R}_{\mathrm{nuc}}$, was defined as the geometrical centre of the beads belonging to the ordered regions of the histone core. Two pairs of beads were carefully selected within the structured histone domains so that their relative positions remained sufficiently stable during the trajectory and provided reproducible internal directions. The resulting vectors were orthogonalized using a Gram–Schmidt procedure to construct two axes, while the third axis was obtained from their cross product, producing an orthonormal, right-handed reference frame $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ attached to the nucleosome.

For the centre $\mathbf{r}_i$ of base pair i, its displacement relative to the nucleosome was defined as $\Delta\mathbf{r}_i = \mathbf{r}_i - \mathbf{R}_{\mathrm{nuc}}$. Its coordinates in the nucleosome-fixed frame were then obtained by projection onto the three axes:

$$X_i = \Delta\mathbf{r}_i \cdot \mathbf{e}_x, \qquad Y_i = \Delta\mathbf{r}_i \cdot \mathbf{e}_y, \qquad Z_i = \Delta\mathbf{r}_i \cdot \mathbf{e}_z. \tag{3.27}$$

Since the reference frame was reconstructed for every saved configuration, these coordinates describe changes in DNA position relative to the histone core independently of the translational and rotational diffusion of the complete complex.

3.8.4.2 Radial positioning and entry/exit opening

The position of each base-pair centre relative to the nucleosome axis was described through the radial coordinate

$$\rho_i(t) = \sqrt{X_i(t)^2 + Y_i(t)^2}, \tag{3.28}$$

where $X_i(t)$ and $Y_i(t)$ are the coordinates of base pair i in the nucleosome-fixed reference frame at time t. The radial coordinate therefore measures the perpendicular distance of the base-pair centre from the z axis defined in the previous subsection.

For every retained frame, $\rho_i(t)$ was calculated separately for each base pair. Its temporal mean and standard deviation were subsequently evaluated at each position along the DNA, averaging the frame-wise distances rather than calculating the distance from the time-averaged coordinates. Following the trajectory selection and equilibration procedure described in Section 3.8.2, the analysis used realization R0 after excluding the first 100 saved frames and was restricted to base pairs 11–137. For the 187-bp chromosome XIII constructs containing linker DNA, base pairs 31–157 were selected and mapped onto positions 11–137 of the corresponding constructs without linkers. The standard deviation describes temporal fluctuations in radial position, not uncertainty in the mean.

The resulting radial profiles were used to identify which regions remained close to the nucleosome axis and which displayed larger outward displacements. Particular attention was paid to the portions of the analysed region closest to the DNA entry and exit sites, where increased radial distances were used as a geometrical indicator of DNA opening. The two ends were compared through their temporal mean profiles to assess differences in their average radial opening. Individual opening and rewrapping events were not counted in this analysis.

3.8.4.3 Effective superhelix radius and pitch

The global geometry of nucleosomal DNA was characterized through an effective superhelix radius and pitch. The analysis used the histone-associated R0 trajectories of 1KX5, Widom 601, the two chromosome XIII sequences without linker DNA, A25 and A125. After excluding the first 100 saved frames, base pairs 11–137 were considered, using their centres expressed in the nucleosome-fixed reference frame. The origin and orientation of this frame were not optimized against the DNA path.

The radial coordinate $\rho_i(t)$ was defined as in the previous subsection. For each retained frame, the azimuthal angle was calculated as

$$\phi_i(t) = \operatorname{atan2}(Y_i(t), X_i(t)) \,. \tag{3.29}$$

The angular coordinates ϕ_i were unwrapped along the DNA contour to remove the artificial discontinuities occurring between $+\pi$ and $-\pi$.

The effective radius of each frame was defined as the mean radial distance of the selected base-pair centres:

$$R(t) = \frac{1}{N} \sum_{i=1}^{N} \rho_i(t), \tag{3.30}$$

where $N = 127$ is the number of base pairs included in the analysis.

The pitch was obtained by an ordinary least-squares fit of the axial coordinates to the linear relation

$$Z_i(t) = a(t) + m(t)\phi_i(t), \tag{3.31}$$

from which

$$P(t) = 2\pi m(t). \tag{3.32}$$

The fit was performed independently for each frame. Under the adopted coordinate convention, negative values of P correspond to left-handed winding. The sign was retained when calculating temporal averages, rather than taking the absolute value of the pitch in each frame.

The coefficient of determination R^2 and the root-mean-square error of the Z residuals were retained to assess how closely the axial coordinates followed the fitted linear relation with ϕ. These diagnostics

describe the axial fit, not the full three-dimensional agreement with an ideal helix. The temporal mean and standard deviation of the frame-wise radius, signed pitch and fit-quality metrics were then calculated over the retained frames. The standard deviations describe temporal fluctuations within the R0 trajectory and do not represent uncertainties in the mean.

3.8.5 DNA–histone proximity profiles

The proximity between the DNA backbone and positively charged histone residues was analysed for the 1KX5 R0 trajectory after excluding the first 100 saved frames. The analysis was based on the minimum bead–bead distance. No distance threshold was introduced to label a phosphate–residue pair as either "in contact" or "not in contact", because the appropriate threshold depends on the molecular representation and on the sites used to measure the distance. In particular, a cutoff defined between specific atoms cannot be transferred directly to the present coarse-grained representation, in which phosphate groups and amino-acid residues are represented by effective beads. The selected residues were separated into three operational groups:

- arginine residues belonging to the non-terminal regions of the histone core;
- lysine residues belonging to the terminal histone tails;
- arginine residues belonging to the terminal histone tails.

For each phosphate j, residue group G and retained frame t, the distance from the nearest bead belonging to that group was calculated as

$$d_{j,G}^{\min}(t) = \min_{k \in G} \left\| \mathbf{r}_j^{\mathrm{P}}(t) - \mathbf{r}_k^{\mathrm{H}}(t) \right\| . \tag{3.33}$$

The minimum distance was calculated before temporal averaging. The identity of the nearest residue was therefore allowed to change between frames, rather than imposing permanent phosphate–residue pairs. This choice is consistent with the dynamic nature of the DNA–histone interface, where DNA breathing, possible sliding and rearrangements of the histone tails can change which residues are closest to a given phosphate.

Accordingly, the numerical minimum distance measured in each frame was retained. The resulting profiles describe how close each phosphate was to each residue group, rather than how often a phosphate–residue pair was classified as a contact. For each phosphate, its temporal mean and standard deviation were subsequently calculated. The two DNA strands were analysed separately.

3.9 Sequence-based crookedness analysis

3.9.1 Definition and calculation of crookedness

The crookedness values used in this work were not extracted from the molecular-dynamics trajectories. They were calculated directly from the nucleotide sequences using the prediction model described in Ref. [17] and made available through the *DNA Mechanics* software repository [37]. The model was developed from MADna simulations of DNA sequences with controlled motif compositions and predicts the crookedness of the preferred relaxed geometry associated with a given sequence.

Crookedness describes how much the path followed by the DNA axis deviates from a straight configuration. Its geometrical definition compares the length measured directly between the endpoints of successive portions of the molecule with the corresponding distance measured along the DNA path. For a straight fragment, these quantities are approximately equal. When the DNA axis is curved, the direct distance between the endpoints becomes shorter than the path connecting them. This relation is expressed through $\cos\beta$, so that $\beta = 0$ represents a straight configuration and larger β values indicate a more crooked molecular path.

In the MADna formulation reported in Ref. [17], the DNA is divided into portions spanning nine consecutive base-pair steps. The direct distances between the endpoints of these portions contribute to the numerator of $\cos\beta$, whereas the sum of the distances between consecutive base-pair centres represents the contour-length term in the denominator.

The sequence-based model estimates these two quantities from the base-pair-step composition:

$$\cos\beta = \frac{\sum_k h_k n_k}{\sum_k l_k n_k}, \tag{3.34}$$

where n_k is the number of occurrences of step type k. The parameters h_k and l_k describe the average contribution of that step type to the direct-extension and contour-length terms, respectively. Their values were obtained from MADna simulations of sequences with controlled motif compositions [17].

Complementary steps were assigned to the same parameter class, producing ten independent classes: AA/TT, GG/CC, AC/GT, CA/TG, AG/CT, GA/TC, AT, TA, CG and GC. For each sequence, the occurrences of these step classes were counted and the crookedness was calculated as

$$\beta = \arccos\left(\frac{\sum_k h_k n_k}{\sum_k l_k n_k}\right). \tag{3.35}$$

The resulting values are expressed in radians. The calculations were performed using the Python implementation distributed through the *DNA Mechanics* repository [37].

The uncertainty in β was obtained by propagating the uncertainties associated with the h_k and l_k parameters. Defining

$$H = \sum_k h_k n_k, \qquad L = \sum_k l_k n_k, \tag{3.36}$$

their uncertainties were calculated as

$$\sigma_H^2 = \sum_k \left(n_k \sigma_{h_k}\right)^2, \qquad \sigma_L^2 = \sum_k \left(n_k \sigma_{l_k}\right)^2. \tag{3.37}$$

They were subsequently propagated through $\beta = \arccos(H/L)$. The resulting uncertainty reflects the parameterization of the prediction model and does not represent temporal fluctuations of the simulated DNA.

3.9.2 Construction of $\Delta\beta$

To compare each target with the corresponding reference sequence, the crookedness difference was defined as

$$\Delta\beta = \beta_{\text{reference}} - \beta_{\text{target}}. \tag{3.38}$$

This definition follows the endpoint convention adopted for the Thermodynamic Integration calculations. The reference was 1KX5 for the A-rich sequences, random controls and Widom 601, and +920 for the chromosome XIII comparison.

Consequently, $\Delta\beta > 0$ means that the reference has a more crooked predicted geometry, $\Delta\beta < 0$ means that the target has a more crooked predicted geometry, and $\Delta\beta$ close to zero indicates similar predicted crookedness values. The uncertainty was propagated in quadrature:

$$\sigma_{\Delta\beta} = \sqrt{\sigma_{\text{reference}}^2 + \sigma_{\text{target}}^2}. \tag{3.39}$$

3.9.3 Comparison with the free-energy differences

The calculated $\Delta\beta$ values were compared with the corresponding relative binding free energies, $\Delta\Delta F_{\mathrm{bind}}$, to examine whether differences in predicted global DNA geometry were associated with the thermodynamic trends obtained from the simulations.

A linear fit was calculated only for the A-rich constructs, because they form a controlled series in which the central adenine tract is progressively extended. The random sequences, Widom 601 and the chromosome XIII systems were included as separate comparisons but were not used in the same fit. The resulting relationships are presented and discussed in Section 4.5.

Chapter 4

Results and Discussion

This chapter presents and discusses the thermodynamic and structural results obtained for the different DNA sequences. The reliability of the free-energy calculations is examined first through convergence, reproducibility, discretization and thermodynamic-cycle consistency checks. The relative nucleosome preferences of the A-rich variants, the random controls, Widom 601 and the chromosome XIII sequences are then analysed, including the comparison between constructs with and without linker DNA. Finally, the thermodynamic trends are compared with sequence-predicted crookedness and with the structural properties extracted from the simulation trajectories, providing complementary information on local DNA geometry, nucleosomal wrapping and DNA–histone interactions.

4.1 Reliability of the free-energy calculations

4.1.1 Convergence of the Thermodynamic Integration profiles

Before interpreting the free-energy results, a series of checks was performed to assess the reliability of the calculations. The first step was to examine the temporal behaviour of the potential-energy derivative. As described in Section 3.7.1, the first 10 ns of each trajectory were excluded as an initial relaxation interval before the derivative was averaged.

Figure 6.16 shows the instantaneous potential-energy derivative, $\partial U/\partial\lambda$, for one representative realization of the 1KX5–R1 transformation at $\lambda = 0.5$. The first 10 ns are omitted from the plot as an initial relaxation interval. Over the retained portion of the trajectory, the instantaneous derivative exhibits substantial fluctuations, as expected for a molecular dynamics simulation, whereas its windowed average fluctuates around an approximately constant value without an evident persistent drift. This supports the stability of the estimated mean derivative for this observable, but does not demonstrate complete equilibration or exhaustive sampling of slower conformational transitions. The windowed average is shown only as a visual convergence check and is not used directly in the free-energy calculation.

Figure 6.17 shows the corresponding Thermodynamic Integration profile. As described in Section 3.7.2, each point represents the final mean value of $\partial U/\partial\lambda$ at a given value of λ, obtained from the ten independent realizations after excluding their initial relaxation intervals. The figure therefore displays the set of mean derivatives that is numerically integrated to obtain the free-energy difference. The profile varies regularly along the alchemical path and does not contain isolated discontinuities or anomalous points, indicating that the evolution of the integrand is consistently resolved by the simulated λ windows.

A preliminary sensitivity analysis was performed for the A147 transformation in the nucleosomal environment to determine whether the selected spacing between consecutive λ windows provided sufficient

resolution along the alchemical path. As shown in Fig. 6.18, increasing the spacing from $\Delta\lambda = 0.1$ to $\Delta\lambda = 0.2$ changed the integrated free energy by approximately 0.5 kcal mol^{-1}, whereas the coarser spacing $\Delta\lambda = 0.5$ produced a larger deviation. The complete set of eleven uniformly spaced windows, corresponding to $\Delta\lambda = 0.1$, was therefore retained as a compromise between numerical resolution and computational cost. Since this analysis was performed for one representative transformation, it supports the selected discretization but does not constitute an independent convergence test for every sequence pair.

4.1.2 Statistical uncertainty and reproducibility

For each value of λ, ten separate molecular dynamics realizations were performed. The A-rich, random-control and Widom 601 systems were initialized from the same coordinates, but evolved independently under the stochastic Langevin dynamics. The dispersion between their mean derivatives therefore measures how strongly the calculated value depends on the evolution of the individual trajectories starting from the same molecular structure.

For the chromosome XIII systems, the ten realizations were instead initialized from ten different starting configurations. This choice was introduced to examine whether sampling a broader range of initial conformations produced an appreciable change in the estimated variability between realizations. The resulting uncertainties are reported together with the chromosome XIII free-energy results in Section 4.4.1. Since these trajectories were also longer than those used for the other sequence groups, the effect of the different starting configurations cannot be separated completely from that of the increased simulation time.

Temporal correlations within the trajectories were also examined in the related work of Estacio Mateos using the same computational framework [31]. A block-bootstrap analysis was used to estimate the uncertainty of the mean potential-energy derivative while accounting for the correlation between successive configurations. The estimates obtained for the individual realizations were combined at each λ value and propagated through the thermodynamic integration. For the representative transformation examined, the resulting statistical uncertainties were small compared with the variation of the mean potential-energy derivative along the alchemical path.

For the calculations reported here, the uncertainties at the individual λ windows were propagated through the numerical integration to obtain the uncertainty of each free-energy difference, as described in Section 3.7.2. The uncertainties of the DNA-only and nucleosomal transformations were then combined when calculating the corresponding relative binding free energy. The reported error bars quantify the statistical uncertainty associated with the sampled trajectories. They do not include every possible source of uncertainty in the final free-energy estimate. In particular, the sensitivity to the spacing of the λ windows and the inconsistency measured through the thermodynamic-cycle closure are assessed separately. Small statistical error bars therefore indicate that the available trajectories provide a well-defined and reproducible estimate within the adopted analysis procedure. However, they do not guarantee that the calculated free energy is equally close to its exact value. Even when the averages obtained from the independent realizations are mutually consistent, the finite simulation time may prevent the system from visiting alternative conformations separated by slow structural transitions. The resulting estimate could therefore describe precisely the region of configurational space sampled by the trajectories without representing the complete equilibrium ensemble. Additional deviations may also arise from the finite discretization of the alchemical path and from other systematic effects revealed by the thermodynamic-cycle consistency test.

4.1.3 Thermodynamic-cycle consistency

After examining temporal convergence and the sensitivity to the λ discretization, a further check was performed to assess the internal consistency of the free-energy calculations. As discussed in Section 2.5.4, the free-energy difference between two states is independent of the thermodynamic path used to connect them. Two alternative alchemical paths connecting the same initial and final

DNA sequences should therefore provide compatible results within their numerical and statistical uncertainties.

This property was tested in the DNA-only environment by calculating the free-energy difference between A147 and 1KX5 through a direct and an indirect path. The direct path consisted of the single transformation A147 → 1KX5. In the indirect path, A147 was first transformed into A50 and A50 was subsequently transformed into 1KX5. All three transformations were evaluated using the same Thermodynamic Integration protocol adopted throughout this work, including the same λ discretization, number of independent realizations and statistical-analysis procedure.

Using the target-to-reference convention adopted in this thesis, the calculated values were -30.4 ± 0.3 kcal mol^{-1} for A147 → 1KX5, -17.6 ± 0.1 kcal mol^{-1} for A147 → A50 and -10.9 ± 0.1 kcal mol^{-1} for A50 → 1KX5. The indirect path therefore gives -28.5 kcal mol^{-1}, compared with -30.4 kcal mol^{-1} for the direct transformation. As shown in Fig. 4.1, the magnitude of the cycle-closure residual is consequently 1.9 ± 0.3 kcal mol^{-1}.

The residual corresponds to approximately 6% of the direct free-energy difference. It is therefore modest relative to the total transformation free energy, but it is not compatible with zero within the propagated statistical uncertainty. The two paths are consequently close, but the cycle does not close within the error bars associated with the individual transformations.

This discrepancy indicates that the statistical uncertainties obtained from the sampled trajectories do not capture every source of variation affecting the complete free-energy calculation. The closure test does not identify a unique origin for the difference, which may contain contributions from finite conformational sampling and from the numerical discretization of the alchemical path.

Since this test was performed for one specific set of transformations, the residual of 1.9 kcal mol^{-1} cannot be treated as a universal error and added directly to every reported free-energy value. Nevertheless, it provides an empirical reference scale for interpreting small differences between sequences. Results separated by substantially less than this value should therefore be discussed cautiously, even when their propagated statistical error bars are smaller.

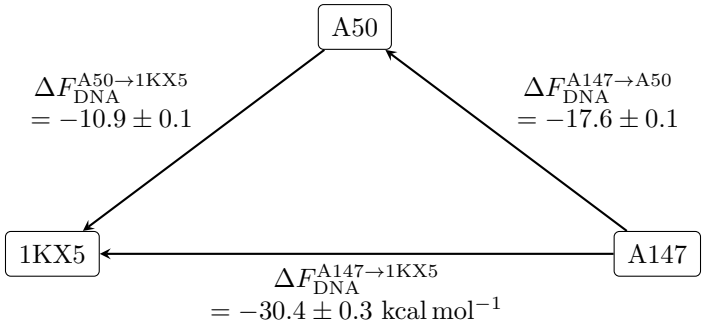


Figure 4.1: Thermodynamic cycle used to test the consistency of the DNA-only free-energy calculations. The direct transformation from A147 to 1KX5 is compared with the indirect path A147 → A50 → 1KX5. Values are reported in kcal mol^{-1} using the endpoint convention adopted in this thesis. The magnitude of the residual cycle-closure difference is 1.9 ± 0.3 kcal mol^{-1}. Diagram reconstructed from the shared calculation reported in Ref. [31].

4.2 A-rich sequences

4.2.1 Free-energy differences of the A-rich variants

After assessing the reliability of the free-energy calculations, the analysis was first applied to the controlled series of A-rich variants. The corresponding results are reported in Table 4.1. For each target, the table presents the transformation free energy calculated in the DNA-only and nucleosomal environments, together with the resulting relative nucleosome-binding free energy.

Table 4.1: Free-energy results for the A-rich target sequences relative to the 1KX5 reference. The transformations follow the target-to-1KX5 convention, and $\Delta\Delta F_{\mathrm{bind}} = \Delta F_{\mathrm{DNA}} - \Delta F_{\mathrm{NUC}}$. All values are expressed in kcal mol^{-1}. The calculations were part of the shared simulation campaign reported in Ref. [31].

Target	ΔF_{DNA}	ΔF_{NUC}	$\Delta\Delta F_{\mathrm{bind}}$
1KX5	0.0	0.0	0.0
A25	-8.31 ± 0.07	-11.31 ± 0.08	3.0 ± 0.1
A50	-10.9 ± 0.1	-16.3 ± 0.1	5.4 ± 0.1
A75	-18.7 ± 0.1	-26.6 ± 0.2	7.9 ± 0.2
A100	-21.1 ± 0.1	-31.6 ± 0.2	10.5 ± 0.2
A125	-29.1 ± 0.2	-41.2 ± 0.2	12.1 ± 0.3
A147	-30.4 ± 0.3	-41.2 ± 0.5	10.8 ± 0.6

All the target-to-1KX5 transformations give negative ΔF values in both environments. According to the convention adopted in this thesis, $\Delta F = F_{\mathrm{1KX5}} - F_{\mathrm{A-rich}}$, so a negative value indicates that 1KX5 has a lower free energy than the corresponding A-rich sequence within the considered molecular environment.

The magnitude of ΔF generally increases with the length of the adenine tract. In the DNA-only simulations, it changes from -8.31 ± 0.07 kcal mol^{-1} for A25 to -30.4 ± 0.3 kcal mol^{-1} for A147. A stronger variation is observed in the nucleosomal environment, where ΔF changes from -11.31 ± 0.08 kcal mol^{-1} for A25 to -41.2 ± 0.5 kcal mol^{-1} for A147.

For every A-rich target, the nucleosomal transformation free energy is more negative than the corresponding DNA-only value. Consequently, their difference produces a positive $\Delta\Delta F_{\mathrm{bind}}$ for all six variants. With the convention used here, this indicates a relative nucleosome-binding preference for 1KX5 over the A-rich target. The dependence of this effect on the length of the adenine tract is examined in the following subsection.

4.2.2 Dependence on A-tract length

Figure 4.2 shows the relative nucleosome-binding free energy as a function of the length of the central adenine tract. The unmodified 1KX5 sequence is shown at zero inserted adenines and defines the reference value.

From A25 to A125, $\Delta\Delta F_{\mathrm{bind}}$ increases systematically with the adenine-tract length. Extending the A-rich region therefore produces an increasing relative nucleosome-binding preference for 1KX5 over the corresponding target.

As discussed in Section 2.2.3, extended A-tracts constitute a strong and well-characterized perturbation of DNA mechanics and are generally associated with unfavourable nucleosome formation. The systematic increase in $\Delta\Delta F_{\mathrm{bind}}$ observed from A25 to A125 shows that the model distinguishes the progressively modified sequences and reproduces the expected qualitative response. Although the consistency checks discussed above indicate that the precise numerical values may be affected by additional uncertainty, the method is sufficiently sensitive to capture the effect of this strong sequence-dependent perturbation.

Alongside this thermodynamic trend, a preliminary inspection of the trajectories in VMD revealed differences in the extent of DNA wrapping. In the trajectories inspected, 1KX5 and Widom 601 remained comparatively compact around the histone core, whereas the A-rich variants explored more open conformations. The departure from the compact wrapped geometry generally appeared more pronounced as the adenine tract was extended. Although these simulations started from a wrapped reference configuration, some of the inspected A-rich trajectories showed opening followed by partial radial returns towards the nucleosome axis, so that their geometry changed over time rather than

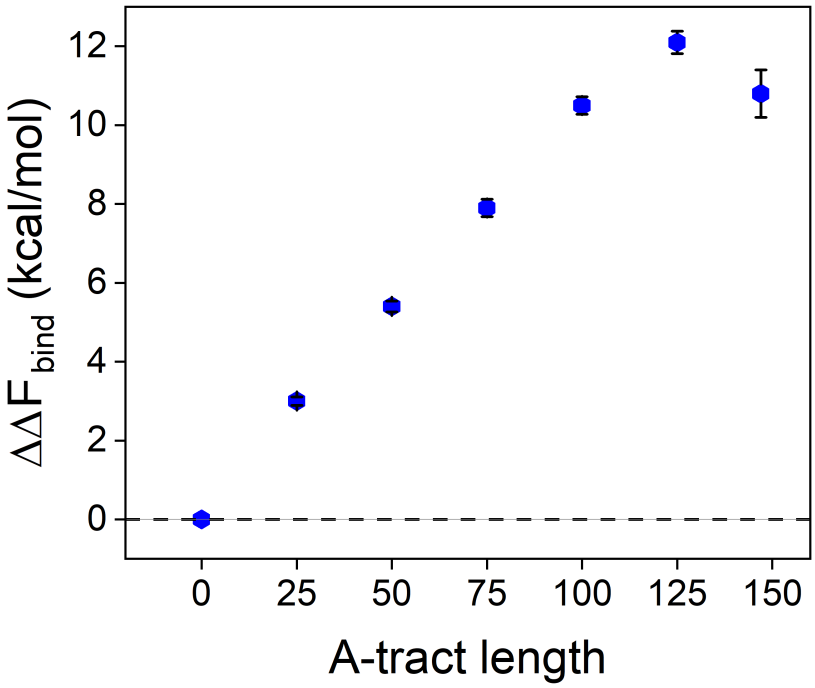


Figure 4.2: Relative nucleosome-binding free energy as a function of A-tract length. The unmodified 1KX5 sequence corresponds to zero inserted adenines and defines the reference value. Positive values indicate that the 1KX5 reference is favoured relative to the corresponding A-rich target in the histone-associated environment.

remaining close to the initial structure.

Figure 6.1 in Appendix 6.1 illustrates this contrast through a compact configuration of 1KX5 and two configurations from the A25 nucleosomal trajectory: the initially wrapped structure and a more open configuration observed later in the simulation. These observations motivated the quantitative analyses of radial positioning and superhelix geometry presented in Sections 4.6.2.1 and 4.6.2.2, which examine where DNA moves away from the histone core and how this affects its overall wrapped path.

For 1KX5, the visual assessment is complemented by the RMSD analysis reported by Estacio Mateos [31] for a representative trajectory from the shared simulation campaign, reproduced in Figure 6.15 in Appendix 6.4. After an initial adjustment, the RMSD of the full nucleosome fluctuates without a sustained increase, while the final configuration retains an assembled nucleosome. Together, these observations support the persistence of the overall nucleosomal organization in this trajectory, although they do not exclude local DNA opening or establish complete conformational sampling.

The observed conformational changes are also relevant to the interpretation of the free-energy results. In some trajectories, opening and partial rewrapping occurred only a few times, raising the possibility that the relative populations of compact and open conformations were not fully sampled. This is particularly relevant when considering the deviation observed for A147. Its relative binding preference is approximately 1.3 kcal mol^{-1} lower than that of A125, a difference smaller than the 1.9 kcal mol^{-1} residual obtained from the thermodynamic-cycle consistency test. Limited sampling of the different conformations may also influence this estimate. The A147 result should therefore not be interpreted as clear evidence for a reversal of the trend without further assessment of the sampling of these structural changes.

4.3 Random controls and Widom 601

4.3.1 Free-energy differences of the random sequences and Widom 601

After examining the strong and systematic response of the A-rich series, the randomized sequences were used to test the response of the model to sequence changes that do not follow a controlled progressive perturbation. The corresponding free-energy results are reported in Table 4.2, while the resulting $\Delta\Delta F_{\mathrm{bind}}$ values are compared in Fig. 4.3.

Table 4.2: Free-energy results for the randomized target sequences and Widom 601 relative to the 1KX5 reference. The transformations follow the target-to-1KX5 convention, and $\Delta\Delta F_{\mathrm{bind}} = \Delta F_{\mathrm{DNA}} - \Delta F_{\mathrm{NUC}}$. All values are expressed in kcal mol^{-1}. The calculations were part of the shared simulation campaign reported in Ref. [31].

Target	ΔF_{DNA}	ΔF_{NUC}	$\Delta\Delta F_{\mathrm{bind}}$
1KX5	0.0	0.0	0.0
R1	0.3 ± 0.2	0.8 ± 0.2	-0.5 ± 0.3
R2	10.8 ± 0.2	9.8 ± 0.2	1.0 ± 0.3
R3	12.0 ± 0.2	13.6 ± 0.2	-1.6 ± 0.3
R4	-0.1 ± 0.2	0.1 ± 0.2	-0.2 ± 0.3
R5	12.5 ± 0.2	12.9 ± 0.2	-0.4 ± 0.3
Widom 601	5.4 ± 0.2	5.2 ± 0.2	0.2 ± 0.3

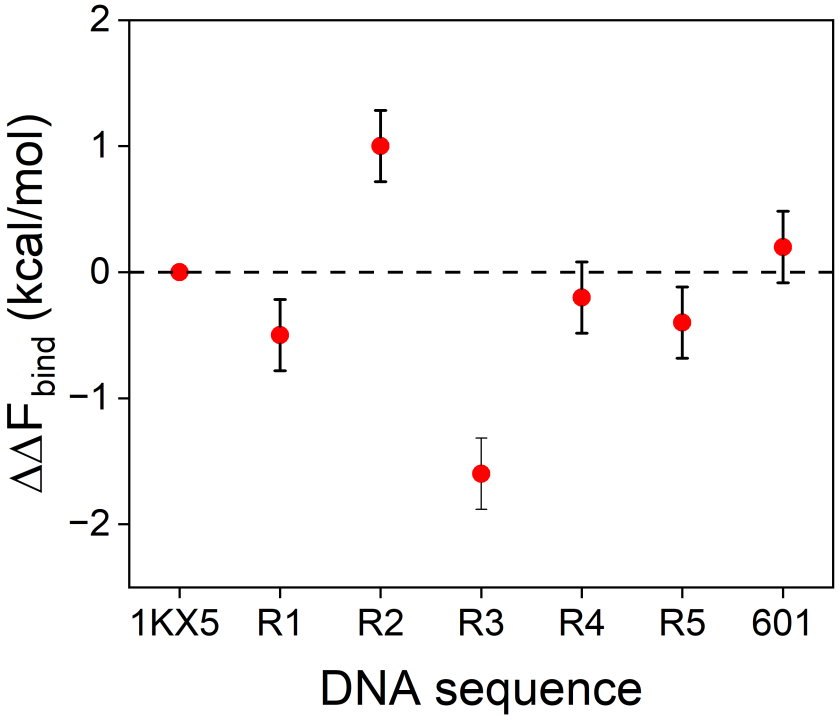


Figure 4.3: Relative nucleosome-binding free energies for the five randomized DNA sequences and Widom 601, using 1KX5 as the reference. The dashed horizontal line indicates $\Delta\Delta F_{\mathrm{bind}} = 0$.

The individual transformation free energies vary substantially among the randomized sequences. R1 and R4 remain close to 1KX5, whereas R2, R3 and R5 show considerably larger positive ΔF values in both the DNA-only and nucleosomal environments. This variation shows that the model captures a sequence-dependent free-energy response also within the randomized set.

For each random sequence, however, the DNA-only and nucleosomal transformation free energies remain relatively close to one another. Their difference consequently produces $\Delta\Delta F_{\mathrm{bind}}$ values between -1.6 and $+1.0$ kcal mol^{-1}, without a common trend across the set. Although some values exceed their propagated statistical error bars, they are all smaller in magnitude than the 1.9 kcal mol^{-1} residual obtained from the thermodynamic-cycle consistency test. The present calculations therefore do not provide sufficient numerical resolution to establish a reliable ordering of the relative nucleosome preferences of the randomized sequences.

Widom 601 similarly gives a small relative free-energy difference, $\Delta\Delta F_{\mathrm{bind}} = 0.2 \pm 0.3$ kcal mol^{-1}, which is compatible with no resolved change in nucleosome preference relative to 1KX5 within the numerical resolution of the present calculations.

4.4 Chromosome XIII

4.4.1 Relative free-energy preference and comparison with experimental data

Two DNA fragments located around the same transcription start site on chromosome XIII were compared as a test against an experimentally known ordering of nucleosome occupancy.

As described in Chapter 3, the chromosome XIII production trajectories were extended from 500 to 1500 ns. In addition, the ten realizations were initialized from ten different molecular configurations rather than from a common starting structure. These choices were introduced to improve conformational sampling, reduce the dependence on a single initial configuration and obtain more stable statistical estimates. They do not eliminate the additional numerical and sampling limitations discussed in Section 4.1, but provide a more extensive sampling protocol than that used for the A-rich and randomized sequences.

The transformation was defined from the -397 target at $\lambda = 0$ to the $+920$ reference at $\lambda = 1$. Table 4.3 reports the results obtained in the DNA-only and nucleosomal environments, both with and without terminal linker DNA.

Table 4.3: Free-energy results for the chromosome XIII transformation from -397 to $+920$, with and without linker DNA. All values are expressed in kcal mol^{-1}. The relative binding preference was calculated as $\Delta\Delta F_{\mathrm{bind}} = \Delta F_{\mathrm{DNA}} - \Delta F_{\mathrm{nucleosome}}$. Derived differences were calculated from the unrounded values and may therefore differ slightly from differences obtained from the displayed entries.

Construct	ΔF_{DNA}	$\Delta F_{\mathrm{nucleosome}}$	$\Delta\Delta F_{\mathrm{bind}}$
Without linkers	-4.7 ± 0.1	-9.8 ± 0.1	5.0 ± 0.2
With linkers	-3.6 ± 0.1	-8.5 ± 0.2	4.8 ± 0.2

As shown in Table 4.3, combining the DNA-only and nucleosomal branches produces a positive $\Delta\Delta F_{\mathrm{bind}}$. With the convention used here, this indicates a relative nucleosome-binding preference for $+920$ over -397. The calculated relative preference is also substantially larger than the 1.9 kcal mol^{-1} residual obtained from the thermodynamic-cycle consistency test. Although its exact magnitude remains subject to the limitations discussed above, its direction is therefore more clearly resolved than the smaller differences observed among the randomized sequences.

The predicted direction agrees with the experimentally observed higher nucleosome occupancy of the $+920$ region. The simulated relative preference is larger than the effective experimental estimate of approximately 2.1 kcal mol^{-1}, but both results have the same sign and lie on the scale of a few kcal mol^{-1}. The comparison should not be interpreted as precise quantitative agreement, because the experimental occupancy difference is not a direct measurement of the binding free energy of an isolated nucleosome and may also depend on the broader genomic and experimental context. Nevertheless, both the simulations and the experimental observation identify $+920$ as the relatively favoured sequence.

This agreement in direction provides further qualitative support for the ability of the coarse-grained model to capture sequence-dependent differences in nucleosome preference.

4.4.2 Effect of linker DNA

The linker constructs were introduced to assess whether the relative free-energy preference calculated for chromosome XIII was influenced by boundary effects. In the 147-bp systems, the DNA terminates approximately at the nucleosome entry and exit regions. This may restrict the motion of the wrapped DNA or make its behaviour sensitive to the proximity of the molecular ends. Adding 20 base pairs to each side provides additional DNA beyond the nucleosomal core and makes it possible to test whether the calculated preference depends on this feature of the original construct.

As shown in Table 4.3, the individual transformation free energies change when the linker regions are added. However, the changes in the DNA-only and nucleosomal branches are similar and largely cancel in their difference. The difference between the two $\Delta\Delta F_{\mathrm{bind}}$ estimates is 0.2 ± 0.3 kcal mol^{-1}, where the uncertainty was obtained by propagating the two reported errors in quadrature. This value is compatible with zero within its propagated uncertainty. The calculations therefore do not provide evidence for a measurable change in the relative sequence preference upon addition of the linker DNA. Both constructs retain a positive relative preference of similar magnitude for +920 over −397.

Within the precision of the calculation, the relative sequence preference is therefore not significantly affected by extending the DNA beyond the nucleosomal core. This provides an additional consistency check for the model and indicates that the thermodynamic conclusion for chromosome XIII is not primarily determined by the finite boundaries of the 147-bp construct.

This result does not exclude more localized effects of the linker regions on DNA geometry or dynamics. The post-simulation analyses performed with the cLibrary-based workflow compare the constructs over equivalent central DNA regions and examine whether the presence of the linkers modifies the local helical parameters, the nucleosomal path or the entry and exit regions. These structural comparisons are presented later in this chapter.

4.5 Crookedness and relative nucleosome preference

After analysing the free-energy results, the crookedness values predicted from the DNA sequences were compared with the corresponding relative binding free energies. The aim was to examine whether this global descriptor of preferred DNA geometry could capture part of the sequence-dependent thermodynamic behaviour observed in the simulations. The values were obtained using the sequence-based model described by Luengo Márquez [17]. In this description, a larger value of β corresponds to a more crooked DNA axis, whereas $\beta = 0$ represents the ideal limit of a straight axis.

For each alchemical comparison, the horizontal coordinate in Figure 4.4 is defined as

$$\Delta\beta = \beta_{\mathrm{reference}} - \beta_{\mathrm{target}}. \tag{4.1}$$

The reference is 1KX5 for the A-rich, random and Widom 601 systems, and the chromosome XIII +920 sequence for the comparison with −397. Consequently, $\Delta\beta > 0$ means that the reference is more crooked than the target, while $\Delta\beta < 0$ means that the target is more crooked. The vertical coordinate follows the free-energy convention used throughout this chapter,

$$\Delta\Delta F_{\mathrm{bind}} = \Delta F_{\mathrm{DNA}}^{\mathrm{target}\to\mathrm{reference}} - \Delta F_{\mathrm{nucleosome}}^{\mathrm{target}\to\mathrm{reference}}. \tag{4.2}$$

Thus, a positive value indicates that the reference sequence has a more favourable relative nucleosome-binding free energy than the target.

Figure 4.4 shows a strong positive correlation within the A-rich series, with $r = 0.91$. As the adenine tract is extended, the crookedness of the target sequence decreases relative to that of 1KX5, and $\Delta\beta$

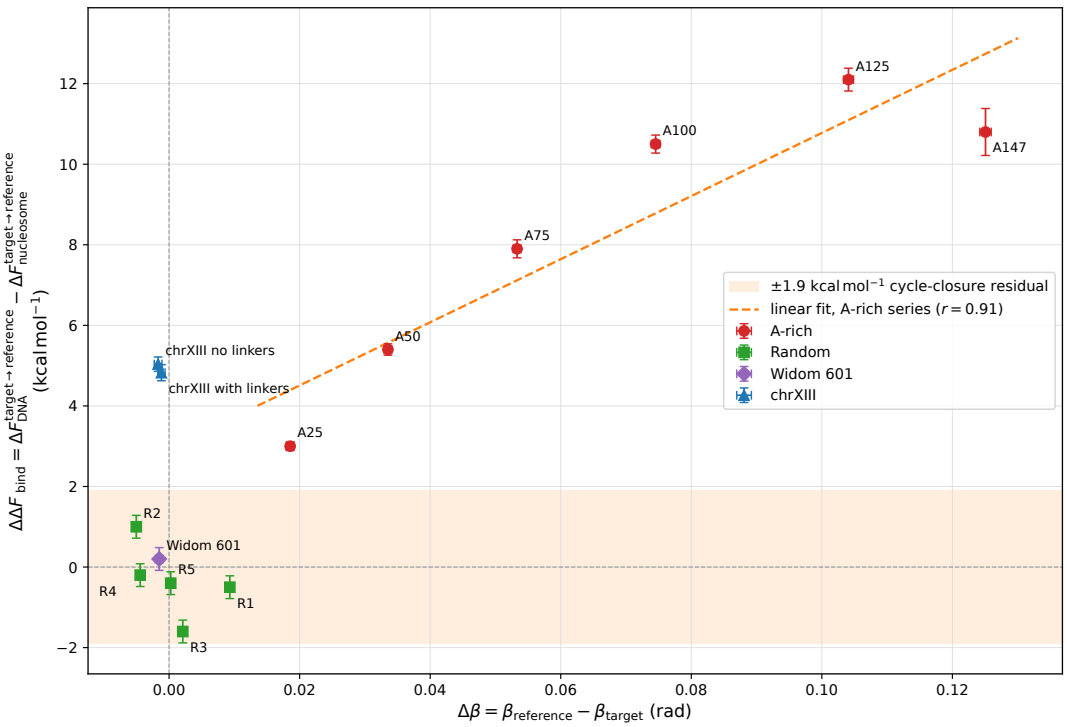


Figure 4.4: Simulated relative nucleosome-binding preference as a function of the crookedness difference between the reference and target sequences. Horizontal and vertical error bars report the propagated uncertainties in $\Delta\beta$ and $\Delta\Delta F_{\mathrm{bind}}$, respectively. The shaded horizontal band marks ± 1.9 kcal mol^{-1}, the magnitude of the thermodynamic-cycle closure residual, centred on the 1KX5 reference value. The dashed line is a linear fit to the controlled A-rich series only; its Pearson correlation coefficient is $r = 0.91$. Random sequences, Widom 601 and the chromosome XIII constructs are shown for comparison but are not included in the fit.

consequently increases. Over the same series, $\Delta\Delta F_{\mathrm{bind}}$ also increases up to A125, indicating a progressively stronger relative nucleosome-binding preference for 1KX5. Crookedness therefore captures the main thermodynamic trend produced by this strong and controlled sequence perturbation.

A147 does not follow the approximately linear trend as closely: despite having the largest $\Delta\beta$, its $\Delta\Delta F_{\mathrm{bind}}$ is lower than that of A125. This is consistent with the caution already discussed for this system, whose free-energy estimate may be affected by both the numerical resolution of the method and incomplete sampling of its more open nucleosomal conformations. The correlation should therefore be interpreted as a descriptive relationship within the A-rich series, rather than as evidence that crookedness alone determines nucleosome stability.

The shaded horizontal band represents the ± 1.9 kcal mol^{-1} residual obtained from the thermodynamic-cycle consistency test. It is not a confidence interval for the linear fit, but an empirical indication of the numerical scale below which differences cannot be interpreted robustly with the present calculations.

The random sequences and Widom 601 are concentrated close to $\Delta\beta = 0$, and their relative binding free energies fall within, or close to, this numerical scale. Consequently, these data do not resolve a clear relationship between crookedness and relative nucleosome preference. This does not demonstrate that crookedness is irrelevant for these sequences; rather, their geometrical and energetic differences are too small compared with the resolution of the calculation to establish an analogous trend.

The chromosome XIII comparison also does not follow the relationship observed for the A-rich series. The -397 sequence is slightly more crooked than $+920$, giving a negative $\Delta\beta$, while the positive $\Delta\Delta F_{\mathrm{bind}}$ indicates a relative nucleosome-binding preference for $+920$. The systems with and without linker DNA have the same $\Delta\beta$ because their central sequences are unchanged, and they produce very similar relative binding free energies.

Overall, the results indicate that crookedness can be a useful proxy for the pronounced structural

and thermodynamic changes produced by progressively extending an A-rich tract. However, it is not sufficient as a standalone predictor across different sequence classes, for which local deformation patterns, mechanical response and DNA–histone interactions may also contribute.

4.6 Structural analysis

4.6.1 Local DNA geometry

The relative nucleosome-binding free energies quantify how each sequence modification affects nucleosome association with respect to the reference sequence, but they do not identify the conformational changes underlying this effect. Crookedness provides a first connection with DNA geometry by describing the sequence-dependent tendency of free DNA to adopt a curved or irregular conformation. However, it reduces the geometry of each sequence to a single global descriptor.

The trajectory-based structural analyses were therefore used to examine the nucleosomal response at different spatial scales, from local base-pair-step adjustments to radial opening and changes in the overall superhelical organization. Because the same sequence was simulated in the DNA-only and histone-associated conditions, its local geometry could be compared directly between the two systems while retaining information about where the changes occur along the DNA and about the conformations sampled during the trajectories. This combination of controlled thermodynamic and spatially resolved structural information would be difficult to obtain from a single experimental approach.

The first level of the analysis concerns the local geometry of DNA. As discussed in Section 2.3.1, the nucleosomal path is produced through coupled translations and rotations distributed along the recurring histone-contact regions. The six base-pair-step parameters introduced in Section 2.1.1 were therefore used to identify where the local geometry changes upon nucleosome association and to assess whether different sequences display similar response patterns or accommodate the wrapped conformation in distinct ways.

The six geometrical parameters were first calculated separately for each sequence in the DNA-only and nucleosomal environments. For completeness, the two sets of profiles obtained for 1KX5 are reported in Appendix 6.1, Figure 6.4. The analysis then focused on the difference between the corresponding temporal means:

$$\Delta p(i) = \overline{p}_{\mathrm{nucleosome}}(i) - \overline{p}_{\mathrm{DNA-only}}(i).$$

This quantity describes how each geometrical parameter changes when passing from DNA-only to the nucleosomal environment. Its sign indicates the direction of the change, while its magnitude indicates how different the two mean profiles are at that position. The resulting difference profiles are shown for 1KX5 and Widom 601 in Figure 4.5, while those obtained for the two chromosome XIII sequences are reported in Appendix 6.1, Figure 6.5. Consistent colours were used for each sequence across the different figures to facilitate comparison.

Across the four sequences, slide, rise and tilt display relatively smooth and recurrent oscillations, whereas roll and twist retain more irregular local variations. Shift shows an intermediate behaviour. The positions of the maxima and minima, however, are not identical between sequences. Different sequences may therefore show a similar variation of a parameter along the DNA even when the corresponding features occur at slightly different base-pair-step indices.

A possible interpretation follows from the geometrical directions described by these parameters. Roll bends DNA towards the grooves and twist changes their rotational orientation relative to the histone surface, making both parameters particularly sensitive to local groove geometry and histone contacts; by contrast, slide, rise and tilt may allow the wrapped DNA to accommodate the recurring nucleosomal constraints through deformation directions that perturb groove-facing contacts less directly, resulting in smoother profiles.

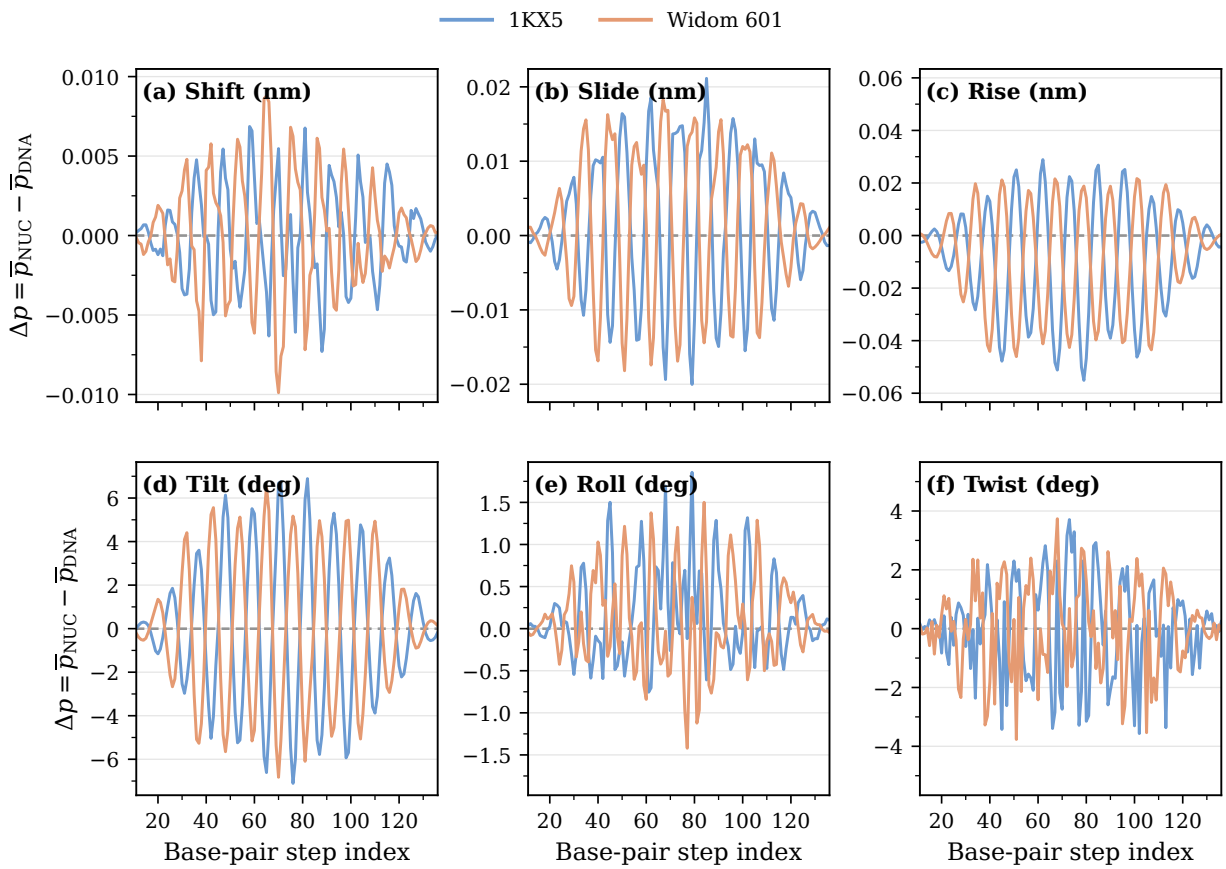


Figure 4.5: Histone-associated changes in the six base-pair-step parameters for 1KX5 and Widom 601. For each parameter p, the plotted quantity is $\Delta p(i) = \overline{p}_{\mathrm{nucleosome}}(i) - \overline{p}_{\mathrm{DNA}}(i)$. Shift, slide and rise are expressed in nanometres; tilt, roll and twist are expressed in degrees. The profiles are temporal means from realization R0, calculated over base-pair steps 11–136 after discarding the first 100 saved frames. The dashed horizontal lines mark zero change.

To determine whether these similar patterns were shifted along the DNA, the profiles of each sequence pair were also compared after introducing a common spatial lag. This procedure shifts one set of profiles along the base-pair-step index and tests whether similar patterns occur at slightly different positions. A single lag was selected for each sequence pair and applied to all six parameters, so that their relative positions within each sequence were preserved. The comparisons before and after this alignment are reported in Appendix 6.1, Figures 6.6 and 6.7.

After the alignment, slide, rise and tilt show spatial Pearson correlations of approximately 0.91–0.96 in both sequence pairs. Shift shows intermediate correlations of approximately 0.73–0.79, whereas the correlations obtained for roll and twist are lower, approximately 0.43–0.68. The similarity between the aligned profiles is therefore particularly pronounced for slide, rise and tilt.

These results identify a recurrent pattern in the calculated local response to the nucleosomal environment, which is observed across DNA sequences with different compositions and biological origins. The regularity may depend on the coarse-grained model and on the procedure used to reconstruct the DNA axis and calculate the six parameters. Further investigation using alternative models and analysis methods would be required to determine how general this behaviour is. Nevertheless, it is a consistent feature emerging from the analysis of all four sequences considered here.

Following the free-energy comparison in Section 4.4.2, a further structural check was performed to determine whether the finite boundaries of the 147-bp constructs affected the local geometry of nucleosomal DNA. The six base-pair-step parameters were compared between the chromosome XIII constructs with and without linker DNA, restricting the analysis to the equivalent central region of the two systems. For both endpoints, the resulting profiles show a close overlap across all six param-

eters, with only small local deviations. Within the representative trajectories analysed here, adding terminal DNA therefore does not produce a pronounced reorganization of the central nucleosomal region. This agreement provides an additional consistency check for the simulation setup. The complete profiles for the −397 and +920 endpoints are reported in Appendix 6.1, Figures 6.8 and 6.9, respectively.

4.6.2 Global geometry and wrapping of nucleosomal DNA

4.6.2.1 Radial positioning and DNA opening

The base-pair-step parameters discussed above describe local changes between consecutive base pairs, but they do not directly show how these changes affect the overall path of the molecule around the histone core. To complement this local description, the global path of nucleosomal DNA was examined first through the radial position of individual base pairs and subsequently through the effective radius and pitch of the superhelix. The radial distance ρ of each base-pair centre from the nucleosome axis was averaged over time. In the resulting radial profiles, the solid line represents this temporal mean and the shaded band represents the standard deviation around the mean, describing how much the radial position fluctuates during the trajectory. These profiles were examined to determine whether the A-rich variants display a different degree and spatial distribution of DNA opening compared with 1KX5 and Widom 601.

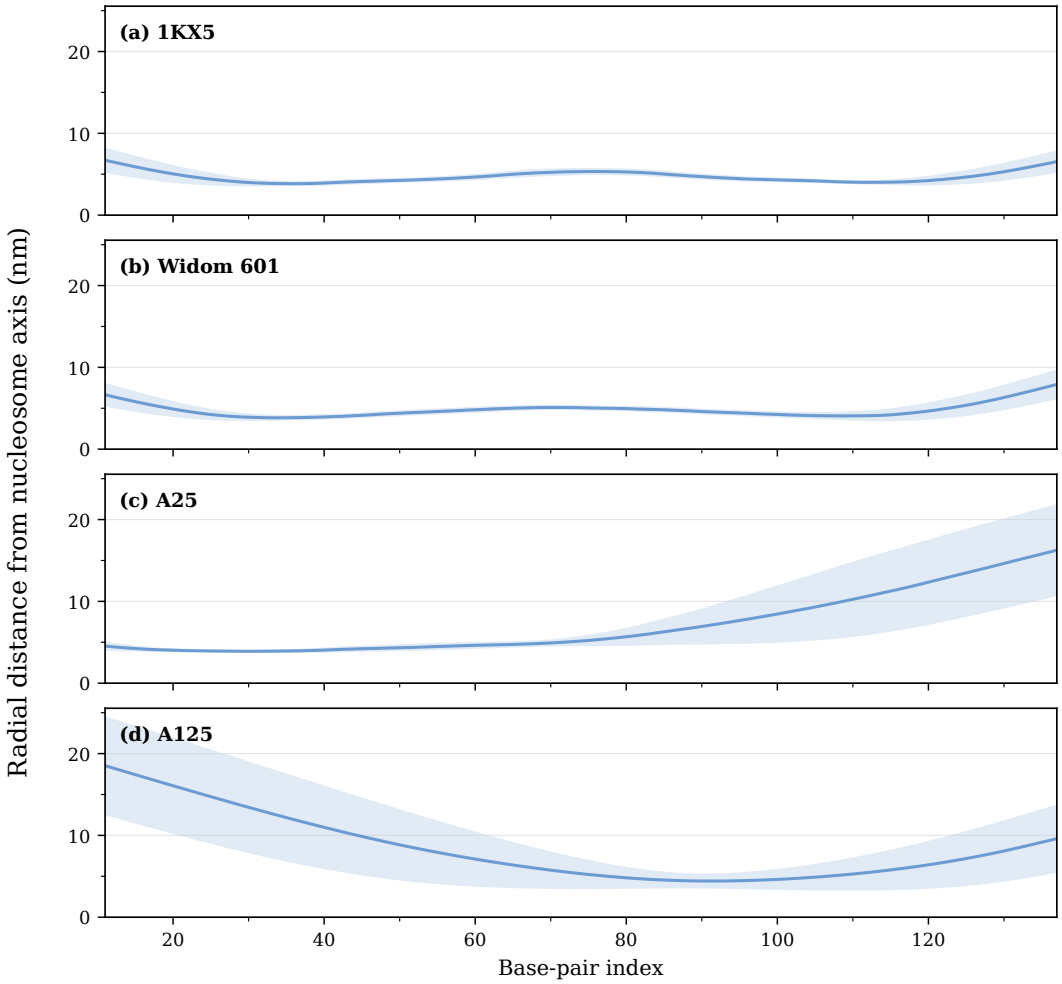


Figure 4.6: Radial-distance profiles for 1KX5, Widom 601, A25 and A125 in the histone-associated simulations. The solid line represents the temporal mean distance of each base-pair centre from the nucleosome axis, while the shaded region represents the standard deviation around the mean. The profiles were obtained from realization R0 over base pairs 11–137 after discarding the first 100 saved frames. The shaded regions describe temporal fluctuations within the analysed trajectory and do not represent uncertainty in the mean.

As discussed in Chapter 2, the mechanical properties of DNA depend on its sequence and contribute to its compatibility with the wrapped nucleosomal path. In particular, extended poly(dA:dT) tracts are generally associated with reduced nucleosome affinity. Against this background, the profiles in Figure 4.6 show a clear structural contrast. Both 1KX5 and Widom 601 retain relatively compact mean profiles, whereas A25 and A125 exhibit substantially larger radial distances over extended portions of the DNA. This difference is qualitatively consistent with the relative preference for 1KX5 over the A-rich variants obtained from the free-energy calculations. The compact Widom 601 profile is also consistent with its high nucleosome affinity reported in the literature [2].

Within the R0 trajectories shown in Figure 4.6, the departure from the compact nucleosomal configuration is more pronounced for A125 than for A25, qualitatively following the increase in A-tract length. In both systems, the largest mean radial distances and the broadest fluctuation bands occur towards the terminal analysed regions, showing that these regions are more open on average and explore a wider range of radial positions than the inner wrapped DNA. The displacement is nevertheless asymmetric: A25 opens mainly towards the high-index end, whereas A125 shows its largest displacement towards the low-index end. Because each profile was obtained from a single realization, this opposite orientation should not be interpreted as evidence that either sequence preferentially opens from a specific end. Instead, it may reflect realization-specific stochastic fluctuations, with one end becoming displaced and remaining open for a substantial fraction of the finite trajectory. The profiles therefore support greater and more variable terminal opening in the A-rich systems, particularly in A125, while the direction of the asymmetry remains realization-dependent.

The time evolution of the radial distances of base pairs 11 and 137 is reported in Appendix 6.1, Figure 6.10. Panel (a) shows the compact 1KX5 R0 reference, whereas panels (b) and (c) show A125 realizations R0 and R1, respectively. This representation complements the time-averaged profile by showing how the terminal displacements develop over the course of the trajectories. Whereas 1KX5 remains comparatively compact, A125 rapidly departs from its initially wrapped configuration and samples substantially larger radial distances. In A125 R0, one terminal analysed region remains widely displaced over extended portions of the simulation, while the other remains closer to the nucleosome axis for longer before also visiting more displaced configurations. The trajectory therefore provides a clear illustration of the greater conformational variability and more persistent departure from the compact wrapped state sampled by A125 relative to 1KX5. For completeness, the second A125 realization in panel (c) provides an additional example consistent with the structural picture described above. A systematic analysis across all ten realizations would be required to quantify the magnitude, frequency and end-specificity of the opening events and to determine how reproducibly the sampled configurations occur. Given the available time, the present analysis was therefore retained as a qualitative structural comparison rather than as a replica-averaged characterization of A125 dynamics.

The radial profiles of the remaining A-rich variants are reported in Appendix 6.1, Figure 6.11, and similarly show greater DNA opening than 1KX5 and Widom 601.

For chromosome XIII, the same analysis was used to assess whether linker DNA altered the geometry of the central nucleosomal region (Appendix 6.1, Figure 6.12). The profiles remain similar over much of the inner region, while differences are more evident near the DNA ends. In the trajectories analysed, adding linkers is associated with larger radial fluctuations at the high-index end of -397 and a larger mean radial distance at the low-index end of $+920$.

4.6.2.2 Superhelix geometry

The radial profiles describe where the DNA moves towards or away from the nucleosome axis, but do not provide a single measure of the overall wrapped geometry. To examine whether the differences between sequences were accompanied by changes in the global nucleosomal path, the effective superhelix radius and pitch were calculated for each retained frame. The main comparison focused on 1KX5, Widom 601 and the two chromosome XIII sequences, which maintained a comparatively compact wrapped configuration.

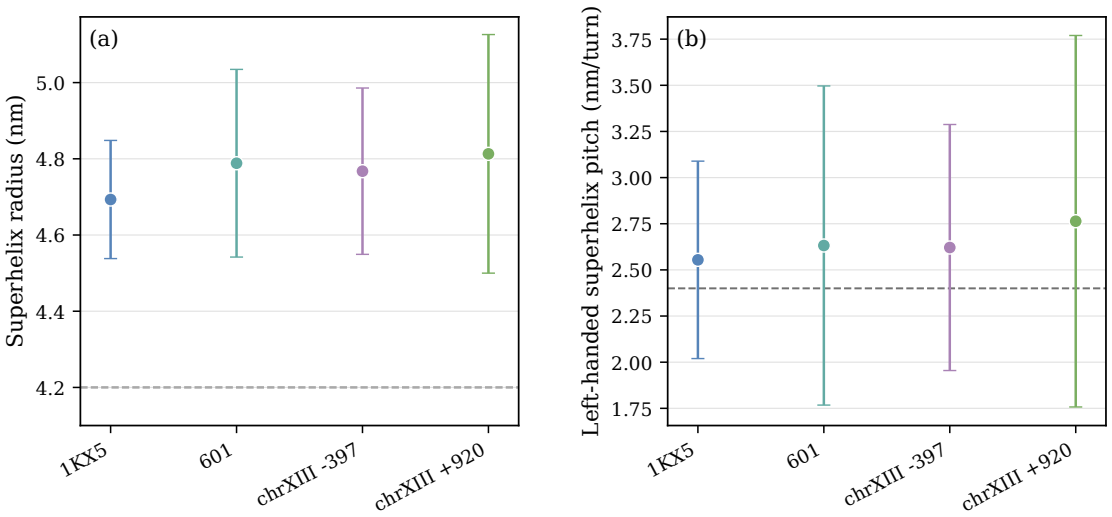


Figure 4.7: Effective superhelix radius (a) and pitch (b) for 1KX5, Widom 601 and the two chromosome XIII sequences without linker DNA. The quantities were calculated for each retained frame of realization R0 using base pairs 11–137 in the nucleosome-fixed reference frame, as described in Section 3.8.4.3. Points show temporal means and bars show the standard deviation, not uncertainty in the mean. In panel (b), the plotted quantity is $-P$, so positive values correspond to left-handed winding under the adopted convention. Dashed lines mark the canonical nucleosome values of 4.2 nm for the superhelix radius and 2.4 nm per turn for the pitch, introduced in Section 2.3.1.

Figure 4.7 shows the temporal means and standard deviations of these quantities. The four systems have similar mean radii, ranging from approximately 4.69 to 4.81 nm, and mean pitches between 2.55 and 2.76 nm per turn, expressed using positive values for left-handed winding. The differences between their mean values are small. The relatively large temporal fluctuations, especially in pitch, reflect the fact that the fit was performed independently on the different conformations sampled during each trajectory. Within the trajectories analysed, these sequences therefore retain broadly similar average global dimensions despite the differences in local geometry discussed above.

These values also provide a comparison with the structural reference introduced in Section 2.3.1, where the ideal-superhelix fit to the central wrapped region gave a radius of 4.2 nm and a pitch of 2.4 nm per turn. The simulated systems reproduce the approximate scale of the nucleosomal superhelix, although their effective radii and pitches are somewhat larger. This agreement is qualitative, since the molecular representation, reference axis and fitting procedure differ from those used for the crystallographic structure.

The same analysis was applied to A25 and A125 as a diagnostic comparison, reported in Figures 6.13 and 6.14 in Appendix 6.1. These systems show larger and more variable effective radii, lower mean R^2 values and larger axial fit residuals. Together with the radial profiles, these observations indicate that DNA opening involves substantial departures from a regular superhelical path. A single radius and pitch therefore provide a less adequate description of their conformations.

The extracted quantities also depend on the axis used in the fit. Here, Z_i and ϕ_i were measured in the nucleosome-fixed coordinate system introduced in Section 3.8.4.1, which was reconstructed from the histone core for every saved configuration. This separates changes in the DNA path from the overall translation and rotation of the nucleosome. However, its z axis is defined from the histone core rather than optimized to the instantaneous DNA path. A difference between this axis and the axis that best describes the DNA superhelix can influence the fitted values, particularly the pitch, which is obtained from the change in axial position along the superhelical path. Radius and pitch should therefore be interpreted as effective descriptors relative to the histone core, with particular caution when interpreting small pitch differences.

This analysis connects the local structural changes to the overall organization of nucleosomal DNA.

Consistent with the radial profiles, the compact sequences retain similar global dimensions, whereas the more open A-rich conformations show a less regular wrapped geometry.

4.6.3 DNA–histone interactions

The local helical parameters describe how DNA is deformed at the base-pair-step level, while the radial and superhelical analyses describe its overall path around the histone octamer. To connect these geometrical observations with the interactions that support the wrapped configuration, the spatial relationship between the DNA backbone and the surrounding histone residues was then examined.

As discussed in Section 2.3.1, structural and simulation studies show that the nucleosomal interface is organized largely through interactions between histone residues and the DNA sugar–phosphate backbone. In the structured histone core, positively charged residues, particularly arginines, approach the backbone at recurrent positions separated approximately by the helical periodicity of DNA. The flexible histone tails, which contain both lysine and arginine residues, can instead approach different regions of nucleosomal and linker DNA while sampling multiple conformations.

In the coarse-grained model, this electrostatic component of the interface is represented by the attraction between the negatively charged phosphate beads and positively charged histone residues. The DNA–histone proximity profiles were therefore examined to determine whether the globally wrapped configuration is accompanied by the characteristic spatial organization of the nucleosomal interface, and to distinguish the contribution of the structured histone core from that of the flexible terminal tails. For this purpose, arginines in the structured core were considered separately from lysines and arginines in the histone tails.

For each phosphate and simulation frame, the minimum distance from each of the three residue groups was calculated, allowing the identity of the nearest residue to change over time. The resulting profiles report the temporal mean of this minimum distance along each DNA strand, while the shaded bands represent its temporal standard deviation. Low values therefore identify phosphate regions that remain, on average, closer to the selected histone group, whereas wide bands indicate greater variation in this proximity during the trajectory.

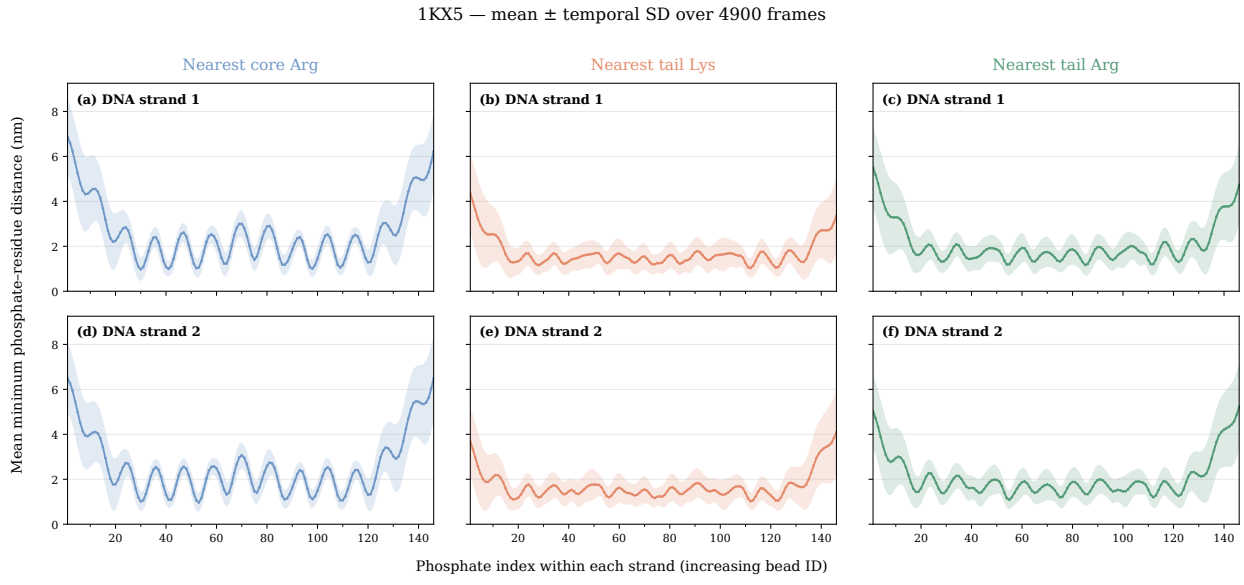


Figure 4.8: DNA–histone proximity profiles for the 1KX5 nucleosome. For each DNA phosphate and retained frame, the minimum distance from an arginine in the structured histone core, a lysine in a terminal histone tail and an arginine in a terminal histone tail was calculated separately. The upper and lower rows refer to the two DNA strands. Solid curves show the temporal mean over 4900 frames of realization R0 after excluding the first 100 saved frames, while the shaded bands show the mean ± temporal standard deviation. The identity of the nearest residue was allowed to change between frames, and no distance cutoff was applied.

The profiles in Figure 4.8 reproduce this characteristic organization. For the core arginines, successive distance minima occur approximately every 10–11 phosphate positions on both DNA strands. This recurrent spacing indicates that the compact 1KX5 trajectory retains the approximately periodic registration between the DNA backbone and the structured histone surface. The distances from the core arginines increase markedly towards the DNA termini, where the DNA is no longer surrounded by the globular core to the same extent as in the inner wrapped region.

A different behaviour is observed for the terminal tails. Histone tails are flexible and contain both lysine and arginine residues, which can approach the phosphate backbone and the DNA grooves while sampling different conformations. Their profiles therefore show a more distributed proximity along the DNA and broad temporal fluctuations rather than the pronounced regular modulation observed for the core arginines. This is consistent with a dynamic association in which different tail residues can approach different regions of the DNA over time.

The three profiles should not, however, be interpreted as a direct comparison of the chemical strength of lysine- and arginine-mediated interactions. The selected groups contain 72 core arginines, 70 tail lysines and 34 tail arginines. The minimum distance is consequently affected by both the spatial distribution and the number of residues available in each group. Moreover, in the coarse-grained model, lysine and arginine both carry charge +1, each residue is represented by a single bead, and directional hydrogen bonds are not explicitly represented. The analysis therefore distinguishes the spatial organization of the structured core and the flexible tails, but it cannot resolve detailed chemical differences between the lysine amino group and the arginine guanidinium group.

Taken together, these profiles show that the compact global geometry of 1KX5 is accompanied by a recognizable organization of the DNA–histone interface. The structured core presents recurrent regions of close approach at approximately the helical periodicity, whereas the terminal tails show a more spatially distributed association and, particularly for lysines, remain comparatively close towards the DNA ends. The temporal variability of the tail profiles is consistent with different tail residues rearranging around the backbone as the trajectory evolves. The analysis therefore connects the overall wrapped path of DNA with the distinct spatial roles of the structured core and the flexible tails. Since minimum-distance profiles do not quantify contact occupancy or contact lifetime, this result should be regarded as a first spatial characterization of the interface. A more direct analysis of its dynamics would require contact occupancies and lifetimes, ideally examined together with changes in the radial opening of DNA.

The recurrent proximity of the core arginines is also consistent with their established structural role in constraining the DNA backbone near inward-facing minor grooves. The coarse-grained trajectory therefore retains the spatial organization associated with the coupling between the histone surface and local DNA geometry.

Chapter 5

Conclusions and Outlook

5.1 Main findings

It is well established in the literature that DNA sequence influences nucleosome formation and that extended A-rich tracts are generally associated with reduced nucleosome affinity. The main contribution of this thesis is the assessment of a combined MADna–CALVADOS 3 framework for describing these effects and connecting relative nucleosome-binding free energies with the structural response of nucleosomal DNA.

Relative nucleosome-binding free energies were obtained through a thermodynamic cycle in which the DNA-only transformation provides the reference branch needed to compare nucleosome association between the target and reference sequences. These thermodynamic results were interpreted together with trajectory-based analyses of local DNA geometry, global nucleosomal wrapping and DNA–histone proximity.

The convergence and numerical stability of the Thermodynamic Integration calculations, together with their sensitivity to the λ discretization, were first assessed. A more stringent test was provided by the DNA-only thermodynamic cycle connecting A147, A50 and 1KX5. The direct and indirect paths differed by 1.9 ± 0.3 kcal mol^{-1}. Since this value is larger than the propagated statistical uncertainties, it provides a practical indication of the current numerical resolution and reveals additional contributions from finite sampling or integration.

The clearest thermodynamic result was obtained for the controlled series of A-rich sequences. From A25 to A125, $\Delta\Delta F_{\mathrm{bind}}$ increased from 3.0 ± 0.1 to 12.1 ± 0.3 kcal mol^{-1}. As the central adenine tract became longer, the A-rich sequences therefore showed progressively lower nucleosome affinity relative to 1KX5. This systematic trend shows that the model is sensitive to a large and controlled modification of DNA sequence and reproduces the expected unfavourable effect of extended A-rich tracts on nucleosome formation. Within the same series, the crookedness difference followed the calculated relative binding preference with a Pearson coefficient of $r = 0.91$, indicating that this descriptor captures an important part of the progressive change produced by the A-rich perturbation. The slightly smaller value obtained for A147 should not be interpreted as a resolved reversal of the trend, because the difference from A125 is small compared with the numerical and sampling uncertainty indicated by the cycle-closure test. Moreover, the trajectories of the longest A-rich constructs display substantial opening of the nucleosomal DNA, suggesting that slow conformational changes may limit the statistical reliability of this comparison.

The randomized sequences produced a different outcome. Their transformation free energies depended on sequence in both environments, but the resulting relative nucleosome-binding preferences ranged only from -1.6 to $+1.0$ kcal mol^{-1}, without a common trend across the set. Widom 601 likewise gave a small relative nucleosome-binding difference with respect to 1KX5, 0.2 ± 0.3 kcal mol^{-1}. Unlike

the controlled A-rich series, these systems did not show a clear relationship between crookedness and the calculated relative binding preference. Two hypotheses can be advanced. First, the free-energy differences may be too small relative to the numerical resolution of the protocol. However, the cycle-closure residual was obtained from the A-rich systems, where large opening and closing motions made conformational sampling particularly demanding. Since comparable transitions were not observed in the more compact random and Widom trajectories, a smaller finite-sampling contribution would be expected for these systems. Alternatively, crookedness alone may be insufficient to describe the local deformation patterns and DNA–histone interactions that distinguish the sequences. The present data do not allow these two explanations to be distinguished.

The chromosome XIII comparison provided an independent test using two genomic sequences with an experimentally known ordering. The simulations favoured the +920 sequence over −397, giving a relative nucleosome-binding preference of 5.0 ± 0.2 kcal mol^{-1} without linker DNA. The direction of this result agrees with the higher experimental nucleosome occupancy of the +920 region. An additional control with linker DNA gave a relative nucleosome-binding preference of 4.8 ± 0.2 kcal mol^{-1}, differing from the no-linker result by only 0.2 ± 0.3 kcal mol^{-1}. For this sequence pair and within the precision of the calculation, the relative binding preference therefore does not appear to be strongly affected by the absence of terminal linker DNA. Since only one genomic sequence pair was examined, the agreement with the experimental ordering should be interpreted qualitatively.

The six base-pair-step parameters were compared between the DNA-only and nucleosomal environments, making it possible to locate the geometrical changes along each sequence rather than reducing the response to a single global value. Across 1KX5, Widom 601 and the two chromosome XIII sequences, slide, rise and tilt showed relatively smooth and recurrent spatial patterns, while roll and twist retained more irregular local variations and shift displayed an intermediate behaviour. After a common spatial alignment, the corresponding profiles were particularly similar for slide, rise and tilt. The occurrence of this pattern in sequences of different composition suggests a common component in the calculated geometrical response to the nucleosomal environment. Its detailed origin, however, cannot yet be assigned uniquely to DNA mechanics, because the regularity may also depend on the coarse-grained representation and on the procedure used to reconstruct the DNA axis and calculate the parameters.

The radial-distance profiles connected these local descriptors to the overall position of DNA around the histone core. In the representative R0 trajectories, 1KX5 and Widom 601 remained comparatively compact, whereas A25 and A125 displayed larger radial distances and broader fluctuations, particularly towards the terminal analysed regions. The perturbation was more pronounced in A125 than in A25, qualitatively consistent with both the increasing relative nucleosome-binding penalty and the expected effect of extending the central adenine tract. Although the preferentially displaced end differed between the two A-rich trajectories and may therefore be realization-dependent, the profiles consistently show that the terminal regions are more open and explore a wider range of radial positions than the inner wrapped DNA. The quantitative profiles thus support the structural contrast initially observed in VMD and identify where along the DNA the departure from the compact configuration occurs.

The superhelix analysis extended this comparison to the overall wrapped geometry. Within the analysed trajectories, 1KX5, Widom 601 and the two chromosome XIII sequences retained similar effective radii and pitches despite their local geometrical differences. A25 and A125 instead showed larger radii and greater departures from the fitted superhelical path, indicating that their opening also affects the overall organization of DNA around the histone core.

For 1KX5, the DNA–histone proximity profiles complemented this geometrical picture by describing the arrangement of the surrounding histone residues along the backbone. Core arginines showed recurrent distance minima at approximately the DNA helical periodicity, whereas tail lysines and arginines showed a less regular spatial distribution of proximity. In particular, tail lysines remained comparatively close to the DNA ends, where distances from core arginines increased. This analysis distinguishes the spatial organization of the core and tail contributions to the interface, without establishing their relative interaction strengths or contact lifetimes.

Taken together, the thermodynamic and structural results provide a coherent picture of the response to sequence modification. The largest resolved free-energy changes, obtained for the A-rich series, are accompanied in the representative trajectories by greater terminal opening and a less regular wrapped geometry. By contrast, the compact systems retain similar overall superhelical dimensions while differing in their local geometry and, in the chromosome XIII comparison, in their calculated relative nucleosome-binding preference. Several independent features—including the compact behaviour of 1KX5 and Widom 601, the greater mobility of terminal nucleosomal DNA and the periodic organization of the core proximity profiles—are qualitatively consistent with previous experimental and computational observations.

The thermodynamic-cycle residual prevents this overall agreement from being interpreted as a complete quantitative validation. Nevertheless, the simulations capture several sequence-dependent features described in the literature across both artificial and genomic sequences. At substantially lower computational cost than an atomistic treatment, the present workflow makes it possible to calculate relative nucleosome-binding free energies through a thermodynamic cycle and to connect them with local and global geometrical changes sampled along the trajectories. It therefore combines thermodynamic, spatial and conformational information that would be difficult to obtain simultaneously from experiments alone. The resulting workflow provides a useful framework for studying sequence-dependent nucleosome behaviour, while the random-sequence comparison and the cycle-closure test define the scale below which the current protocol cannot yet provide a robust ranking.

5.2 Limitations and future developments

The thermodynamic-cycle test shows that the propagated statistical uncertainties do not capture the full numerical uncertainty of the calculations. Because the cycle was deliberately constructed from the A-rich transformations, which displayed the largest conformational changes and represented the most demanding sampling case, its residual should not be transferred unchanged to every sequence class. Repeating the closure test for compact systems, including randomized, Widom and genomic sequences, would determine whether their more limited conformational rearrangements are associated with smaller errors.

The present analysis of the A-rich trajectories provides a qualitative structural comparison that complements the thermodynamic results, showing greater terminal opening and larger conformational variability than in the compact reference systems. A systematic analysis across all ten realizations would be a natural extension of this work, allowing the reproducibility of this contrast to be tested and the magnitude, persistence and end-specificity of the opening events to be quantified. Combining replica-averaged radial profiles with the corresponding individual time series would help separate the common response to increasing A-tract length from realization-specific fluctuations. Longer trajectories would additionally improve the sampling of changes between more compact and more open configurations and help determine whether the non-monotonic A147 result reflects limited conformational sampling or a genuine change at the longest tract length.

The partial agreement found for crookedness indicates that it can be a useful proxy for large, controlled sequence perturbations, but that it does not capture all contributions to nucleosome preference. Future analyses could combine it with local curvature, groove geometry, DNA opening and DNA–histone proximity to distinguish sequences with similar global crookedness but different local geometries or interactions.

The recurrent patterns observed in the six base-pair-step parameters should also be tested using alternative procedures for reconstructing the DNA axis. Applying the same analysis to another coarse-grained model and, for a smaller number of representative systems, to atomistic trajectories would help determine which features are general and which depend on the molecular representation or analysis method.

The molecular interpretation of the DNA–histone interface is also limited by the coarse-grained resolution. In CALVADOS 3, each amino acid is represented by a single bead and the net charge of

a charged residue is assigned to that effective interaction site. Consequently, the atomistic spatial distribution of charge within the side chains and directional contacts such as hydrogen bonds are not resolved. Within this resolution, a time-resolved analysis of the DNA–histone interface could nevertheless provide additional information on contact occupancies and lifetimes and their relationship with DNA breathing and rewrapping. Such an analysis would first require a contact definition appropriate for the coarse-grained representation, since the choice of a distance threshold is not unambiguous and atomistic cutoffs cannot be transferred directly to distances between effective beads.

Another important development would be to investigate nucleosome sliding by following the complete pattern of DNA–histone contacts over time. A sliding event would appear as a coordinated shift of this pattern from one set of base-pair indices to neighbouring positions, indicating that the DNA has moved along its contour relative to the histone core. A change in the residue nearest to an individual phosphate would not be sufficient to establish sliding, since it could also result from histone-tail rearrangement or local DNA opening. Preliminary inspection of the present trajectories did not indicate pronounced sliding, but the finite length of the simulated DNA restricts the extent of nucleosome repositioning that can be sampled. Systems containing longer flanking DNA would therefore provide a more suitable setting for studying this process.

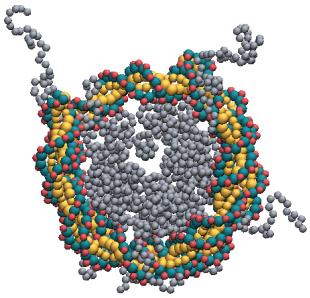
Finally, the chromosome XIII analysis should be extended to additional genomic regions with experimentally characterized nucleosome occupancy. Together with improved sampling and cycle-closure tests for compact systems, these comparisons would provide a stronger assessment of whether the workflow can resolve the subtler sequence-dependent preferences that remain uncertain in the present calculations.

Chapter 6

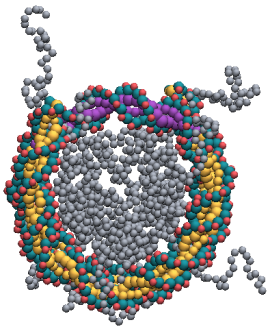
Appendix

6.1 Appendix A: Supplementary Figures

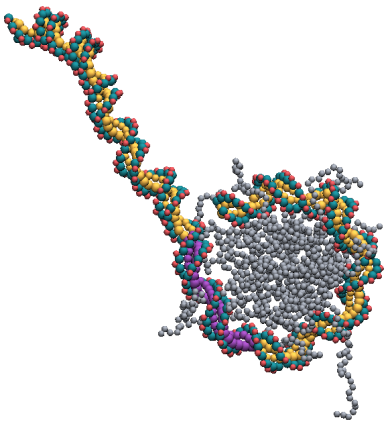
Representative nucleosomal configurations



(a) 1KX5 reference endpoint ($\lambda = 1$).



(b) Initial A25 configuration at frame 0 ($\lambda = 0$).



(c) A25 configuration at frame 4999 ($\lambda = 0$).

Figure 6.1: Illustrative coarse-grained snapshots from the histone-associated simulations: (a) the 1KX5 reference endpoint; (b) the initial closed A25 configuration; and (c) the open A25 configuration at the final saved frame. Histones are shown in grey; DNA phosphates, sugars and bases are shown in coral, teal and gold, respectively, with the central 25-bp A:T tract highlighted in purple. The trajectories were visualized in this work using VMD [21]. Sphere sizes were selected for clarity and do not represent a common physical scale for DNA and protein beads.

Representative DNA-only configurations

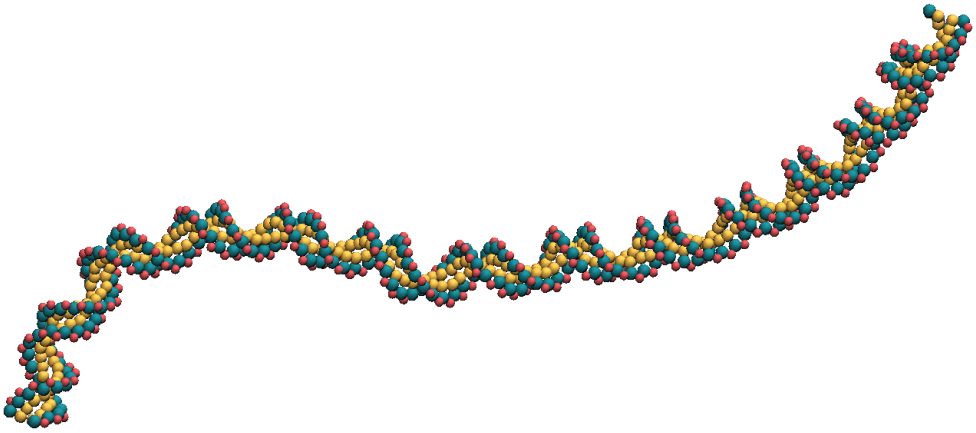
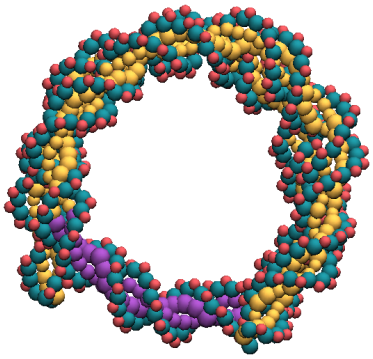
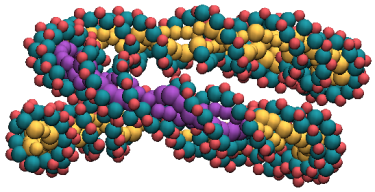


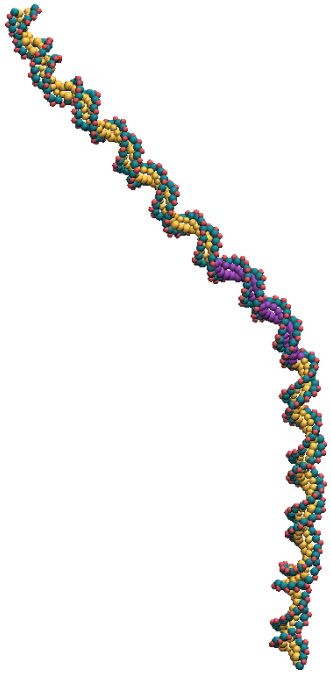
Figure 6.2: DNA-only configuration at frame 1000 of the $\lambda = 1$ endpoint, corresponding to the 1KX5 reference sequence. The simulation was initialized from the nucleosome-derived DNA geometry and subsequently evolved in the absence of histones. DNA phosphates, sugars and bases are shown in coral, teal and gold, respectively. The configuration was visualized in this work using VMD [21] and captured from the OpenGL display. The selected frame provides a structural illustration and is not, by itself, a test of equilibration or conformational preference.



(a) Initial configuration, first view.



(b) Initial configuration, second view.



(c) Frame 1000, extended configuration.

Figure 6.3: Illustrative configurations from the A25 DNA-only simulation at the $\lambda = 0$ endpoint. Panels (a) and (b) show the same closed initial configuration from two viewing directions, whereas panel (c) shows the more extended configuration observed at frame 1000. DNA phosphates, sugars and bases are shown in coral, teal and gold, respectively, with the central 25-bp A:T tract highlighted in purple. The configurations were visualized in this work using VMD [21] and captured from the OpenGL display. The snapshots illustrate the conformational change but do not, by themselves, establish equilibration or a preferred DNA shape.

Absolute local helical profiles for 1KX5

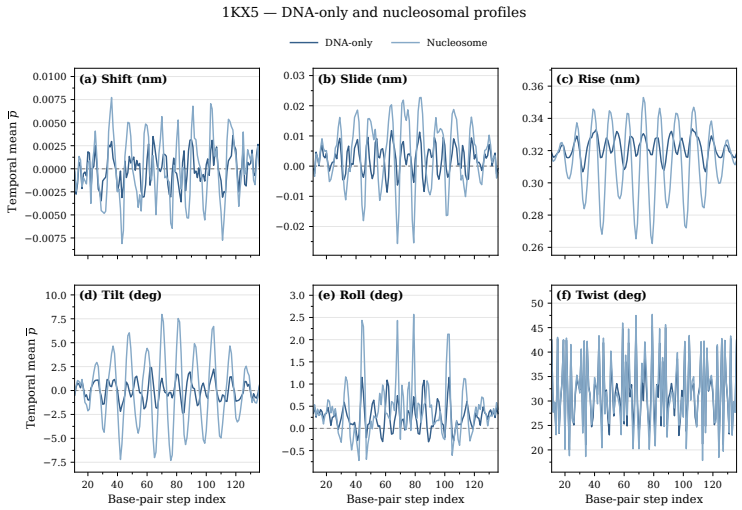


Figure 6.4: Temporally averaged absolute profiles of the six base-pair-step parameters for the 1KX5 sequence in the DNA-only and histone-associated environments. The dark-blue curves represent the DNA-only simulation and the light-blue curves the nucleosomal simulation; both curves therefore refer to the same sequence and differ only in the simulated environment. Shift, slide and rise are expressed in nanometres, while tilt, roll and twist are expressed in degrees. The profiles were obtained from realization R0 over base-pair steps 11–136 after discarding the first 100 saved frames. These profiles are shown separately from the difference profiles reported in the main text.

Local helical response of the chromosome XIII sequences

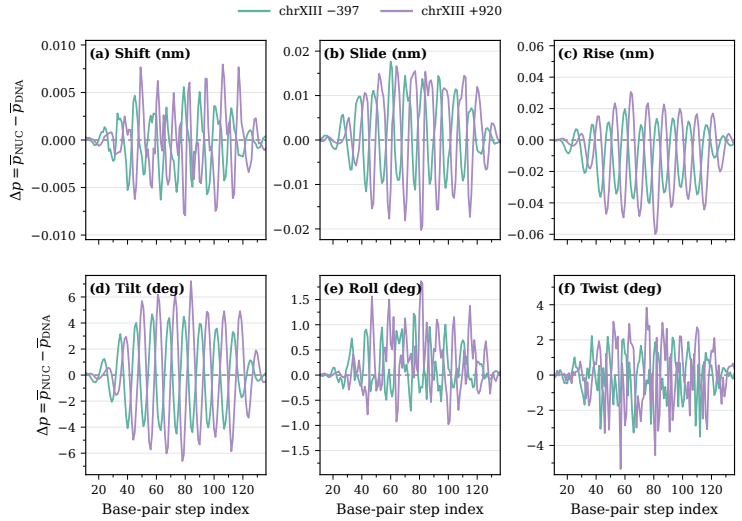


Figure 6.5: Histone-associated changes in the six base-pair-step parameters for the chromosome XIII −397 and +920 sequences in the 147-bp constructs without linker DNA. For each parameter p, the plotted quantity is $\Delta p(i) = \overline{p}_{\mathrm{nucleosome}}(i) - \overline{p}_{\mathrm{DNA}}(i)$. Shift, slide and rise are expressed in nanometres; tilt, roll and twist are expressed in degrees. The profiles are temporal means from realization R0, calculated over base-pair steps 11–136 after discarding the first 100 saved frames. The dashed horizontal lines mark zero change.

Spatially aligned helical-response profiles

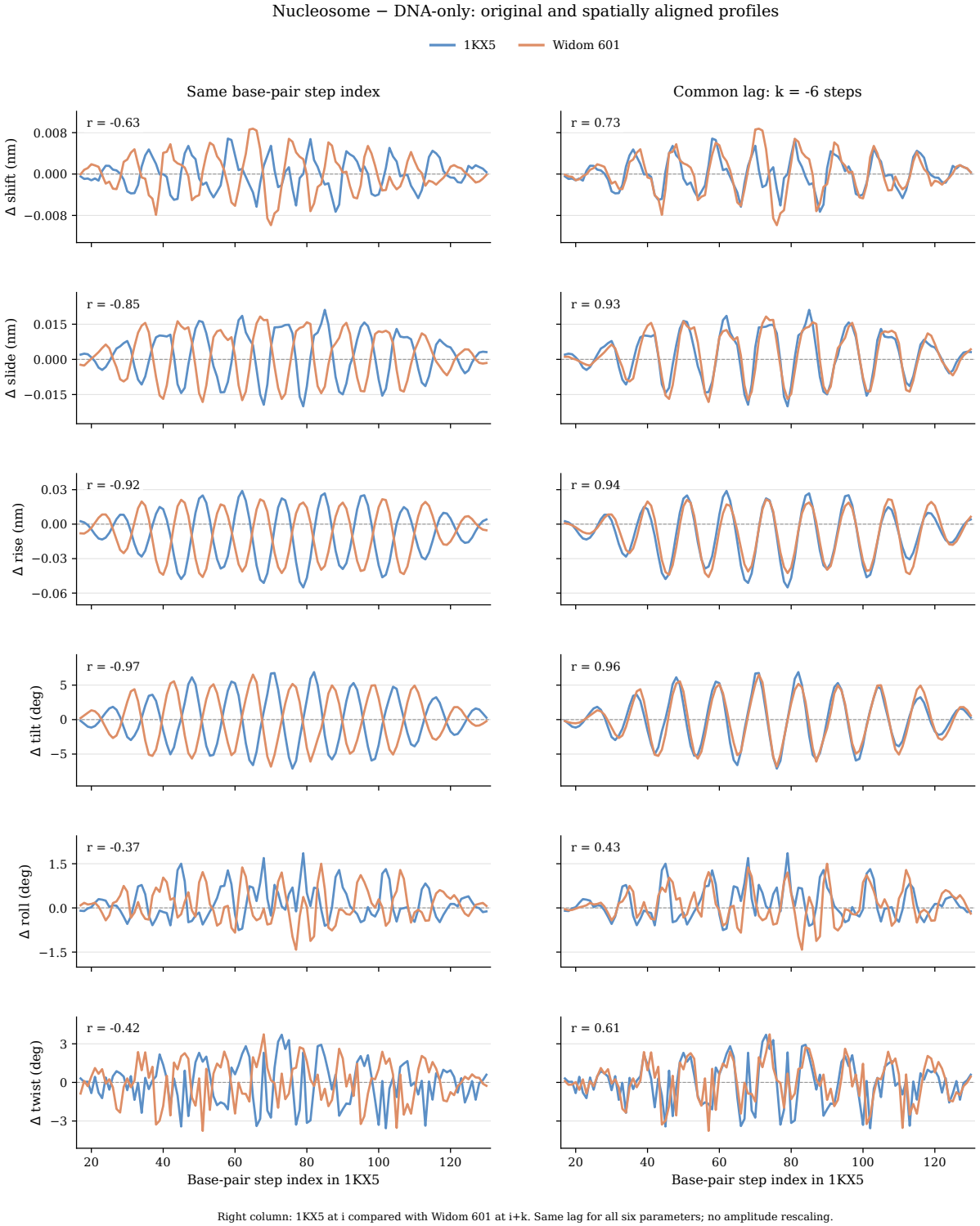


Figure 6.6: Comparison of the six histone-associated response profiles for 1KX5 and Widom 601 before and after a common spatial alignment. The plotted quantity is $\Delta p(i) = \overline{p}_{\mathrm{nucleosome}}(i) - \overline{p}_{\mathrm{DNA}}(i)$. The left column compares the profiles at the same base-pair-step index; in the right column, the Widom 601 profile is shifted by the common lag selected from the six signed spatial Pearson correlations. The same lag is used for all six parameters.

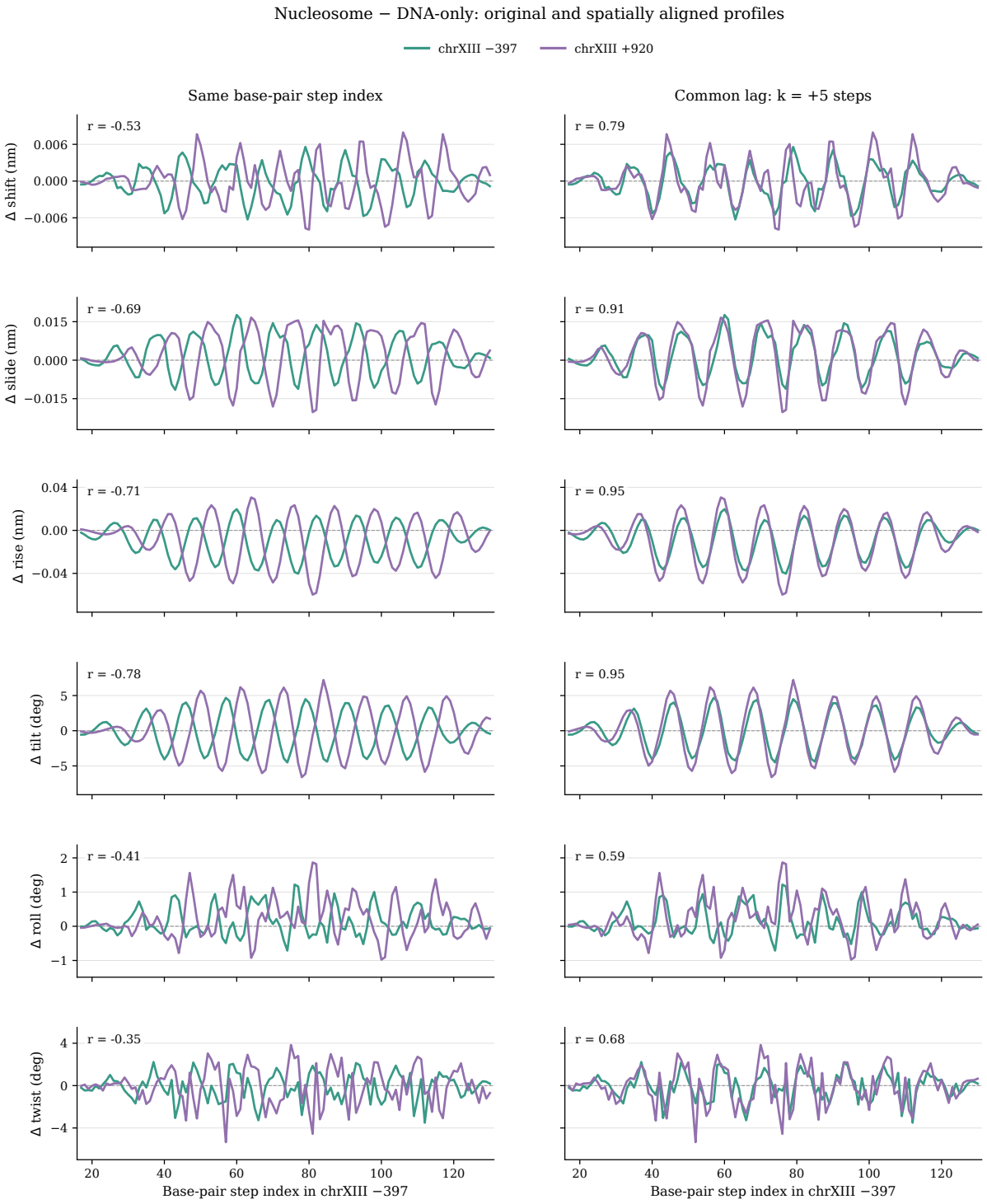


Figure 6.7: Comparison of the six histone-associated response profiles for the chromosome XIII −397 and +920 sequences before and after a common spatial alignment. The plotted quantity is $\Delta p(i) = \overline{p}_{\text{nucleosome}}(i) - \overline{p}_{\text{DNA}}(i)$. The left column compares the profiles at the same base-pair-step index; in the right column, the +920 profile is shifted by the common lag selected from the six signed spatial Pearson correlations. The same lag is used for all six parameters.

Effect of linker DNA on the local geometry of chromosome XIII

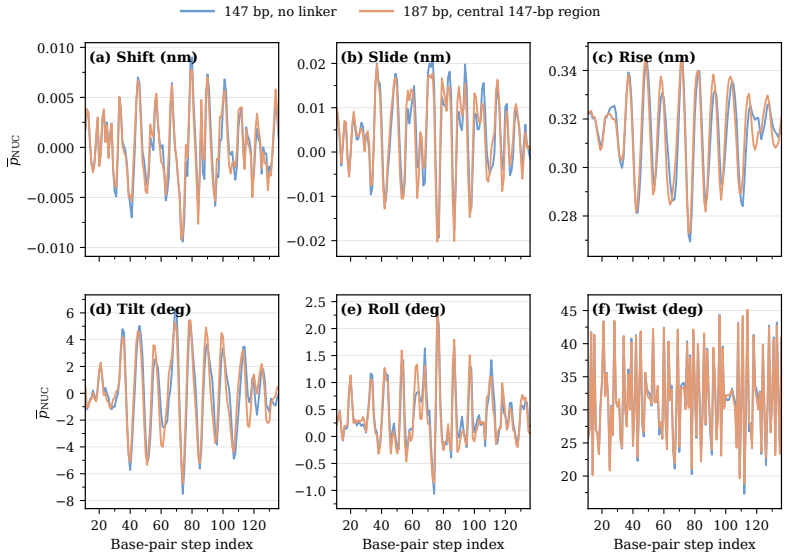


Figure 6.8: Comparison of the temporally averaged six base-pair-step parameters for the chromosome XIII −397 endpoint ($\lambda = 0$) in the histone-associated constructs with and without linker DNA. The blue profiles represent the 147-bp construct, whereas the orange profiles represent the equivalent central region of the 187-bp construct. For the latter, source steps 31–156 were mapped onto positions 11–136. The profiles were calculated from realization R0 after discarding the first 100 saved frames. Shift, slide and rise are expressed in nanometres; tilt, roll and twist are expressed in degrees.

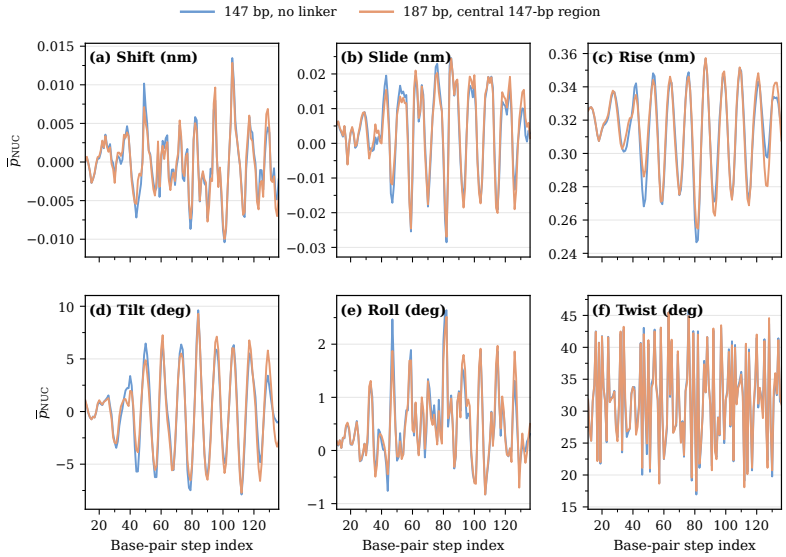


Figure 6.9: Comparison of the temporally averaged six base-pair-step parameters for the chromosome XIII +920 endpoint ($\lambda = 1$) in the histone-associated constructs with and without linker DNA. The blue profiles represent the 147-bp construct, whereas the orange profiles represent the equivalent central region of the 187-bp construct. For the latter, source steps 31–156 were mapped onto positions 11–136. The profiles were calculated from realization R0 after discarding the first 100 saved frames. Shift, slide and rise are expressed in nanometres; tilt, roll and twist are expressed in degrees.

Temporal variation of radial opening in A125

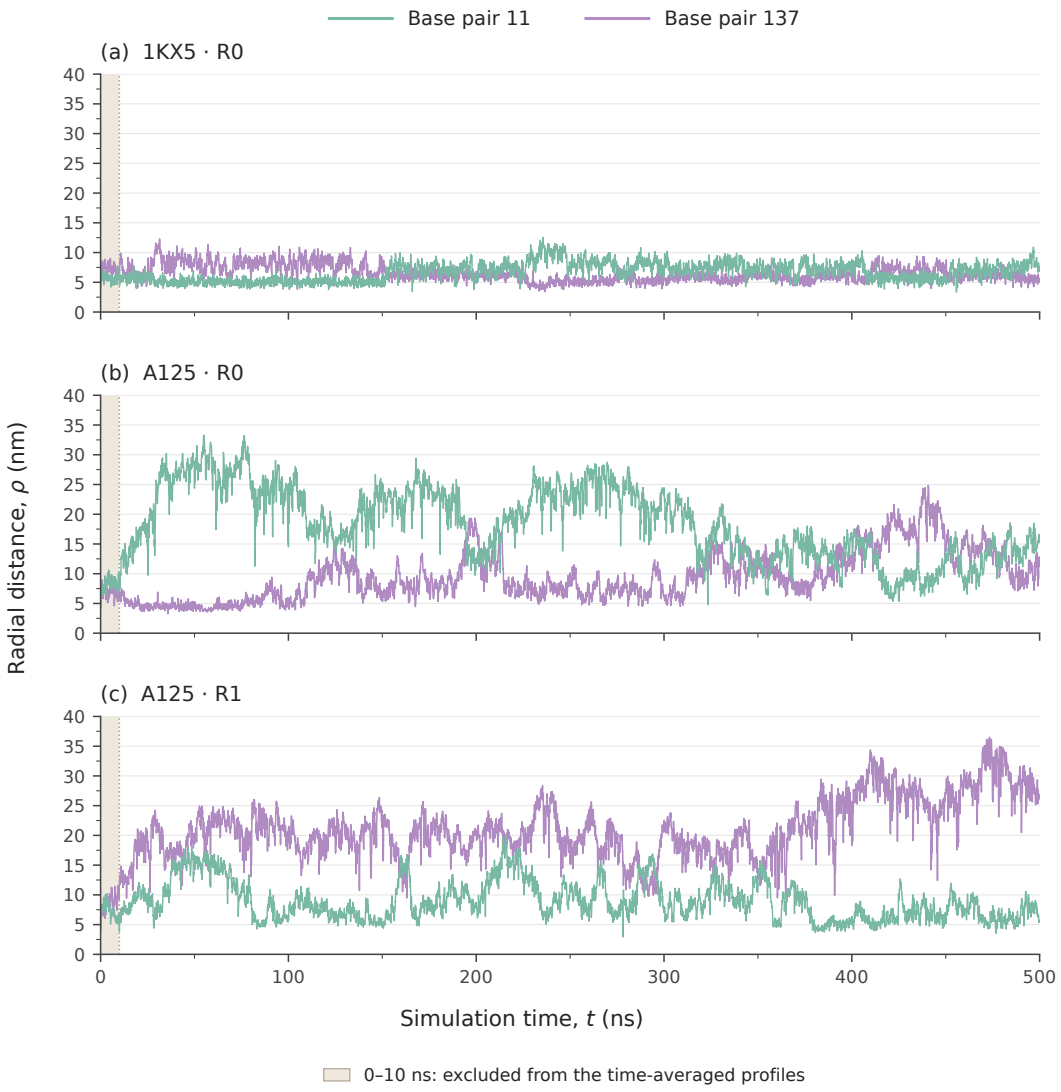


Figure 6.10: Time evolution of the radial distance of base pairs 11 and 137 from the nucleosome axis for individual realizations: (a) 1KX5 R0, (b) A125 R0 and (c) A125 R1. All saved frames are shown without smoothing. The shaded interval from 0 to 10 ns was excluded from the time-averaged radial profiles but is included here to show the initial structural relaxation. Base pairs 11 and 137 delimit the analysed interval and are not the physical termini of the 147-bp DNA fragment. The traces illustrate individual realizations and should not be interpreted as averages over the ten replicas or as estimates of equilibrium populations.

Radial positioning across the A-rich variants

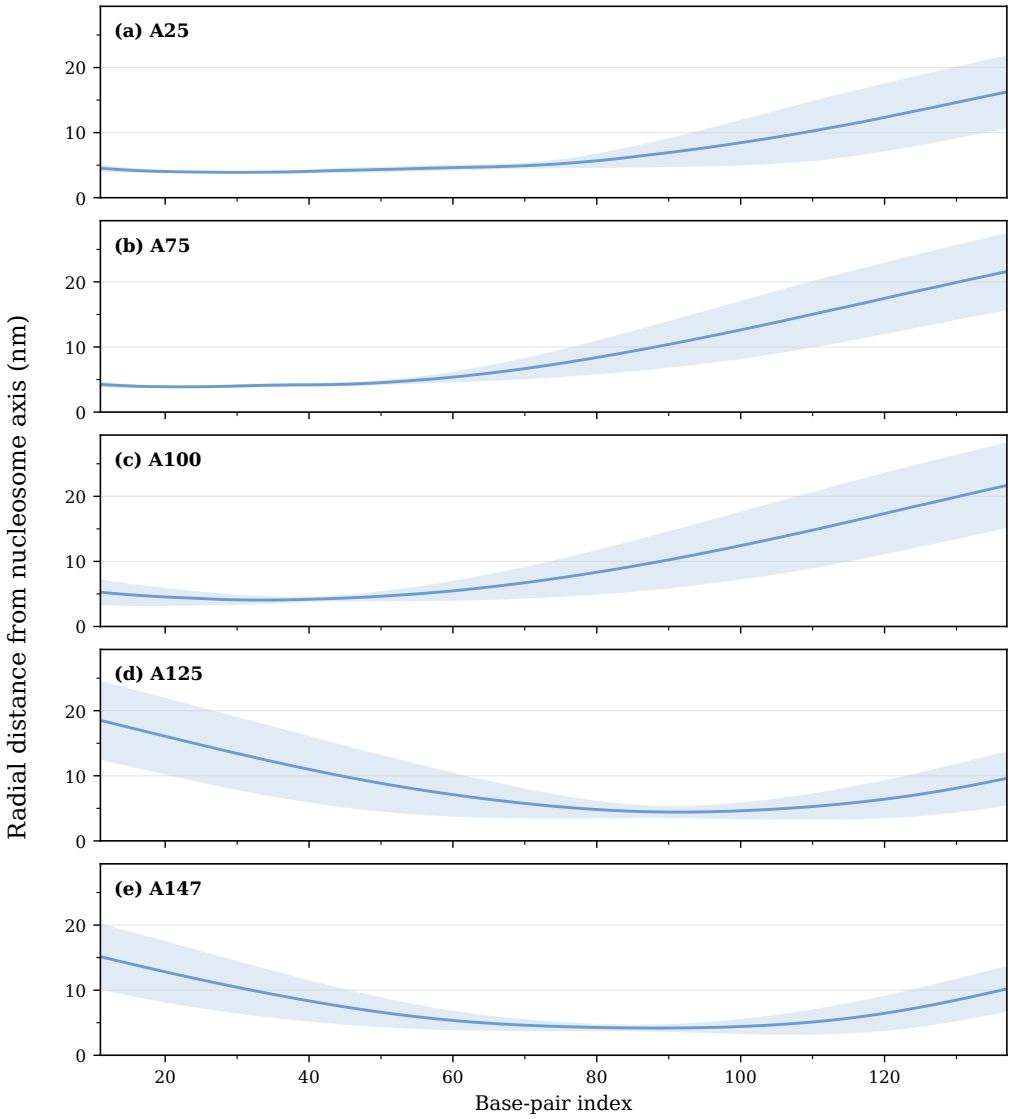


Figure 6.11: Radial-distance profiles for A25, A75, A100, A125 and A147 in the histone-associated simulations. The solid lines show the temporal mean distance of each base-pair centre from the nucleosome axis, and the shaded bands indicate the mean ± standard deviation. The profiles were obtained from realization R0 over base pairs 11–137 after discarding the first 100 saved frames. All panels use the same axis limits. The bands describe temporal fluctuations within each trajectory, not uncertainty in the mean. The 1KX5 and Widom 601 reference profiles are shown in Figure 4.6.

Radial positioning in chromosome XIII constructs with and without linker DNA

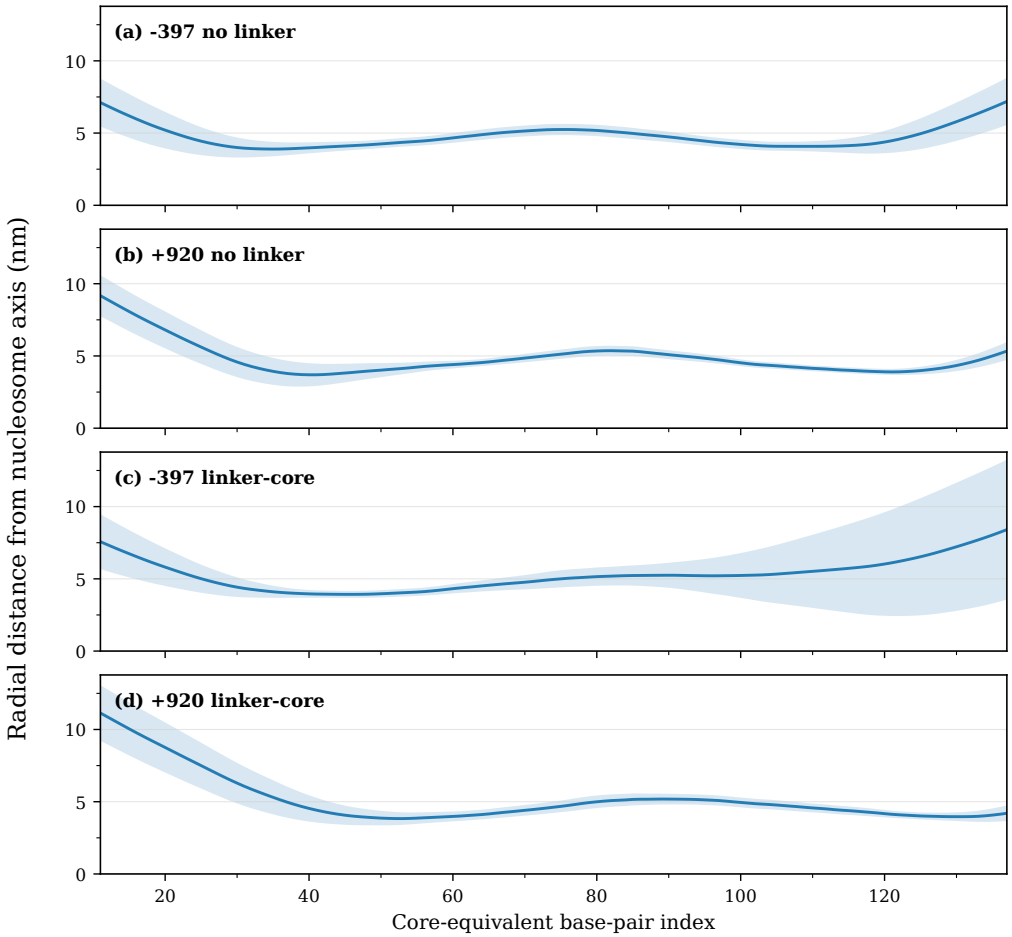


Figure 6.12: Radial-distance profiles for the chromosome XIII −397 and +920 endpoints in the histone-associated constructs with and without linker DNA. In the panel labels, "linker-core" denotes the central 147-bp region of the corresponding 187-bp linker construct, mapped onto the equivalent numbering of the 147-bp construct without linkers. The solid line represents the temporal mean distance of each base-pair centre from the nucleosome axis, while the shaded region represents the standard deviation around the mean. The profiles were obtained from realization R0 after discarding the first 100 saved frames; the shaded regions describe temporal fluctuations and not uncertainty in the mean.

Superhelix geometry and fit diagnostics

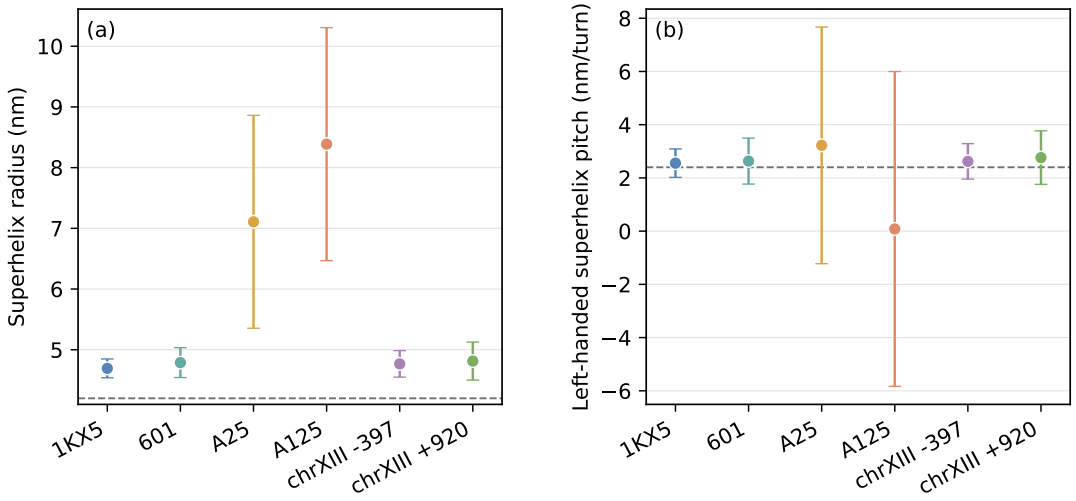


Figure 6.13: Effective superhelix radius (a) and pitch (b), including A25 and A125 alongside the four compact systems. The analysis used base pairs 11–137 of realization R0 after discarding the first 100 saved frames. Points show temporal means and bars show the standard deviation. The plotted pitch is $-P$, with positive values corresponding to left-handed winding under the adopted convention; its sign was retained before temporal averaging. A mean near zero can therefore result from cancellation between positive and negative fitted pitches and does not establish a planar DNA path. Dashed lines mark the canonical nucleosome values of 4.2 nm for the superhelix radius and 2.4 nm per turn for the pitch. The A-rich estimates should be read together with the fit diagnostics in Figure 6.14.

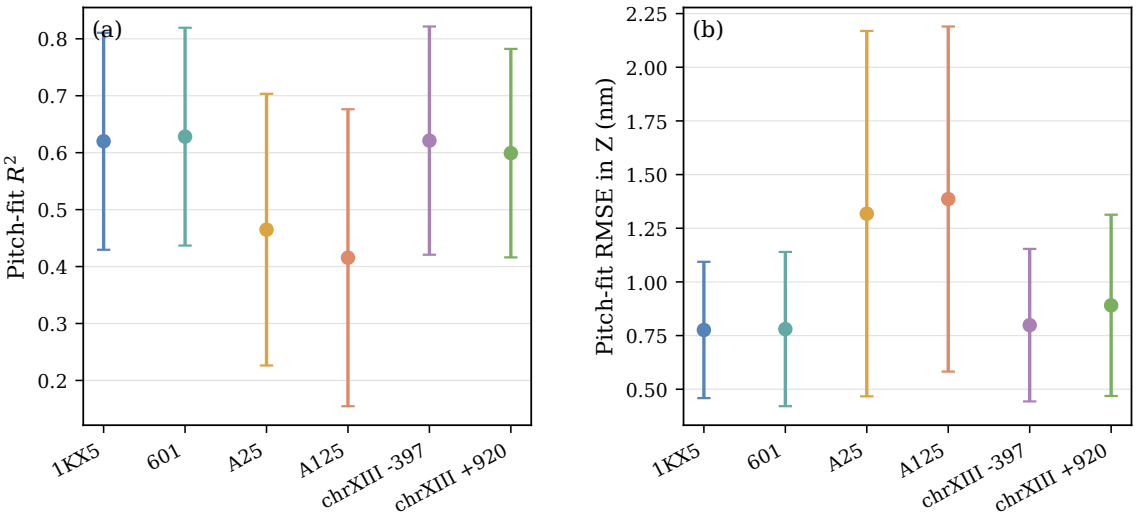


Figure 6.14: Diagnostics of the frame-wise linear fit $Z_i = a + m\phi_i$ used to estimate the effective superhelix pitch: (a) coefficient of determination R^2 and (b) root-mean-square error of the axial residuals, in nanometres. Points show temporal means and bars show the standard deviation over the retained R0 frames, using the same base-pair range and equilibration exclusion as Figure 6.13. These metrics describe the axial fit relative to the nucleosome-fixed reference frame, not a full three-dimensional fit of an ideal helix.

6.2 Appendix B: Supplementary Tables

Table 6.1: Reference, A-rich, randomized and Widom 601 DNA sequences used in the thermodynamic-integration simulations. All sequences have a total length of 147 bp. The labels A25–A125 indicate sequences obtained by replacing a central region of the 1KX5 reference sequence with A-rich tracts of increasing length, while A147 denotes the all-A sequence. R1–R5 are random control sequences, and 601 corresponds to the Widom 601 positioning sequence.

Label	DNA sequence
1KX5	ATCAATATCCACCTGCAGATACTACCAAAAGTGTATTTGGAAACTGCTCCATCAAAAGGCATGTTCAGCTGGAATCCAGCT GAACATGCCTTTTGATGGAGCAGTTTCCAAATACACTTTTGGTAGTATCTGCAGGTGGATATTGAT
A25	ATCAATATCCACCTGCAGATACTACCAAAAGTGTATTTGGAAACTGCTCCATCAAAAGGCAAAAAAAAAAAAAAAAAAAA AAAAATGCCTTTTGATGGAGCAGTTTCCAAATACACTTTTGGTAGTATCTGCAGGTGGATATTGAT
A50	ATCAATATCCACCTGCAGATACTACCAAAAGTGTATTTGGAAACTGCTCAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAATGGAGCAGTTTCCAAATACACTTTTGGTAGTATCTGCAGGTGGATATTGAT
A75	ATCAATATCCACCTGCAGATACTACCAAAAGTGTATAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATACACTTTTGGTAGTATCTGCAGGTGGATATTGAT
A100	ATCAATATCCACCTGCAGATACTAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATAGTATCTGCAGGTGGATATTGAT
A125	ATCAATATCCAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATGGATATTGAT
A147	AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
R1	AGTATGCCTGAAATTCGTACGAATTCAGCATTCGATCCGGAAATTCCGGGTACTGCAAGTCCTAGAAATACACAGAGGGCT CAATACGCATACAGCTATCGCCAATCCGCTCGTTTAAACGCGCCAAATTTTCACGGTACGGCATTA
R2	CCATTGCATACGAGATCGGTAGATTGATAAGGGACACAGAATATCCCCGGACGCAATAGACGGACAGCTTGGTATCCTAAG CACAGTCGCGCGTCCGAACCTAGCTCTACTTTAGAGGCCCCGGATTCTGGTGCTCGTAGACCGCAG
R3	AAATAGAGAGCTTTGATATCTGCATGTCTAGCTTTAGAATTCAGTTTAGCGCGCTGATCTGAGTCGAGATAAAATCACCAG TACCCAAGACCAGGGGGGCTCGCCACGTTGGCTAATCCTGGTACATCTTGTAATCAATATTCAGTA
R4	GAAAATTTGTGTTAGAAGGACGAGTCACCATGTACCAATAGCGATAACGATCGGTCGGACTATTCATTGTGGTGATGACGC TGGGTTTACGTGGGAAAGGTGCTTGTGTCCCGACAGGCTAGGATATAATGCTGAGGCCCTTCCCCA
R5	AGCGTTCAGCGTGGGATTTGCTACAACTTCCGAGTCCTACATGTGCGTGTTCATGTTATGTATGCACAAGGCCGAGAATAG GACGTAGCCTTCGAGTTAGTACGTAGCGTGGTCGCACAAGCACAGTAGATCCTCCCCGCGCATCCT
601	ATCGAGAATCCCGGTGCCGAGGCCGCTCAATTGGTCGTAGACAGCTCTAGCACCGCTTAAACGCACGTACGCGCTGTCCCC CGCGTTTTAACCGCCAAGGGGATTACTCCCTAGTCTCCAGGCACGTGTCAGATATATACATCCGAT

Table 6.2: Genomic chromosome XIII sequences used in the thermodynamic-integration simulations. The labels identify the centres of the two 147-bp fragments relative to the transcription start site of profile `chrXIII_NM_001182668.1`. For the extended constructs, 20 adenines were added to each end of the corresponding genomic fragment, giving a total length of 187 bp.

Label	DNA sequence
chrXIII −397	ATGATCCTGTTGGTACCCAGATTCTAGGTCTCCATTGAACAATGATTGAAAAATTTCCCACCATTCCAGTAAAAACG TGTCATTAGCGTCCATCAGCATTTTCGAAGGTAATAGCCCATCTTCACTAACAATGTCACCCTTAGATGC
chrXIII +920	AGAACCTAGCTCCAAATATATTTGTTTTCCAATGTCCCCTTTGGACATTGTAGATTTATTGGGCGTGGATCCTGAGC TACTATGGGAACTCACCAGGTCAGACGCATTTTCTACATCAGACGAAGCGGCTAAAACTTCGTTGTTATT

6.3 Appendix C: Thermodynamic-Integration Derivatives

The free-energy difference between the target and reference sequences was calculated by Thermodynamic Integration,

$$\Delta F = \int_0^1 \left\langle \frac{\partial U}{\partial \lambda} \right\rangle_\lambda d\lambda. \tag{6.1}$$

This appendix reports the analytical expressions used to evaluate the instantaneous derivative $\partial U/\partial\lambda$ from each saved configuration. The resulting values were averaged separately at every simulated value of λ, as described in Sections 3.6 and 3.7.

Interpolation of sequence-dependent parameters

Each sequence-dependent parameter x was linearly interpolated between the target sequence, corresponding to $\lambda = 0$, and the reference sequence, corresponding to $\lambda = 1$:

$$x(\lambda) = (1-\lambda)x^{(0)} + \lambda x^{(1)}. \tag{6.2}$$

Its derivative is therefore constant along the alchemical path:

$$x' \equiv \frac{dx}{d\lambda} = x^{(1)} - x^{(0)}. \tag{6.3}$$

At a fixed molecular configuration, the derivative of the potential energy was decomposed as

$$\frac{\partial U}{\partial \lambda} = \frac{\partial U_{\text{bond}}}{\partial \lambda} + \frac{\partial U_{\text{angle}}}{\partial \lambda} + \frac{\partial U_{\text{dihedral}}}{\partial \lambda} + \frac{\partial U_{\text{WCA}}}{\partial \lambda}. \tag{6.4}$$

These are the terms whose sequence-dependent parameters change explicitly along the alchemical transformation.

Bonded contributions

For the harmonic bond potential

$$U_{\text{bond}} = \sum_b k_b (r_b - r_{0,b})^2, \tag{6.5}$$

the corresponding derivative is

$$\frac{\partial U_{\text{bond}}}{\partial \lambda} = \sum_b \left[k_b'(r_b - r_{0,b})^2 - 2k_b(r_b - r_{0,b})r_{0,b}' \right]. \tag{6.6}$$

Here, r_b is the instantaneous bond length, whereas k_b and $r_{0,b}$ are, respectively, the interpolated force constant and equilibrium bond length.

For the angular potential

$$U_{\text{angle}} = \sum_a k_a (\theta_a - \theta_{0,a})^2, \tag{6.7}$$

the derivative is

$$\frac{\partial U_{\text{angle}}}{\partial \lambda} = \sum_a \left[k'_a(\theta_a - \theta_{0,a})^2 - 2k_a(\theta_a - \theta_{0,a})\theta'_{0,a} \right]. \tag{6.8}$$

In this expression, θ_a is the instantaneous angle and $\theta_{0,a}$ is its interpolated equilibrium value.

For the dihedral potential

$$U_{\text{dihedral}} = \sum_d k_d \left[1 + \cos(\phi_d - \phi_{0,d})\right], \tag{6.9}$$

differentiation at fixed instantaneous dihedral angle ϕ_d gives

$$\frac{\partial U_{\text{dihedral}}}{\partial \lambda} = \sum_d \left\{ k'_d \left[1 + \cos(\phi_d - \phi_{0,d})\right] + k_d \sin(\phi_d - \phi_{0,d})\phi'_{0,d} \right\}. \tag{6.10}$$

Excluded-volume contribution

The effective interaction distance for a pair of beads i, j is

$$\sigma_{ij} = R_i + R_j, \qquad \sigma'_{ij} = R'_i + R'_j, \tag{6.11}$$

where R_i and R_j are the corresponding bead radii. For an interacting pair satisfying $r_{ij} < \sigma_{ij}$, the WCA contribution to the derivative is

$$\frac{\partial U_{\text{WCA},ij}}{\partial \lambda} = 12\epsilon \frac{\sigma'_{ij}}{\sigma_{ij}} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right], \tag{6.12}$$

and is zero outside the interaction range. The total WCA derivative is obtained by summing Eq. 6.12 over all relevant non-excluded pairs.

Terms independent of λ

Electrostatic interactions and the terms defining the histone model do not contribute explicitly to $\partial U/\partial\lambda$, because their parameters remain unchanged during the alchemical transformation. Nevertheless, they determine the configurations sampled in each thermodynamic window and therefore influence the ensemble average $\langle \partial U/\partial\lambda \rangle_\lambda$.

6.4 Appendix D: Validation and convergence figures

Structural persistence of the 1KX5 nucleosome

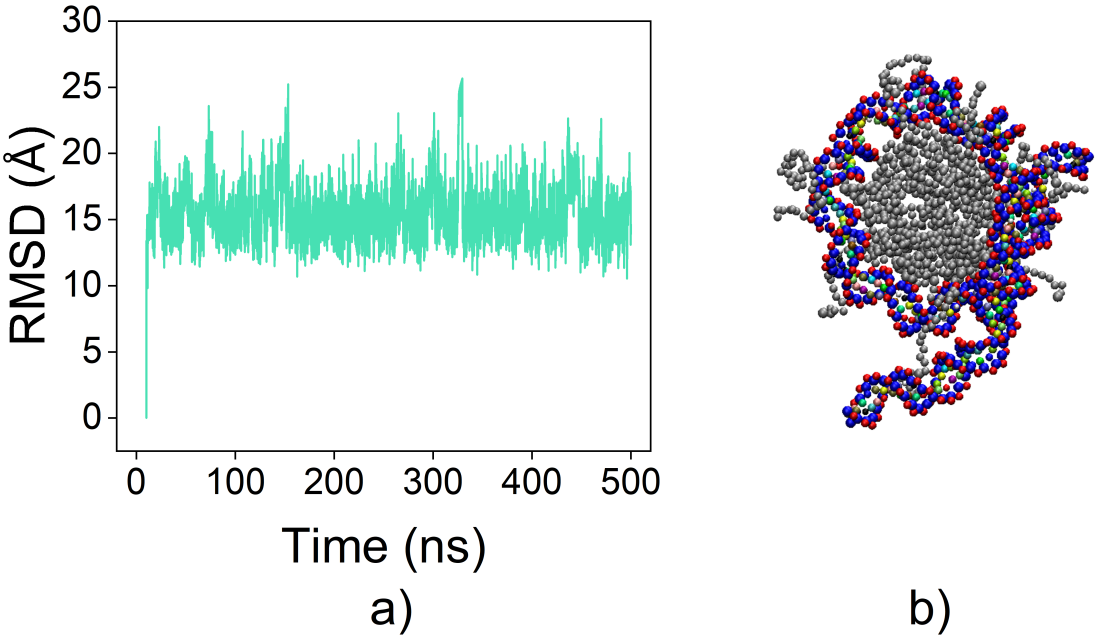


Figure 6.15: Structural check for a representative 1KX5 nucleosome trajectory. (a) Root-mean-square deviation (RMSD) calculated using all coarse-grained nucleosome beads after optimal rotational alignment. The first 10 ns were excluded and the first frame after this interval was used as the reference. (b) Configuration at the final saved frame of the same trajectory. The figure is reproduced from Figure 5 of Estacio Mateos [31], from the shared simulation campaign.

Representative time-convergence profile

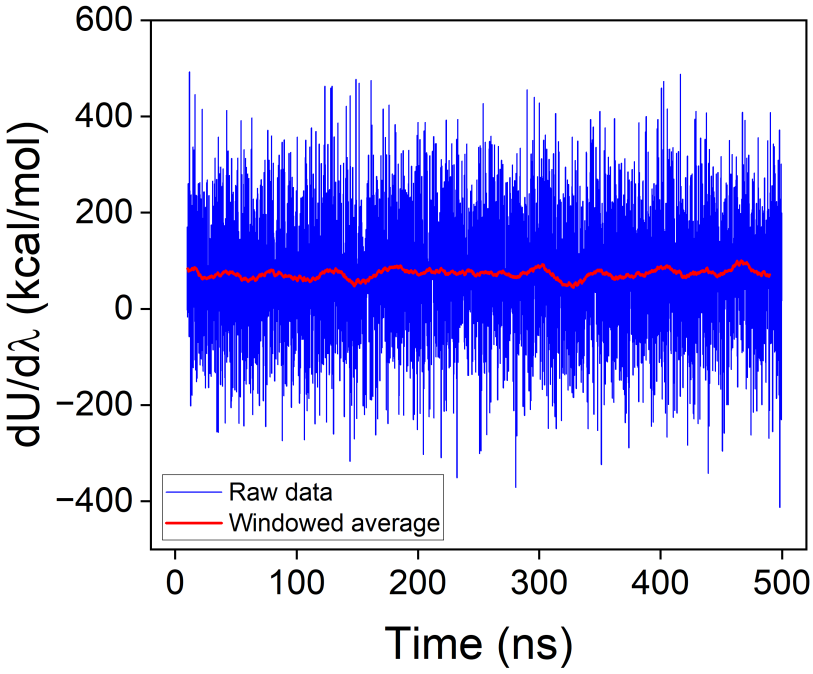


Figure 6.16: Temporal evolution of $\partial U/\partial\lambda$ at $\lambda = 0.5$ for one representative realization of the 1KX5–R1 transformation. The raw time series is shown in blue and its windowed average in red. The figure is reproduced from the shared analysis reported by Estacio Mateos [31].

Representative Thermodynamic Integration profile

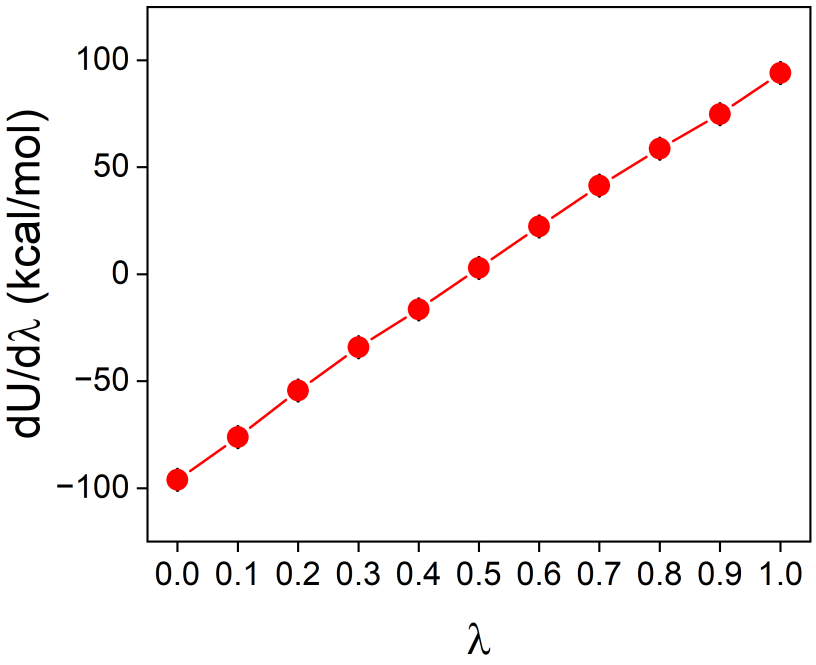


Figure 6.17: Representative Thermodynamic Integration profile for the 1KX5–R1 transformation, showing the ensemble-averaged derivative $\langle\partial U/\partial\lambda\rangle_\lambda$ at the eleven simulated values of λ. The figure is reproduced from the shared analysis reported by Estacio Mateos [31].

Sensitivity to the spacing of the alchemical windows

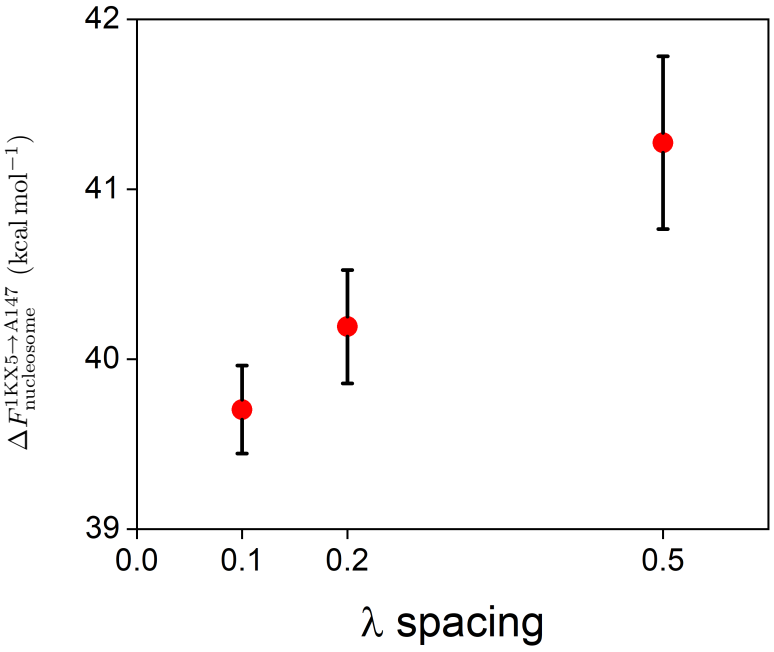


Figure 6.18: Dependence of the integrated free-energy difference on the spacing between consecutive λ windows for the 1KX5 $\rightarrow$ A147 transformation in the histone-associated system. The original figure uses the transformation direction adopted in the shared analysis by Estacio Mateos [31].

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